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ORGANIC ELECTROLUMINESCENT MATERIALS AND DEVICES

Abstract

The present disclosure provides a compound having a first ligand L.sub.A comprising a structure of Formula I:

##STR00001##

In Formula I, each of ring A and ring B is a 5- or 6-membered ring; each of X.sup.1 to X.sup.3 is C or N; Y.sup.1 and Y.sup.2 are C, Si, or Ge; each  custom-character is a single bond or a double bond; L is a linking group; each R.sup.1, R.sup.2, R.sup.3, R.sup.4, R, R', R.sup.A, and R.sup.B is hydrogen or a General Substituent; and L.sub.A is coordinated to a metal M. Formulations, OLEDs, and consumer products containing the compound are also provided.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/563,727, filed on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure generally relates to organic or metal coordination compounds and formulations and their various uses including as emitters, sensitizers, charge transporters, or exciton transporters in devices such as organic light emitting diodes and related electronic devices and consumer products.

BACKGROUND

[0003] Opto-electronic devices that make use of organic materials are becoming increasingly desirable for various reasons. Many of the materials used to make such devices are relatively inexpensive, so organic opto-electronic devices have the potential for cost advantages over inorganic devices. In addition, the inherent properties of organic materials, such as their flexibility, may make them well suited for particular applications such as fabrication on a flexible substrate. Examples of organic opto-electronic devices include organic light emitting diodes/devices (OLEDs), organic phototransistors, organic photovoltaic cells, organic scintillators, and organic photodetectors. For OLEDs, the organic materials may have performance advantages over conventional materials.

[0004] OLEDs make use of thin organic films that emit light when voltage is applied across the device. OLEDs are becoming an increasingly interesting technology for use in applications such as displays, illumination, and backlighting.

[0005] One application for emissive molecules is a full color display. Industry standards for such a display call for pixels adapted to emit particular colors, referred to as “saturated” colors. In particular, these standards call for saturated red, green, and blue pixels. Alternatively, the OLED can be designed to emit white light. In conventional liquid crystal displays emission from a white backlight is filtered using absorption filters to produce red, green and blue emission. The same technique can also be used with OLEDs. The white OLED can be either a single emissive layer (EML) device or a stack structure. Color may be measured using CIE coordinates, which are well known to the art.

SUMMARY

[0006] In one aspect, the present disclosure provides a compound having a first ligand L.sub.A comprising a structure of Formula I:

##STR00002##

In Formula I:

[0007] each of ring A and ring B is independently a 5-membered or 6-membered carbocyclic or heterocyclic ring; [0008] each of X.sup.1 to X.sup.3 is independently C or N; [0009] each of Y.sup.1 and Y.sup.2 is independently C, Si, or Ge; [0010] each  independently represents a single bond or a double bond; [0011] L is selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO.sub.2, CRR', SiRR', and GeRR'; [0012] each of R.sup.A and R.sup.B independently represents mono to the maximum allowable substitutions, or no substitutions; [0013] each R.sup.1, R.sup.2, R.sup.3,

R.sup.4, R, R', R.sup.A, and R.sup.B is independently hydrogen, or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and combinations thereof; [0014] L.sub.A is coordinated to a metal M selected from the group consisting of Ir, Rh, Re, Ru, Os, Pt, Pd, Ag, Au, and Cu; [0015] M can be coordinated to other ligands; [0016] L.sub.A can be a monodentate, bidentate, tridentate, tetradentate, pentadentate, or hexadentate ligand; and any two substituents can be joined or fused to form a ring.

[0017] In another aspect, the present disclosure provides a formulation comprising a compound having a first ligand L.sub.A comprising a structure of Formula I as described herein.

[0018] In yet another aspect, the present disclosure provides an OLED having an organic layer comprising a compound having a first ligand L.sub.A comprising a structure of Formula I as described herein.

[0019] In yet another aspect, the present disclosure provides a consumer product comprising an OLED with an organic layer comprising a compound having a first ligand L.sub.A comprising a structure of Formula I as described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 shows an organic light emitting device.

[0021] FIG. 2 shows an inverted organic light emitting device that does not have a separate electron transport layer.

DETAILED DESCRIPTION

A. Terminology

[0022] Unless otherwise specified, the below terms used herein are defined as follows:

[0023] As used herein, “top” means furthest away from the substrate, while “bottom” means closest to the substrate. Where a first layer is described as “disposed over” a second layer, the first layer is disposed further away from substrate. There may be other layers between the first and second layer, unless it is specified that the first layer is “in contact with” the second layer. For example, a cathode may be described as “disposed over” an anode, even though there are various organic layers in between.

[0024] As used herein, “solution processable” means capable of being dissolved, dispersed, or transported in and/or deposited from a liquid medium, either in solution or suspension form.

[0025] As used herein, and as would be generally understood by one skilled in the art, a first “Highest Occupied Molecular Orbital” (HOMO) or “Lowest Unoccupied Molecular Orbital” (LUMO) energy level is “greater than” or “higher than” a second HOMO or LUMO energy level if the first energy level is closer to the vacuum energy level. Since ionization potentials (IP) are measured as a negative energy relative to a vacuum level, a higher HOMO energy level corresponds to an IP having a smaller absolute value (an IP that is less negative). Similarly, a higher LUMO energy level corresponds to an electron affinity (EA) having a smaller absolute value (an EA that is less negative). On a conventional energy level diagram, with the vacuum level at the top, the LUMO energy level of a material is higher than the HOMO energy level of the same material. A “higher” HOMO or LUMO energy level appears closer to the top of such a diagram than a “lower” HOMO or LUMO energy level.

[0026] As used herein, and as would be generally understood by one skilled in the art, a first work function is “greater than” or “higher than” a second work function if the first work function has a higher absolute value. Because work functions are generally measured as negative numbers relative

to vacuum level, this means that a “higher” work function is more negative. On a conventional energy level diagram, with the vacuum level at the top, a “higher” work function is illustrated as further away from the vacuum level in the downward direction. Thus, the definitions of HOMO and LUMO energy levels follow a different convention than work functions.

[0027] Layers, materials, regions, and devices may be described herein in reference to the color of light they emit. In general, as used herein, an emissive region that is described as producing a specific color of light may include one or more emissive layers disposed over each other in a stack.

[0028] As used herein, a “NIR”, “red”, “green”, “blue”, “yellow” layer, material, region, or device refers to a layer, a material, a region, or a device that emits light in the wavelength range of about 700-1500 nm, 580-700 nm, 500-600 nm, 400-500 nm, 540-600 nm, respectively, or a layer, a material, a region, or a device that has a highest peak in its emission spectrum in the respective wavelength region. In some arrangements, separate regions, layers, materials, or devices may provide separate “deep blue” and “light blue” emissions. As used herein, the “deep blue” emission component refers to an emission having a peak emission wavelength that is at least about 4 nm less than the peak emission wavelength of the “light blue” emission component. Typically, a “light blue” emission component has a peak emission wavelength in the range of about 465-500 nm, and a “deep blue” emission component has a peak emission wavelength in the range of about 400-470 nm, though these ranges may vary for some configurations.

[0029] In some arrangements, a color altering layer that converts, modifies, or shifts the color of the light emitted by another layer to an emission having a different wavelength is provided. Such a color altering layer can be formulated to shift wavelength of the light emitted by the other layer by a defined amount, as measured by the difference in the wavelength of the emitted light and the wavelength of the resulting light. In general, there are two classes of color altering layers: color filters that modify a spectrum by removing light of unwanted wavelengths, and color changing layers that convert photons of higher energy to lower energy. For example, a “red” color filter can be present in order to filter an input light to remove light having a wavelength outside the range of about 580-700 nm. A component “of a color” refers to a component that, when activated or used, produces or otherwise emits light having a particular color as previously described. For example, a “first emissive region of a first color” and a “second emissive region of a second color different than the first color” describes two emissive regions that, when activated within a device, emit two different colors as previously described.

[0030] As used herein, emissive materials, layers, and regions may be distinguished from one another and from other structures based upon light initially generated by the material, layer or region, as opposed to light eventually emitted by the same or a different structure. The initial light generation typically is the result of an energy level change resulting in emission of a photon. For example, an organic emissive material may initially generate blue light, which may be converted by a color filter, quantum dot or other structure to red or green light, such that a complete emissive stack or sub-pixel emits the red or green light. In this case the initial emissive material, region, or layer may be referred to as a “blue” component, even though the sub-pixel is a “red” or “green” component.

[0031] In some cases, it may be preferable to describe the color of a component such as an emissive region, sub-pixel, color altering layer, or the like, in terms of 1931 CIE coordinates. For example, a yellow emissive material may have multiple peak emission wavelengths, one in or near an edge of the “green” region, and one within or near an edge of the “red” region as previously described. Accordingly, as used herein, each color term also corresponds to a shape in the 1931 CIE coordinate color space. The shape in 1931 CIE color space is constructed by following the locus between two color points and any additional interior points. For example, interior shape parameters for red, green, blue, and yellow may be defined as shown below:

TABLE-US-00001 Color CIE Shape Parameters Central Red Locus: [0.6270, 0.3725]; [0.7347, 0.2653]; Interior: [0.5086, 0.2657] Central Green Locus: [0.0326, 0.3530]; [0.3731, 0.6245];

Interior: [0.2268, 0.3321 Central Blue Locus: [0.1746, 0.0052]; [0.0326, 0.3530]; Interior: [0.2268, 0.3321] Central Yellow Locus: [0.3731, 0.6245]; [0.6270, 0.3725]; Interior: [0.3700, 0.4087]; [0.2886, 0.4572]

[0032] The terms “halo,” “halogen,” and “halide” are used interchangeably and refer to fluorine, chlorine, bromine, and iodine.

[0033] The term “acyl” refers to a substituted carbonyl group (—C(O)—R.sub.s).

[0034] The term “ester” refers to a substituted oxycarbonyl (—O—C(O)—R.sub.s or —C(O)—O—R.sub.s) group.

[0035] The term “ether” refers to an —OR.sub.s group.

[0036] The terms “sulfanyl” or “thio-ether” are used interchangeably and refer to a —SR.sub.s group.

[0037] The term “selenyl” refers to a —SeR.sub.s group.

[0038] The term “sulfinyl” refers to a —S(O)—R.sub.s group.

[0039] The term “sulfonyl” refers to a —SO.sub.2—R.sub.s group.

[0040] The term “phosphino” refers to a group containing at least one phosphorus atom bonded to the relevant structure. Common examples of phosphino groups include, but are not limited to, groups such as a —P(R.sub.s).sub.2 group or a $\text{—PO(R.sub.s).sub.2}$ group, wherein each R.sub.s can be same or different.

[0041] The term “silyl” refers to a group containing at least one silicon atom bonded to the relevant structure. Common examples of silyl groups include, but are not limited to, groups such as a $\text{—Si(R.sub.s).sub.3}$ group, wherein each R.sub.s can be same or different.

[0042] The term “germyl” refers to a group containing at least one germanium atom bonded to the relevant structure. Common examples of germyl groups include, but are not limited to, groups such as a $\text{—Ge(R.sub.s).sub.3}$ group, wherein each R.sub.s can be same or different.

[0043] The term “boryl” refers to a group containing at least one boron atom bonded to the relevant structure. Common examples of boryl groups include, but are not limited to, groups such as a —B(R.sub.s).sub.2 group or its Lewis adduct —B(R.sub.s).sub.3 group, wherein R.sub.s can be same or different.

[0044] In each of the above, R.sub.s can be hydrogen or a substituent selected from the group consisting of the general substituents as defined in this application. Preferred R.sub.s is selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, and combination thereof. More preferably R.sub.s is selected from the group consisting of alkyl, cycloalkyl, aryl, heteroaryl, and combination thereof.

[0045] The term “alkyl” refers to and includes both straight and branched chain alkyl groups having an alkyl carbon atom bonded to the relevant structure. Preferred alkyl groups are those containing from one to fifteen carbon atoms, preferably one to nine carbon atoms, and includes methyl, ethyl, n-propyl, iso-propyl, n-butyl, sec-butyl, iso-butyl, tert-butyl, n-pentyl, 2-methylbutyl, 3-methylbutyl, 2,2-dimethylpropyl, 1,3-dimethylpropyl, 1,1-dimethylpropyl, 2-ethylpropyl, 1,2-dimethylpropyl, n-hexyl, 2-methylpentyl, 3-methylpentyl, 2,2-dimethylbutyl, 2,3-dimethylbutyl, n-heptyl, 2-methylhexyl, 3-methylhexyl, 2,2-dimethylpentyl, 2,3-dimethylpentyl, 2,4-dimethylpentyl, 3,3-dimethylpentyl, 3-ethylpentyl, 2,2,3-trimethylbutyl, and the like. Additionally, the alkyl group can be further substituted.

[0046] The term “cycloalkyl” refers to and includes monocyclic, polycyclic, and spiro alkyl groups having a ring alkyl carbon atom bonded to the relevant structure. Preferred cycloalkyl groups are those containing 3 to 12 ring carbon atoms and includes cyclopropyl, cyclopentyl, cyclohexyl, bicyclo[3.1.1]heptyl, spiro[4.5]decyl, spiro[5.5]undecyl, adamantyl, and the like. Additionally, the cycloalkyl group can be further substituted.

[0047] The terms “heteroalkyl” or “heterocycloalkyl” refer to an alkyl or a cycloalkyl group, respectively, having at least one carbon atom replaced by a heteroatom. Optionally the at least one

heteroatom is selected from O, S, N, P, B, Si, Ge and Se, preferably, O, S or N. Additionally, the heteroalkyl or heterocycloalkyl group can be further substituted.

[0048] The term “alkenyl” refers to and includes both straight and branched chain alkene groups. Alkenyl groups are essentially alkyl groups that include at least one carbon-carbon double bond in the alkyl chain with one carbon atom from the carbon-carbon double bond that is bonded to the relevant structure. Cycloalkenyl groups are essentially cycloalkyl groups that include at least one carbon-carbon double bond in the cycloalkyl ring. The term “heteroalkenyl” as used herein refers to an alkenyl group having at least one carbon atom replaced by a heteroatom. Optionally the at least one heteroatom is selected from O, S, N, P, B, Si, Ge, and Se, preferably, O, S, or N. Preferred alkenyl, cycloalkenyl, or heteroalkenyl groups are those containing two to fifteen carbon atoms. Additionally, the alkenyl, cycloalkenyl, or heteroalkenyl group can be further substituted.

[0049] The term “alkynyl” refers to and includes both straight and branched chain alkyne groups. Alkynyl groups are essentially alkyl groups that include at least one carbon-carbon triple bond in the alkyl chain with one carbon atom from the carbon-carbon triple bond that is bonded to the relevant structure. Preferred alkynyl groups are those containing two to fifteen carbon atoms. Additionally, the alkynyl group can be further substituted.

[0050] The terms “aralkyl” or “arylalkyl” are used interchangeably and refer to an aryl-substituted alkyl group having an alkyl carbon atom bonded to the relevant structure. Additionally, the aralkyl group can be further substituted.

[0051] The term “heterocyclic group” refers to and includes aromatic and non-aromatic cyclic groups containing at least one heteroatom. Optionally the at least one heteroatom is selected from O, S, Se, N, P, B, Si, Ge, and Se, preferably, O, S, N, or B. Hetero-aromatic cyclic groups may be used interchangeably with heteroaryl. Preferred hetero-non-aromatic cyclic groups are those containing 3 to 10 ring atoms, preferably those containing 3 to 7 ring atoms, which includes at least one hetero atom, and includes cyclic amines such as morpholino, piperidino, pyrrolidino, and the like, and cyclic ethers/thio-ethers, such as tetrahydrofuran, tetrahydropyran, tetrahydrothiophene, and the like. Additionally, the heterocyclic group can be further substituted or fused.

[0052] The term “aryl” refers to and includes both single-ring and polycyclic aromatic hydrocarbonyl groups. The polycyclic rings may have two or more rings in which two carbons are common to two adjoining rings (the rings are “fused”). Preferred aryl groups are those containing six to thirty carbon atoms, preferably six to twenty-four carbon atoms, six to eighteen carbon atoms, and more preferably six to twelve carbon atoms. Especially preferred is an aryl group having six carbons, ten carbons, twelve carbons, fourteen carbons, or eighteen carbons. Suitable aryl groups include phenyl, biphenyl, triphenyl, triphenylene, tetraphenylene, naphthalene, anthracene, phenalene, phenanthrene, pyrene, chrysene, perylene, and azulene, preferably phenyl, biphenyl, triphenyl, triphenylene, and naphthalene. Additionally, the aryl group can be further substituted or fused, such as, without limitation, fluorene.

[0053] The term “heteroaryl” refers to and includes both single-ring aromatic groups and polycyclic aromatic ring systems that include at least one heteroatom. The heteroatoms include, but are not limited to O, S, Se, N, P, B, Si, Ge, and Se. In many instances, O, S, N, or B are the preferred heteroatoms. Hetero-single ring aromatic systems are preferably single rings with 5 or 6 ring atoms, and the ring can have from one to six heteroatoms. The hetero-polycyclic ring systems can have two or more aromatic rings in which two atoms are common to two adjoining rings (the rings are “fused”) wherein at least one of the rings is a heteroaryl. The hetero-polycyclic aromatic ring systems can have from one to six heteroatoms per ring of the polycyclic aromatic ring system. Preferred heteroaryl groups are those containing three to thirty carbon atoms, preferably three to twenty-four carbon atoms, three to eighteen carbon atoms, and more preferably three to twelve carbon atoms. Suitable heteroaryl groups include dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, triazole, oxazole, thiazole,

oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, benzofuropyridine, furodipyrindine, benzothienopyridine, thienodipyrindine, benzoselenophenopyridine, selenophenodipyrindine, azaborine, borazine, 5 λ .sup.2,9 λ .sup.2-diaza-13b-boranaphtho[2,3,4-de]anthracene, 5 λ .sup.2-benzo[d]benzo[4,5]imidazo[3,2-a]imidazole, and 5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene; preferably dibenzothiophene, dibenzofuran, dibenzoselenophene, carbazole, indolocarbazole, imidazole, pyridine, triazine, benzimidazole, 5 λ .sup.2,9 λ .sup.2-diaza-13b-boranaphtho[2,3,4-de]anthracene, 5 λ .sup.2-benzo[d]benzo[4,5]imidazo[3,2-a]imidazole, and 5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene. Additionally, the heteroaryl group can be further substituted or fused.

[0054] Of the aryl and heteroaryl groups listed above, the groups of triphenylene, naphthalene, anthracene, dibenzothiophene, dibenzofuran, dibenzoselenophene, carbazole, indolocarbazole, imidazole, pyridine, pyrazine, pyrimidine, triazine, benzimidazole, 5 λ .sup.2,9 λ .sup.2-diaza-13b-boranaphtho[2,3,4-de]anthracene, 5 λ .sup.2-benzo[d]benzo[4,5]imidazo[3,2-a]imidazole, 5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene, and the respective aza-analogs of each thereof are of particular interest.

[0055] In many instances, the General Substituents are selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, selenyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

[0056] In some instances, the Preferred General Substituents are selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, heteroalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, aryl, heteroaryl, nitrile, isonitrile, sulfanyl, and combinations thereof.

[0057] In some instances, the More Preferred General Substituents are selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, aryl, heteroaryl, nitrile, sulfanyl, and combinations thereof.

[0058] In some instances, the Even More Preferred General Substituents are selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, silyl, aryl, heteroaryl, nitrile, and combinations thereof.

[0059] In yet other instances, the Most Preferred General Substituents are selected from the group consisting of deuterium, alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof.

[0060] The terms “substituted” and “substitution” refer to a substituent other than H that is bonded to the relevant position, e.g., a carbon or nitrogen. For example, when R^{sup.1} represents mono-substitution, then one R^{sup.1} must be other than H (i.e., a substitution). Similarly, when R^{sup.1} represents di-substitution, then two of R^{sup.1} must be other than H. Similarly, when R^{sup.1} represents zero or no substitution, R^{sup.1}, for example, can be a hydrogen for all available valencies of ring atoms, as in carbon atoms for benzene and the nitrogen atom in pyrrole, or simply represents nothing for ring atoms with fully filled valencies, e.g., the nitrogen atom in pyridine. The maximum number of substitutions possible in a ring structure will depend on the total number of available valencies in the ring atoms.

[0061] As used herein, “combinations thereof” indicates that one or more members of the applicable list are combined to form a known or chemically stable arrangement that one of ordinary skill in the art can envision from the applicable list. For example, an alkyl and deuterium can be combined to form a partial or fully deuterated alkyl group; a halogen and alkyl can be combined to form a halogenated alkyl substituent; and a halogen, alkyl, and aryl can be combined to form a halogenated arylalkyl. In one instance, the term substitution includes a combination of two to four

of the listed groups. In another instance, the term substitution includes a combination of two to three groups. In yet another instance, the term substitution includes a combination of two groups. Preferred combinations of substituent groups are those that contain up to fifty atoms that are not hydrogen or deuterium, or those which include up to forty atoms that are not hydrogen or deuterium, or those that include up to thirty atoms that are not hydrogen or deuterium. In many instances, a preferred combination of substituent groups will include up to twenty atoms that are not hydrogen or deuterium.

[0062] The “aza” designation in the fragments described herein, i.e. aza-dibenzofuran, aza-dibenzothiophene, etc. means that one or more of the C—H groups in the respective aromatic ring can be replaced by a nitrogen atom, for example, and without any limitation, azatriphenylene encompasses both dibenzo[f,h]quinoxaline and dibenzo[f,h]quinoline. One of ordinary skill in the art can readily envision other nitrogen analogs of the aza-derivatives described above, and all such analogs are intended to be encompassed by the terms as set forth herein.

[0063] As used herein, “deuterium” refers to an isotope of hydrogen. Deuterated compounds can be readily prepared using methods known in the art. For example, U.S. Pat. No. 8,557,400, Patent Pub. No. WO 2006/095951, and U.S. Pat. Application Pub. No. US 2011/0037057, which are hereby incorporated by reference in their entireties, describe the making of deuterium-substituted organometallic complexes. Further reference is made to Ming Yan, et al., *Tetrahedron* 2015, 71, 1425-30 and Atzrodt et al., *Angew. Chem. Int. Ed. (Reviews)* 2007, 46, 7744-65, which are incorporated by reference in their entireties, describe the deuteration of the methylene hydrogens in benzyl amines and efficient pathways to replace aromatic ring hydrogens with deuterium, respectively.

[0064] As used herein, any specifically listed substituent, such as, without limitation, methyl, phenyl, pyridyl, etc. includes undeuterated, partially deuterated, and fully deuterated versions thereof. Similarly, classes of substituents such as, without limitation, alkyl, aryl, cycloalkyl, heteroaryl, etc. also include undeuterated, partially deuterated, and fully deuterated versions thereof. Unless otherwise specified, atoms in chemical structures without valences fully filled by H or D should be considered to include undeuterated, partially deuterated, and fully deuterated versions thereof. For example, the chemical structure of

##STR00003##

implies to include C.sub.6H.sub.6, C.sub.6D.sub.6, C.sub.6H.sub.3D.sub.3, and any other partially deuterated variants thereof. Some common basic partially or fully deuterated group include, without limitation, CD.sub.3, CD.sub.2C(CH.sub.3).sub.3, C(CD.sub.3).sub.3, and C.sub.6D.sub.5.

[0065] It is to be understood that when a molecular fragment is described as being a substituent or otherwise attached to another moiety, its name may be written as if it were a fragment (e.g. phenyl, phenylene, naphthyl, dibenzofuryl) or as if it were the whole molecule (e.g. benzene, naphthalene, dibenzofuran). As used herein, these different ways of designating a substituent or attached fragment are considered to be equivalent.

[0066] In some instances, a pair of substituents in the molecule can be optionally joined or fused into a ring. The preferred ring is a five to nine-membered carbocyclic or heterocyclic ring, including both instances where the portion of the ring formed by the pair of substituents is saturated and where the portion of the ring formed by the pair of substituents is unsaturated. In yet other instances, a pair of adjacent substituents can be optionally joined or fused into a ring. As used herein, “adjacent” means that the two substituents involved can be on the same ring next to each other, or on two neighboring rings having the two closest available substitutable positions, such as 2, 2' positions in a biphenyl, or 1, 8 position in a naphthalene.

B. The Compounds of the Present Disclosure

[0067] In one aspect, the present disclosure provides a compound having a first ligand L.sub.A comprising a structure of Formula I:

##STR00004##

In Formula I:

[0068] each of ring A and ring B is independently a 5-membered or 6-membered carbocyclic or heterocyclic ring; [0069] each of X.sup.1 to X.sup.3 is independently C or N; [0070] each of Y.sup.1 and Y.sup.2 is independently C, Si, or Ge; [0071] each  custom-character independently represents a single bond or a double bond; [0072] L is selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO.sub.2, CRR', SiRR', and GeRR'; [0073] each of R.sup.A and R.sup.B independently represents mono to the maximum allowable substitutions, or no substitutions; [0074] each R.sup.1, R.sup.2, R.sup.3, R.sup.4, R, R', R.sup.A, and R.sup.B is independently hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and combinations thereof; [0075] L.sub.A is coordinated to a metal M selected from the group consisting of Ir, Rh, Re, Ru, Os, Pt, Pd, Ag, Au, and Cu; [0076] M can be coordinated to other ligands; [0077] L.sub.A can be a monodentate, bidentate, tridentate, tetradentate, pentadentate, or hexadentate ligand; and [0078] any two substituents can be joined or fused to form a ring.

[0079] In some embodiments when M is Ir, Ir is not directly coordinated to ring A or ring B. In some embodiments when M is Ir, at least one of R.sup.1 to R.sup.4 is not H. In some embodiments when M is Ir, at least one of Y.sup.1 or Y.sup.2 is Si or Ge. In some embodiments when M is Ir, L is not CRR'.

[0080] In some embodiments, if M is Ir, then L.sub.A does not comprise

##STR00005##

[0081] In some embodiments, the compound is not

##STR00006##

[0082] In some embodiments, at least one R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.A, or R.sup.B is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.A is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.B is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.1 is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.2 is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.3 is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.4 is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.A, or R.sup.B is selected from the group consisting of the Preferred General Substituents defined herein.

[0083] In some embodiments of Formula I, at least one R.sup.1, R.sup.2, R.sup.3, R.sup.4, R, R', R.sup.A, and R.sup.B is partially or fully deuterated. In some embodiments, at least one R.sup.A is partially or fully deuterated. In some embodiments, at least one R.sup.B is partially or fully deuterated. In some embodiments, at least one of R.sup.1, R.sup.2, R.sup.3, or R.sup.4 is partially or fully deuterated. In some embodiments, at least one R or R' is partially or fully deuterated.

[0084] In some embodiments, each of ring A and ring B is independently a 5-membered or 6-membered aryl or heteroaryl ring.

[0085] In some embodiments, each R.sup.1, R.sup.2, R.sup.3, R.sup.4, R, R', R.sup.A, and R.sup.B is independently hydrogen, or a substituent selected from the group consisting of the Preferred General Substituents defined herein. In some embodiments, each R.sup.1, R.sup.2, R.sup.3, R.sup.4, R, R', R.sup.A, and R.sup.B is independently hydrogen, or a substituent selected from the group consisting of the More Preferred General Substituents defined herein. In some embodiments, each R.sup.1, R.sup.2, R.sup.3, R.sup.4, R, R', R.sup.A, and R.sup.B is independently hydrogen, or

a substituent selected from the group consisting of the Even More Preferred General Substituents defined herein. In some embodiments, each R.sup.1, R.sup.2, R.sup.3, R.sup.4, R, R', R.sup.A, and R.sup.B is independently hydrogen, or a substituent selected from the group consisting of the Most Preferred General Substituents defined herein.

[0086] In some embodiments, both Y.sup.1 and Y.sup.2 are carbon. In some embodiments, one of Y.sup.1 or Y.sup.2 is Si or Ge, and the other is carbon.

[0087] In some embodiments, metal M is Ir. In some embodiments, metal M is Pt or Pd. In some embodiments, metal M is Pt. In some embodiments, metal M is Pd.

[0088] In some embodiments, at least one of ring A or ring B is not aromatic. In some embodiments, exactly one of ring A or ring B is not aromatic. In some embodiments, at least one of ring A or ring B is a 5-membered, non-aromatic ring. In some embodiments, at least one of ring A or ring B is a 6-membered, non-aromatic ring.

[0089] In some embodiments, at least one of ring A or ring B is aromatic. In some embodiments, exactly one of ring A or ring B is aromatic.

[0090] In some embodiments, each of ring A and ring B is carbocyclic.

[0091] In some embodiments, at least one of ring A or ring B is heterocyclic. In some embodiments, exactly one of ring A or ring B is heterocyclic.

[0092] In some embodiments, each of ring A and ring B is heterocyclic.

[0093] In some embodiments, each of ring A and ring B is a 5-membered ring. In some embodiments, each of ring A and ring B is a 6-membered ring. In some embodiments, one of ring A or ring B is a 5-membered ring, and the other one of ring A or ring B is a 6-membered ring.

[0094] In some embodiments, each of ring A and ring B is independently selected from the group consisting of benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, and triazole. In some embodiments, each of ring A and ring B is independently selected from the group consisting of benzene, pyridine, pyrimidine, pyrrole, imidazole, and thiazole. In some embodiments, at least one of ring A or ring B is benzene. In some embodiments, each of ring A and ring B is benzene.

[0095] In some embodiments, at least one of ring A or ring B comprises at least one N atom. In some embodiments, at least one of ring A or ring B comprises at least two N atoms. In some embodiments, the one of ring A or ring B is a 5-membered ring, while the one of ring A or ring B is a 6-membered ring in other embodiments,

[0096] In some embodiments, at least one of ring A or ring B comprises a heteroatom other than N. In some embodiments, at least one of ring A or ring B comprises a heteroatom selected from S or O. In some such embodiments, the heteroatom is S. In some such embodiments, the heteroatom is O.

[0097] In some embodiments, each of X.sup.1, X.sup.2, and X.sup.3 is C.

[0098] In some embodiments, at least one of X.sup.1, X.sup.2, or X.sup.3 is N. In some embodiments, at least one of X.sup.1 or X.sup.2 is N. In some embodiments, each of X.sup.1 and X.sup.2 is N. In some embodiments, X.sup.3 is C. In some embodiments, X.sup.3 is N. In some embodiments, at least one of ring A or ring B is not aromatic.

[0099] In some embodiments, at least one  is a double bond in a Lewis dot structure. In some embodiments, the  bond formed with X.sup.1 is a double bond. In some embodiments, the  bond formed with X.sup.2 is a double bond. In some embodiments, the  bond formed with X.sup.3 is a double bond.

[0100] In some embodiments, exactly one  is a double bond. In some such embodiments, the  bonded to X.sup.3 is double bond.

[0101] In some embodiments, L is a direct bond.

[0102] In some embodiments, L is selected from the group consisting of O, S, and Se. In some embodiments, L is selected from the group consisting of BR, NR, and PR. In some embodiments, L is selected from the group consisting of P(O)R, C=O, C=S, C=Se, C=NR', C=CRR', S=O, and

SO.sub.2.

[0103] In some embodiments, L is selected from the group consisting of BRR', CRR', SiRR', and GeRR'. In some embodiments, L is BRR'. In some embodiments, L is SiRR'. In some embodiments, L is GeRR'. In some such embodiments, R and R' are both H. In other embodiments, at least one of R or R' comprises at least one C atom. In other embodiments, each of R and R' comprises at least one C atom. In some embodiments, R and R' are joined or fused to form a ring, while R and R' are not joined or fused.

[0104] In some embodiments, L is CRR'

[0105] In some embodiments, the first ligand L.sub.A comprises a structure of Formula II:

##STR00007##

wherein: [0106] each of moiety C and moiety D is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring; [0107] each of X^{sup.4} to X^{sup.7}, Z^{sup.1}, and Z^{sup.2} is independently C or N; [0108] L^{sup.X} is selected from the group consisting of a direct bond BR, BRR', NR, PR, P(O)R, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO.sub.2, CR, CRR', SiRR', and GeRR'; [0109] K^{sup.1} and K^{sup.2} are each independently selected from the group consisting of a direct bond, O, S, N(R^{sup.α}), P(R^{sup.α}), B(R^{sup.α}), C(R^{sup.α})(R^{sup.β}), and Si(R^{sup.α})(R^{sup.β}); [0110] each of R^{sup.C} and R^{sup.D} independently represents mono to the maximum allowable substitutions, or no substitutions; [0111] each R, R', R^{sup.α}, R^{sup.β}, R^{sup.C}, and R^{sup.D} is independently hydrogen, or a substituent selected from the group consisting of the General Substituents defined herein; [0112] any two substituents can be joined or fused to form a ring; [0113] L.sub.A is coordinated to the metal M by the dashed lines; [0114] M may be coordinated to other ligands; [0115] L.sub.A may be joined with other ligands to comprise a tridentate, tetradentate, pentadentate, or hexadentate ligand; and [0116] at least one of the following conditions is true: [0117] (1) at least one R^{sup.C} or R^{sup.D} comprises a structure of Formula I; [0118] (2) moiety C comprises a structure of Formula I; [0119] (3) moiety D comprises a structure of Formula I; and [0120] (4) at least one R^{sup.C} and at least one R^{sup.D} are joined to form a structure comprising Formula I.

[0121] In some embodiments of Formula II, at least one R^{sup.1}, R^{sup.2}, R^{sup.3}, R^{sup.4}, R^{sup.A}, R^{sup.B}, R^{sup.C}, or R^{sup.D} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.A} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.B} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.C} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.D} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.1} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.2} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.3} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.4} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.1}, R^{sup.2}, R^{sup.3}, R^{sup.4}, R^{sup.A}, R^{sup.B}, R^{sup.C}, or R^{sup.D} is selected from the group consisting of the Preferred General Substituents defined herein.

[0122] In some embodiments Formula II, at least one R^{sup.1}, R^{sup.2}, R^{sup.3}, R^{sup.4}, R^{sup.A}, R^{sup.B}, R^{sup.C}, or R^{sup.D} is partially or fully deuterated. In some embodiments, at least R^{sup.1} is partially or fully deuterated. In some embodiments, at least R^{sup.2} is partially or fully deuterated. In some embodiments, at least R^{sup.3} is partially or fully deuterated. In some embodiments, at least R^{sup.4} is partially or fully deuterated. In some embodiments, at least one R^{sup.A} is partially or fully deuterated. In some embodiments, at least one R^{sup.B} is partially or

fully deuterated. In some embodiments, at least one R.sup.C is partially or fully deuterated. In some embodiments, at least one R.sup.D is partially or fully deuterated.

[0123] In some embodiments, moiety C and moiety D are fused together by a moiety that includes a structure of Formula I.

[0124] In some embodiments of Formula II, each R, R', R.sup.α, R.sup.β, R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.A, R.sup.B, R.sup.C, and R.sup.D is independently a hydrogen or a substituent selected from the group consisting of the Preferred General Substituents.

[0125] In some embodiments of Formula II, each of moiety C and moiety D is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered or 6-membered carbocyclic or heterocyclic ring.

[0126] In some embodiments of Formula II, each of moiety C and moiety D is independently a 5-membered or 6-membered aryl or heteroaryl ring.

[0127] In some embodiments of Formula II, at least one of R.sup.C or R.sup.D comprises a structure of Formula I. In some embodiments of Formula II, at least one R.sup.C comprises a structure of Formula I. In some embodiments of Formula II, at least one R.sup.D comprises a structure of Formula I.

[0128] In some embodiments of Formula II, moiety C or moiety D comprises a structure of Formula I. In some embodiments of Formula II, moiety C comprises a structure of Formula I. In some embodiments of Formula II, moiety D comprises a structure of Formula I.

[0129] In some embodiments of Formula II, two R.sup.C or two R.sup.D are joined to form a structure of Formula I.

[0130] In some embodiments of Formula II, metal M is Ir. In some embodiments of Formula II, metal M is Pt or Pd. In some embodiments of Formula II, metal M is Pt. In some embodiments of Formula II, metal M is Pd.

[0131] In some embodiments of Formula II, each of moiety C and moiety D is independently selected from the group consisting of the following Cyclic Moiety List: benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, imidazole derived carbene. Pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, triazole, naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, phenanthro[3,2-b]benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, benzimidazole derived carbene, aza-benzimidazole derived carbene, benzobenzimidazole, aza-benzobenzimidazole, carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene. In some embodiments, the aza variant includes one N on a benzo ring. In some embodiments, the aza variant includes one N on a benzo ring and the N is bonded to the metal M.

[0132] In some embodiments of Formula II, moiety C is a monocyclic ring. In some embodiments of Formula II, moiety C is selected from the group consisting of benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, imidazole derived carbene, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, and triazole. In some embodiments of Formula II, moiety C is pyridine or imidazole. In some such embodiments, moiety C is an imidazole that forms an N-M bond. In some embodiments of Formula II, moiety C is an imidazole that forms a carbene-M bond.

[0133] In some embodiments of Formula II, when moiety D comprises or consists essentially of a structure of Formula I, then moiety C is a monocyclic imidazole or a fused imidazole moiety with the imidazole portion coordinated to the metal. In such embodiments, Z.sup.1 may be N. In some embodiments, when moiety D comprises a structure of formula I, Z.sup.1 is an imidazole derived carbene carbon and coordinated to the metal.

[0134] In some embodiments of Formula II, when moiety D comprises a structure of Formula I,

then moiety C is a polycyclic fused ring system comprises three or more fused rings wherein two of the three or more fused rings are 5-membered rings. In some embodiments of Formula II, when moiety D comprises a structure of Formula I, then moiety C is a polycyclic fused ring system comprises four or more fused rings.

[0135] In some embodiments of Formula II, when Z^{sup.1} is N, moiety C is monocyclic or unfused pyridine or pyrimidine and L^{sup.X} is a direct bond, then Formula I is fused to moiety D.

[0136] In some embodiments, when Z^{sup.1} is N, moiety C is pyridine or pyrimidine and two R^{sup.C} are joined to form a ring fused to the pyridine or pyrimidine and L^{sup.X} is a direct bond, then Formula I is fused to moiety D.

[0137] In some embodiments of Formula II, when moiety D comprises Formula I, L^{sup.X} is a direct bond, and ring A or ring B is directly coordinated to the metal, then at least one of ring A or ring B is a 5-membered ring. In some embodiments of Formula II, when moiety D comprises Formula I, L^{sup.X} is a direct bond, and ring A or ring B is directly coordinated to the metal, then the ring coordinated to the metal contains at least one ring N atom. In some embodiments of Formula II, when moiety D comprises Formula I, L^{sup.X} is a direct bond, and ring A or ring B is directly coordinated to the metal, then the ring not coordinated to the metal contains at least one substituent selected from the group consisting of the General Substituents defined herein.

[0138] In some embodiments of Formula II, moiety C is a polycyclic fused ring system. In some embodiments, moiety C comprises at least 3 rings. In some embodiments, moiety C comprises at least 4 rings. In some embodiments, moiety C comprises at least 5 rings.

[0139] In some embodiments of Formula II, moiety C is selected from the group consisting of naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, phenanthro[3,2-b]benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, benzimidazole derived carbene, aza-benzimidazole derived carbene, benzobenzimidazole, aza-benzobenzimidazole, carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene.

[0140] In some embodiments of Formula II, moiety C is quinoline, isoquinoline, or benzimidazole. In some embodiments, the benzimidazole can form an N-M bond. In some embodiments, the benzimidazole forms a carbene-M bond. In some embodiments, the quinoline or isoquinoline forms an N-M bond.

[0141] In some embodiments, at least one of moiety C and moiety D can independently be a polycyclic fused ring structure. In some embodiments, at least one of moiety C and moiety D can independently be a polycyclic fused ring structure comprising at least two fused rings. In some embodiments, the polycyclic fused ring structure has one 6-membered ring and one 5-membered ring. In some such embodiments, either the 5-membered ring or the 6-membered ring can coordinate to the metal. In some embodiments, the polycyclic fused ring structure has two 6-membered rings. In some embodiments, at least one of moiety C and moiety D can independently be selected from the group consisting of benzofuran, benzothiophene, benzoselenophene, naphthalene, and aza-variants thereof.

[0142] In some embodiments, at least one of moiety C and moiety D can independently be a polycyclic fused ring structure comprising at least three fused rings. In some embodiments, the polycyclic fused ring structure has two 6-membered rings and one 5-membered ring. In some such embodiments, the 5-membered ring is fused to the ring coordinated to metal M and the second 6-membered ring is fused to the 5-membered ring. In some embodiments, at least one of moiety C and moiety D can independently be selected from the group consisting of dibenzofuran, dibenzothiophene, dibenzoselenophene, and aza-variants thereof. In some such embodiments, at least one of moiety C and moiety D can independently be further substituted at the ortho- or meta-

position of the O, S, or Se atom by a substituent selected from the group consisting of deuterium, fluorine, nitrile, alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof. In some such embodiments, the aza-variants contain exactly one N atom at the 6-position (ortho to the O, S, or Se) with a substituent at the 7-position (meta to the O, S, or Se).

[0143] In some embodiments, at least one of moiety C or moiety D can independently be a polycyclic fused ring structure comprising at least four fused rings. In some embodiments, the polycyclic fused ring structure comprises three 6-membered rings and one 5-membered ring. In some such embodiments, the 5-membered ring is fused to the ring coordinated to metal M, the second 6-membered ring is fused to the 5-membered ring, and the third 6-membered ring is fused to the second 6-membered ring. In some such embodiments, the third 6-membered ring is further substituted by a substituent selected from the group consisting of deuterium, fluorine, nitrile, alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof.

[0144] In some embodiments, at least one of moiety C or moiety D can independently be a polycyclic fused ring structure comprising at least five fused rings. In some embodiments, the polycyclic fused ring structure comprises four 6-membered rings and one 5-membered ring or three 6-membered rings and two 5-membered rings. In some embodiments comprising two 5-membered rings, the 5-membered rings are fused together. In some embodiments comprising two 5-membered rings, the 5-membered rings are separated by at least one 6-membered ring. In some embodiments with one 5-membered ring, the 5-membered ring is fused to the ring coordinated to metal M, the second 6-membered ring is fused to the 5-membered ring, the third 6-membered ring is fused to the second 6-membered ring, and the fourth 6-membered ring is fused to the third 6-membered ring.

[0145] In some embodiments, at least one of moiety C or moiety D can independently be an aza version of the polycyclic fused rings described above. In some such embodiments, at least one of moiety C and moiety D can independently contain exactly one aza N atom. In some such embodiments, at least one of moiety C and moiety D contains exactly two aza N atoms, which can be in one ring, or in two different rings. In some such embodiments, the ring having aza N atom is separated by at least two other rings from the metal M atom. In some such embodiments, the ring having aza N atom is separated by at least three other rings from the metal M atom. In some such embodiments, each of the ortho positions of the aza N atom is substituted.

[0146] In some embodiments, moiety C is or comprises the structure of Formula I. In some embodiments, the structure of Formula I is fused to a moiety selected from the group consisting of the Cyclic Moiety List defined herein. In some such embodiments, the structure of Formula I is fused to a moiety selected from the group consisting of pyridine, imidazole, benzimidazole, quinoline, and isoquinoline.

[0147] In some embodiments, moiety D is a monocyclic ring. In some embodiments, moiety D is selected from the group consisting of benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, imidazole derived carbene, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, and triazole. In some embodiments, moiety D is benzene.

[0148] In some embodiments, moiety D is a polycyclic fused ring system. In some embodiments, moiety D comprises at least 3 rings. In some embodiments, moiety D comprises at least 4 rings. In some embodiments, moiety D comprises at least 5 rings.

[0149] In some embodiments, moiety D is selected from the group consisting of naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, benzimidazole derived carbene, aza-benzimidazole derived carbene, benzobenzimidazole, aza-benzobenzimidazole, carbazole, aza-carbazole, dibenzofuran, phenanthro[3,2-b]benzofuran, aza-dibenzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene. In some embodiments, moiety D is naphthalene or dibenzofuran.

[0150] In some embodiments, moiety D is or comprises the structure of Formula I. In some embodiments, the structure of Formula I is fused to a moiety selected from the group consisting of the Cyclic Moiety List defined herein. In some such embodiments, moiety D comprises dibenzofuran where one benzene ring is coordinated to the metal M and the structure of Formula I is fused to the other benzene ring.

[0151] In some embodiments, L.sup.X is a direct bond. In some embodiments, L.sup.X is selected from the group consisting of O, S, and Se. In some embodiments, L.sup.X is selected from the group consisting of P(O)R, C=O, C=S, C=Se, C=NR', C=CRR', S=O, and SO.sub.2. In some embodiments, L.sup.X is selected from the group consisting of BRR', CRR', SiRR', and GeRR'. In some embodiments, L.sup.X is CR.

[0152] In some embodiments, L.sup.X is selected from the group consisting of BR, NR, and PR. In some embodiments, L.sup.X is BR. In some embodiments, L.sup.X is NR. In some embodiments, L.sup.X is PR. In some of the foregoing embodiments, R is aryl or heteroaryl. In some such embodiments, R is benzene. In some such embodiments, R is joined or fused with one R.sup.C or R.sup.D to form a ring. In some embodiments, the ring formed is a 5-membered ring. In some such embodiments, the ring formed is a pyrrole ring. In some embodiments, R is joined or fused to one R.sup.D to form a ring. In some embodiments, the R has a structure of Formula I.

[0153] In some embodiments, Z.sup.1 is N or carbene C and Z.sup.2 is C. In some embodiments, Z.sup.1 is N. In some embodiments, Z.sup.1 is carbene C.

[0154] In some embodiments, X.sup.4 and X.sup.5 are N, and X.sup.6 and X.sup.8 are C. In some embodiments, each of X.sup.4, X.sup.5, X.sup.6, and X.sup.8 is C.

[0155] In some embodiments, K.sup.1 is a direct bond. In some embodiments, K.sup.1 is O or S. In some embodiments, K.sup.1 is O. In some embodiments, K.sup.1 is S. In some embodiments, K.sup.1 is N(R.sup.α), P(R.sup.α), or B(R.sup.α). In some embodiments, K.sup.1 is C(R.sup.α)(R.sup.β) or Si(R.sup.α)(R.sup.β).

[0156] In some embodiments, K.sup.2 is a direct bond. In some embodiments, K.sup.2 is O or S. In some embodiments, K.sup.2 is O. In some embodiments, K.sup.2 is S. In some embodiments, K.sup.2 is N(R.sup.α), P(R.sup.α), or B(R.sup.α). In some embodiments, K.sup.2 is C(R.sup.α)(R.sup.β) or Si(R.sup.α)(R.sup.β).

[0157] In some embodiments, each of K.sup.1 and K.sup.2 is a direct bond. In some embodiments, K.sup.1 is a direct bond, and K.sup.2 is not a direct bond.

[0158] In some embodiments, the first ligand L.sub.A comprises an electron-withdrawing group selected from the group consisting of the following EWG1 LIST: F, CF.sub.3, CN, COCH.sub.3, CHO, COCF.sub.3, COOMe, COOCF.sub.3, NO.sub.2, SF.sub.3, SiF.sub.3, PF.sub.4, SF.sub.3, OCF.sub.3, SCF.sub.3, SeCF.sub.3, SOCF.sub.3, SeOCF.sub.3, SO.sub.2F, SO.sub.2CF.sub.3, SeO.sub.2CF.sub.3, OseO.sub.2CF.sub.3, OCN, SCN, SeCN, NC, .sup.+N(R.sup.k2).sub.3, (R.sup.k2).sub.2CCN, (R.sup.k2).sub.2CCF.sub.3, CNC(CF.sub.3).sub.2, BR.sup.k3R.sup.k2, substituted or unsubstituted dibenzoborole, 1-substituted carbazole, 1,9-substituted carbazole, substituted or unsubstituted carbazole, substituted or unsubstituted pyridine, substituted or unsubstituted pyrimidine, substituted or unsubstituted pyrazine, substituted or unsubstituted pyridoxine, substituted or unsubstituted triazine, substituted or unsubstituted oxazole, substituted or unsubstituted benzoxazole, substituted or unsubstituted thiazole, substituted or unsubstituted benzothiazole, substituted or unsubstituted imidazole, substituted or unsubstituted benzimidazole, ketone, carboxylic acid, ester, nitrile, isonitrile, sulfinyl, sulfonyl, partially and fully fluorinated alkyl, partially and fully fluorinated aryl, partially and fully fluorinated heteroaryl, cyano-containing alkyl, cyano-containing aryl, cyano-containing heteroaryl, isocyanate,

##STR00008## ##STR00009## [0159] wherein each R.sup.k1 represents mono to the maximum allowable substitution, or no substitutions; [0160] wherein Y.sup.G is selected from the group consisting of BR.sub.e, NR.sub.e, PR.sub.e, O, S, Se, C=O, S=O, SO.sub.2, CR.sub.eR.sub.f, SiR.sub.eR.sub.f, and GeR.sub.eR.sub.f; and [0161] wherein each of R.sup.k1, R.sup.k2, R.sup.k3,

R.sub.e, and R.sub.f is independently a hydrogen, or a substituent selected from the group consisting of the General Substituents defined herein.

[0162] In some embodiments, the first ligand L.sub.A comprises an electron-withdrawing group selected from the group consisting of the structures of the following EWG2 List:

##STR00010## ##STR00011## ##STR00012## ##STR00013## ##STR00014## ##STR00015##
##STR00016## ##STR00017## ##STR00018## ##STR00019## ##STR00020## ##STR00021##

[0163] In some embodiments, the first ligand L.sub.A comprises an electron-withdrawing group selected from the group consisting of the structures of the following EWG3 LIST:

##STR00022## ##STR00023## ##STR00024## ##STR00025## ##STR00026##

[0164] In some embodiments, the first ligand L.sub.A comprises an electron-withdrawing group selected from the group consisting of the structures of the following EWG4 LIST:

##STR00027## ##STR00028##

[0165] In some embodiments, the first ligand L.sub.A comprises a π -electron deficient electron-withdrawing group selected from the group consisting of the structures of the following Pi-EWG LIST: CN, COCH.sub.3, CHO, COCF.sub.3, COOMe, COOCF.sub.3, NO.sub.2, SF.sub.3, SiF.sub.3, PF.sub.4, SF.sub.3, OCF.sub.3, SCF.sub.3, SeCF.sub.3, SOCF.sub.3, SeOCF.sub.3, SO.sub.2F, SO.sub.2CF.sub.3, SeO.sub.2CF.sub.3, OSeO.sub.2CF.sub.3, OCN, SCN, SeCN, NC, +N(R.sup.k2).sub.3, BR.sup.k2R.sup.k3, substituted or unsubstituted dibenzoborole, 1-substituted carbazole, 1,9-substituted carbazole, substituted or unsubstituted carbazole, substituted or unsubstituted pyridine, substituted or unsubstituted pyrimidine, substituted or unsubstituted pyrazine, substituted or unsubstituted pyridazine, substituted or unsubstituted triazine, substituted or unsubstituted oxazole, substituted or unsubstituted benzoxazole, substituted or unsubstituted thiazole, substituted or unsubstituted benzothiazole, substituted or unsubstituted imidazole, substituted or unsubstituted benzimidazole, ketone, carboxylic acid, ester, nitrile, isonitrile, sulfinyl, sulfonyl, partially and fully fluorinated aryl, partially and fully fluorinated heteroaryl, cyano-containing aryl, cyano-containing heteroaryl, isocyanate,

##STR00029##

wherein the variables are the same as previously defined.

[0166] In some embodiments, the compound comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, the compound comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, the compound comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, the compound comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, the compound comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0167] In some embodiments, the structure of Formula I comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, the structure of Formula I comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, the structure of Formula I comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, the structure of Formula I comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, the structure of Formula I comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0168] In some embodiments, at least one R.sup.A is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R.sup.A is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R.sup.A is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R.sup.A is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R.sup.A is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0169] In some embodiments, at least one R.sup.B is or comprises an electron-withdrawing group

from the EWG1 LIST as defined herein. In some embodiments, at least one R.sup.B is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R.sup.B is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R.sup.B is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R.sup.B is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0170] In some embodiments, at least one R.sup.C is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R.sup.C is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R.sup.C is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R.sup.C is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R.sup.C is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0171] In some embodiments, at least one R.sup.D is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R.sup.D is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R.sup.D is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R.sup.D is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R.sup.D is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0172] In some embodiments, at least one R.sup.α or R.sup.β is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R.sup.α or R.sup.β is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R.sup.α or R.sup.β is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R.sup.α or R.sup.β is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R.sup.α or R.sup.β is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0173] In some embodiments, at least one R or R' is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R or R' is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R or R' is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R or R' is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R or R' is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0174] In some embodiments, at least one of R.sup.1, R.sup.2, R.sup.3, or R.sup.4 is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one of R.sup.1, R.sup.2, R.sup.3, or R.sup.4 is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one of R.sup.1, R.sup.2, R.sup.3, or R.sup.4 is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one of R.sup.1, R.sup.2, R.sup.3, or R.sup.4 is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one of R.sup.1, R.sup.2, R.sup.3, or R.sup.4 is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0175] In some embodiments, metal M is directly coordinated to ring A or ring B. In some embodiments, the ring containing Z.sup.1 is ring A or ring B. In some embodiments, the ring containing Z.sup.2 is ring A or ring B.

[0176] In some embodiments, metal M is not directly coordinated to ring A or ring B.

[0177] In some embodiments, moiety C comprises the structure of Formula I and ring A or ring B is fused to the ring containing Z.sup.1. In some embodiments, moiety C comprises the structure of

Formula I and ring A and ring B are each separated from the ring containing Z^{sup.1} by at least one ring. In some embodiments, moiety C comprises the structure of Formula I and ring A and ring B are each separated from the ring containing Z^{sup.1} by at least two rings.

[0178] In some embodiments, moiety D comprises the structure of Formula I and ring A or ring B is fused to the ring containing Z^{sup.2}. In some embodiments, moiety D comprises the structure of Formula I and ring A and ring B are each separated from the ring containing Z^{sup.2} by at least one ring. In some embodiments, moiety D comprises the structure of Formula I and ring A and ring B are each separated from the ring containing Z^{sup.2} by at least two rings.

[0179] In some embodiments, R^{sup.C} comprises a structure of Formula I. In some embodiments, R^{sup.D} comprises a structure of Formula I.

[0180] In some embodiments, one of R^{sup.A} or R^{sup.B} is bonded to moiety C or moiety D.

[0181] In some embodiments, one of R^{sup.A} or R^{sup.B} is bonded to moiety C. In some embodiments, R^{sup.A} is bonded to moiety C. In some embodiments, R^{sup.B} is bonded to moiety C. In some embodiments, R^{sup.A} and R^{sup.B} are bonded to moiety C.

[0182] In some embodiments, one of R^{sup.A} or R^{sup.B} is bonded to moiety D. In some embodiments, R^{sup.A} is bonded to moiety D. In some embodiments, R^{sup.B} is bonded to moiety D. In some embodiments, R^{sup.A} and R^{sup.B} are bonded to moiety D.

[0183] In some embodiments, L is bonded to moiety C or moiety D. In some embodiments, L is bonded to moiety C. In some embodiments, L is bonded to moiety D.

[0184] In some embodiments, one of R^{sup.1}, R^{sup.2}, R^{sup.3}, or R^{sup.4} is bonded to moiety C or moiety D. In some embodiments, one of R^{sup.1}, R^{sup.2}, R^{sup.3}, or R^{sup.4} is bonded to moiety C. In some embodiments, one of R^{sup.1}, R^{sup.2}, R^{sup.3}, or R^{sup.4} is bonded to moiety D.

[0185] In some embodiments, the structure of Formula I is not fused to the structure of Formula II.

[0186] In some embodiments, at least one of ring A or ring B is part of moiety C or moiety D, but neither ring A nor ring B is the ring containing Z^{sup.1} or the ring containing Z^{sup.2}. In some embodiments, at least one of ring A or ring B is part of moiety C, but neither ring A nor ring B is the ring containing Z^{sup.1}. In some embodiments, at least one of ring A or ring B is part of moiety D, but neither ring A nor ring B is the ring containing Z^{sup.2}.

[0187] In some embodiments, at least one of ring A or ring B is fused to the ring containing Z^{sup.1} or the ring containing Z^{sup.2}. In some embodiments, at least one of ring A or ring B is fused to the ring containing Z. In some embodiments, at least one of ring A or ring B is fused to the ring containing Z^{sup.2}.

[0188] In some embodiments, moiety C or moiety D comprise ring A and ring B, and each of ring A and ring B is fused to a ring other than the ring containing Z^{sup.1} or the ring containing Z^{sup.2}. In some embodiments, moiety C comprises ring A and ring B, and each of ring A and ring B is fused to a ring other than the ring containing Z^{sup.1}. In some embodiments, moiety D comprises ring A and ring B, and each of ring A and ring B is fused to a ring other than the ring containing Z^{sup.2}.

[0189] In some embodiments, Formula I has a structure selected from the group consisting of the structures of the following LIST 1:

##STR00030##

wherein: [0190] each of X^{sup.10} to X^{sup.15} is independently C or N; and [0191] any two substituents can be joined or fused to form a ring.

[0192] In some embodiments, the structure of LIST 1 may be a substituent of the compound comprising a first ligand L_{sub.A}. In some embodiments, moiety C comprises the structure of LIST 1. In some embodiments, moiety D comprises the structure of LIST 1.

[0193] In some embodiments of LIST 1, two neighboring substituents may be fused with a ring of the first ligand L_{sub.A}. In some embodiments, two R^{sup.A} may be fused with a ring of the first ligand L_{sub.A}. In some embodiments, two R^{sup.B} may be fused with a ring of the first ligand

L.sub.A. In some embodiments, one R.sup.A and one R.sup.1 or R.sup.2 may be fused with a ring of the first ligand L.sub.A. In some embodiments, one R.sup.B and one R.sup.3 or R.sup.4 may be fused with a ring of the first ligand L.sub.A.

[0194] In some embodiments, Formula I has a structure selected from the group consisting of the following LIST 2:

##STR00031## ##STR00032## ##STR00033## ##STR00034## ##STR00035## ##STR00036##
##STR00037## ##STR00038## ##STR00039## ##STR00040## ##STR00041## ##STR00042##

[0195] In some embodiments, the structure of Formula II may be as a substituent of the compound comprising a first ligand L.sub.A. In such embodiments, one hydrogen or deuterium may be removed and replaced with the bond leading to moiety C or moiety D.

[0196] In some embodiments, two or more hydrogen or deuterium atoms may be removed in order to fuse the structure of LIST 2 to one or more adjacent rings. In some embodiments, ring A is fused to an adjacent ring. In some embodiments, ring B is fused to an adjacent ring. In some embodiments, ring A is fused to a first adjacent ring and ring B is fused to a second adjacent ring. In some of the foregoing embodiments, the structure of Formula I is part of moiety C or moiety D.

[0197] It should also be understood that for each structure of LIST 2, the structure may be partially or fully deuterated in the compound comprising the first ligand L.sub.A.

[0198] In some embodiments, the ligand L.sub.A is selected from the group consisting of the structures of the following LIST 3:

##STR00043## ##STR00044## ##STR00045## ##STR00046## ##STR00047## ##STR00048##
##STR00049## ##STR00050## ##STR00051## ##STR00052## ##STR00053## ##STR00054##
##STR00055## ##STR00056##
##STR00057## ##STR00058## ##STR00059## ##STR00060## ##STR00061## ##STR00062##
##STR00063## ##STR00064## ##STR00065## ##STR00066## ##STR00067## ##STR00068##
##STR00069## ##STR00070## ##STR00071## ##STR00072##

wherein: [0199] T is selected from the group consisting of B, Al, Ga, and In; [0200] X.sub.1 to X.sub.58 are each independently C or N; [0201] K.sup.1' is selected from the group consisting of a direct bond, BR.sub.e, NR.sub.e, PR.sub.e, O, S, Se, C=O, S=O, SO.sub.2, CR.sub.eR.sub.f, SiR.sub.eR.sub.f, and GeR.sub.eR.sub.f; [0202] each occurrence of Y', Y'', Y.sub.1, Y.sup.A, Y.sup.B, and Y.sup.C is independently selected from the group consisting of BR.sub.e, NR.sub.e, PR.sub.e, O, Se, C=O, S=O, SO.sub.2, CR.sub.eR.sub.f, SiR.sub.sR.sub.f, and GeR.sub.eR.sub.f; [0203] each of Y.sub.1 to Y.sub.23 is independently selected from the group consisting of carbon and nitrogen; [0204] W.sup.1 is selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO.sub.2, CRR', P(O)R, SiRR', and GeRR'; [0205] W.sup.2 is B, N, or P; [0206] each R.sup.A1, R.sup.B1, R.sup.B2, R.sup.B3, R.sub.a, R.sub.b, R.sub.c, and R.sub.d independently represent from mono to the maximum possible number of substitutions, or no substitution; [0207] each R, R', R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1, R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sub.e, R.sub.f, R.sup.A1, R.sup.B1, R.sup.B2, and R.sup.B3 is independently a hydrogen, or a substituent selected from the group consisting of the General Substituents defined herein; and [0208] any two substituents can be joined or fused to form a ring; and [0209] one of the following is true: [0210] (1) at least one of R, R', R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1, R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sup.A1, R.sup.B1, R.sup.B2, R.sup.B3, R.sub.e, and R.sub.f comprises a structure of Formula I or is selected from the structures of LIST 1 or LIST 2 defined herein; [0211] (2) at least one pair selected from (X.sub.50, X.sub.51), (X.sub.52, X.sub.53), (X.sub.51, X.sub.54), (X.sub.55, X.sub.56), or (X.sub.57, X.sub.58) is joined by a moiety selected from the group consisting of the structures of the following LIST 4:

##STR00073## [0212] (3) two R.sup.A1, R.sup.B1, R.sup.B2, R.sup.B3, R.sub.a, R.sub.b, R.sub.c, or R.sub.d are fused or fused to form a structure of Formula I, or a structure of LIST 1 or LIST 2.

[0213] In some embodiments where ligand L.sub.A is a structure selected from LIST 3, X.sub.s may be C and is connected to the top ring having N coordinated to the metal. In some

embodiments, X.sub.9 may be C and is connected to the top ring having N coordinated to the metal. In some embodiments, X.sub.10 may be C and is connected to the top ring having N coordinated to the metal. In some embodiments, X.sub.11 may be C and is connected to the top ring having N coordinated to the metal.

[0214] In some embodiments where ligand L.sub.A is a structure having R.sup.A1 from LIST 3, two R.sup.A1 are joined to form a fused ring. In some such embodiments, the fused ring may be benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, or thiazole. In some such embodiments, the fused ring may be benzene.

[0215] In some embodiments where ligand L.sub.A is a structure having Y.sup.A from LIST 3, Y.sup.A may be NR* and R* is or comprises a structure of Formula XA,

##STR00074##

wherein: [0216] ring F' is a 5-membered to 10-membered carbocyclic or heterocyclic ring; [0217] R.sup.F' represents mono to tri-substitutions, or no substitutions; and [0218] each R.sup.1', R.sup.2', and R.sup.F' is independently hydrogen or a substituent selected from the group consisting of deuterium, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, ether, ester, phosphino, selenyl, and combinations thereof, and combinations thereof.

[0219] In some embodiments, ring F' is a 5-membered or 6-membered carbocyclic or heterocyclic ring. In some embodiments, ring F' is a 5-membered or 6-membered aryl or heteroaryl ring.

[0220] In some embodiments, R.sup.1' and R.sup.2' are each independently hydrogen, deuterium, or R.sup.1' and R.sup.2' are joined with an R.sup.F' to form a ring. In some embodiments, at least one of R.sup.1' and R.sup.2' is not hydrogen or deuterium.

[0221] In some embodiments, each of R.sup.1' and R.sup.2' is independently selected from the group consisting of alkyl, cycloalkyl, silyl, germyl, and combinations thereof.

[0222] In some embodiments, R.sup.1' and R.sup.2' are the same. In some embodiments, R.sup.1' and R.sup.2' are different.

[0223] In some embodiments, each of R.sup.1' and R.sup.2' comprises at least one carbon atom. In some embodiments, each of R.sup.1' and R.sup.2' comprises at least two carbon atoms. In some embodiments, each of R.sup.1' and R.sup.2' comprises at least three carbon atoms. In some embodiments, each of R.sup.1' and R.sup.2' comprises at least four carbon atoms.

[0224] In some embodiments, at least one RF' is not hydrogen or deuterium. In some embodiments, at least one RF is selected from the group consisting of alkyl, cycloalkyl, silyl, germyl, and combinations thereof.

[0225] In some embodiments, ring F' is selected from the group consisting of benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, imidazole-derived carbene, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, and triazole.

[0226] In some embodiments where ligand L.sub.A is a structure having Y.sup.A from LIST 3, Y.sup.A may be NR* and R* is or comprises a structure of Formula XB,

##STR00075##

wherein each of X.sub.1a, X.sub.2a, and X.sub.3a is independently C or N.

[0227] In some embodiments, the RF bonded to X.sub.2a is selected from the group consisting of alkyl, cycloalkyl, silyl, germyl, and combinations thereof. In some embodiments, the RF' bonded to X.sub.2a is alkyl. In some embodiments, the RF' bonded to X.sub.2a is aryl or heteroaryl. In some embodiments, the RF' bonded to X.sub.2a is silyl. In some embodiments, the RF' bonded to X.sub.2a is germyl.

[0228] In some embodiments, each of X.sub.1a, X.sub.2a, and X.sub.3a is C.

[0229] In some embodiments where ligand L.sub.A is a structure having Y.sup.A from LIST 3, Y.sup.A may be NR* and R* is or comprises a structure of Formula XC,

##STR00076##

[0230] In some embodiments of Formula XC, the R.sup.F' meta to both R.sup.1' and R.sup.2' has a

structure of Formula XC.

[0231] In some embodiments of Formula XC, at least one of R.sup.1' or R.sup.2' is not hydrogen or deuterium. In some embodiments of Formula XC, each of R.sup.1' and R.sup.2' is independently selected from the group consisting of alkyl, cycloalkyl, aryl, heteroaryl, silyl, germyl, and combinations thereof. In some such embodiments, R.sup.1' and R.sup.2' are the same, while R.sup.1' and R.sup.2' are different in other embodiments.

[0232] In some embodiments of Formula XC, each of R.sup.1' and R.sup.2' is hydrogen or deuterium.

[0233] In some embodiments of Formula XC, at least one R.sup.F' is not hydrogen or deuterium.

[0234] In some embodiments of Formula XC, at least one R.sup.F' is selected from the group consisting of alkyl, cycloalkyl, aryl, heteroaryl, silyl, germyl, and combinations thereof. In some embodiments of Formula XC, the R.sup.F' meta to both R.sup.1' and R.sup.2' is selected from the group consisting of alkyl, cycloalkyl, aryl, heteroaryl, silyl, germyl, and combinations thereof.

[0235] In some embodiments of Formula XC, the R.sup.F' meta to both R.sup.1' and R.sup.2' is selected from the group consisting of aryl, alkyl, silyl, germyl, and combinations thereof. In some embodiments of Formula XC, the R.sup.F' meta to both R.sup.1' and R.sup.2' is selected from the group consisting of phenyl, t-butyl, Si(Me).sub.3, Si(Ph).sub.3, Ge(Me).sub.3, Ge(Ph).sub.3, and combinations thereof. In some embodiments of Formula XC, the R.sup.F' meta to both R.sup.1' and R.sup.2' is phenyl, t-butyl, Si(Me).sub.3, Si(Ph).sub.3, Ge(Me).sub.3, or Ge(Ph).sub.3.

[0236] In some embodiments of Formula XC, the R.sup.F' meta to both R.sup.1' and R.sup.2' is phenyl. In some embodiments of Formula XC, the R.sup.F' meta to both R.sup.1' and R.sup.2' is t-butyl. In some embodiments of Formula XC, the R.sup.F' meta to both R.sup.1' and R.sup.2' is Si(Me).sub.3 or Si(Ph).sub.3. In some embodiments of Formula XC, the R.sup.F' meta to both R.sup.1' and R.sup.2' is Ge(Me).sub.3 or Ge(Ph).sub.3.

[0237] In some embodiments where ligand L.sub.A is selected from LIST 3, at least one R.sup.A1, R.sup.B1, R.sup.B2, R.sup.B3, R.sub.a, R.sub.b, R.sub.c, or R.sub.d is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.A1 is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.B1 is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.B2 is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.B3 is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sub.a is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sub.b is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sub.c is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sub.d is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.A1, R.sup.B1, R.sup.B2, R.sup.B3, R.sub.a, R.sub.b, R.sub.c, or R.sub.d is selected from the group consisting of the Preferred General Substituents defined herein.

[0238] In some embodiments where ligand L.sub.A is selected from LIST 3, at least one of R.sup.A1, R.sup.B1, R.sup.B2, R.sup.B3, R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1, R.sub.a, R.sub.b, R.sub.c, R.sub.d is partially or fully deuterated. In some embodiments, at least one R.sup.A1 is partially or fully deuterated. In some embodiments, at least one R.sup.B1 is partially or fully deuterated. In some embodiments, at least one R.sup.B2 is partially or fully deuterated. In some embodiments, at least one R.sup.B3 is partially or fully deuterated. In some embodiments, at least one R.sub.a is partially or fully deuterated. In some embodiments, at least one R.sub.b is partially or fully deuterated. In some embodiments, at least one R.sub.c is partially or fully deuterated. In some embodiments, at least one R.sub.d is partially or fully deuterated. In some embodiments, at least one R.sub.a1, R.sub.b1, R.sub.c1, or R.sub.d1 is partially or fully deuterated.

[0239] In some embodiments where ligand L.sub.A is selected from LIST 3, at least one R.sup.A1

LIST as defined herein. In some embodiments, at least one R.sub.c is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R.sub.c is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R.sub.c is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0246] In some embodiments where ligand L.sub.A is selected from LIST 3, at least one R.sub.d is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R.sub.d is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R.sub.d is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R.sub.d is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R.sub.d is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0247] In some embodiments where ligand L.sub.A is selected from LIST 3, at least one of R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1 is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one of R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1 is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one of R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1 is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one of R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1 is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one of R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1 is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0248] In some embodiments, the ligand L.sub.A is selected from the group consisting of the structures of the following LIST 5:

##STR00077## ##STR00078## ##STR00079## ##STR00080## ##STR00081## ##STR00082##
##STR00083## ##STR00084## ##STR00085## ##STR00086## ##STR00087## ##STR00088##
##STR00089## ##STR00090## ##STR00091## ##STR00092## ##STR00093## ##STR00094##
##STR00095## ##STR00096## ##STR00097## ##STR00098## ##STR00099## ##STR00100##
##STR00101## ##STR00102## ##STR00103## ##STR00104## ##STR00105##
##STR00106## ##STR00107## ##STR00108## ##STR00109## ##STR00110## ##STR00111##
##STR00112## ##STR00113## ##STR00114## ##STR00115## ##STR00116## ##STR00117##
##STR00118## ##STR00119## ##STR00120## ##STR00121## ##STR00122## ##STR00123##
##STR00124## ##STR00125## ##STR00126## ##STR00127## ##STR00128## ##STR00129##
##STR00130## ##STR00131## ##STR00132## ##STR00133## ##STR00134## ##STR00135##
##STR00136## ##STR00137## ##STR00138##
##STR00139## ##STR00140## ##STR00141## ##STR00142## ##STR00143##

wherein: [0249] K.sup.1' is selected from the group consisting of a direct bond, NR.sub.e, PR.sub.e, O, S, and Se; [0250] each of X.sub.50 to X.sub.59, and Y.sup.2 to Y.sup.7 is independently selected from the group consisting of carbon and nitrogen; [0251] each occurrence of X, Y, Y', Y'', and W.sup.1 is independently selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO.sub.2, CRR', SiRR', and GeRR'; [0252] each of R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sup.AA, R.sup.BB, and R.sup.C independently represents mono to the maximum allowable substitutions, or no substitutions; [0253] each R, R', R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sup.AA, R.sup.BB, R.sup.C, and R.sup.N is independently hydrogen or a substituent selected from the group consisting of the General Substituents defined herein; [0254] the remaining variables are the same as previously defined; [0255] any two substituents can be fused or joined to form a ring; and [0256] any two substituents can be joined or fused to form a ring; and [0257] one of the following is true: [0258] (1) at least one of R, R', R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sup.AA, R.sup.BB, R.sup.C, and

R.sup.N comprises a structure of Formula I or is selected from a structure of LIST 1 or LIST 2 defined herein; or [0259] (2) at least one pair of (X.sub.50, X.sub.51), (X.sub.52, X.sub.53), (X.sub.51, X.sub.54), (X.sub.55, X.sub.56), (X.sub.57, X.sub.58), or (X.sub.53, X.sub.59) is joined to a moiety selected from the structures of the following LIST 6:

##STR00144## or [0260] (3) two R.sup.AA, R.sup.BB, R.sup.C, R.sub.a, R.sub.b, R.sub.c, or R.sub.d are joined or fused to form a structure of Formula I, or to a structure of LIST 1 or LIST 2 defined herein.

[0261] In some embodiments where ligand L.sub.A is a structure selected from LIST 5, each of X.sub.50, X.sub.51, X.sub.52, X.sub.53, X.sub.54, X.sub.55, and X.sub.56 is independently C. In some embodiments, X.sub.50 is N. In some embodiments, X.sub.51 is N. In some embodiments, X.sub.52 is N. In some embodiments, X.sub.53 is N. In some embodiments, X.sub.54 is N. In some embodiments, X.sub.55 is N. In some embodiments, X.sub.56 is N. In some embodiments, X and Y are each independently O. In some embodiments, R.sup.N has a Formula XA, XB, or XC as previously defined. In such embodiments, all the Formula XA, XB, and XC related embodiments can be equally applied here.

[0262] In some embodiments where ligand L.sub.A is a structure having R.sup.AA from LIST 5, two R.sup.AA are joined to form a fused ring. In some such embodiments, the fused ring may be benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, or thiazole. In some such embodiments, the fused ring may be benzene.

[0263] In some embodiments where ligand L.sub.A is selected from LIST 5, at least one R.sup.AA, R.sup.BB, R.sup.C, R.sub.a, R.sub.b, R.sub.c, or R.sub.d is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.AA is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.BB is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.C is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sub.a is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sub.b is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sub.c is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sub.d is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R.sup.AA, R.sup.BB, R.sup.C, R.sub.a, R.sub.b, R.sub.c, or R.sub.d is selected from the group consisting of the Preferred General Substituents defined herein.

[0264] In some embodiments, where ligand L.sub.A is selected from LIST 5, at least one of R.sup.N, R.sup.AA, R.sup.BB, R.sup.C, R.sub.a, R.sub.b, R.sub.c, or R.sub.d is partially or fully deuterated. In some embodiments, at least one R.sup.N is partially or fully deuterated. In some embodiments, at least one R.sup.AA is partially or fully deuterated. In some embodiments, at least one R.sup.BB is partially or fully deuterated. In some embodiments, at least one R.sub.a is partially or fully deuterated. In some embodiments, at least one R.sub.b is partially or fully deuterated. In some embodiments, at least one R.sub.c is partially or fully deuterated. In some embodiments, at least one R.sub.d is partially or fully deuterated.

[0265] In some embodiments where ligand L.sub.A is selected from LIST 5, at least one R.sup.AA is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R.sup.AA is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R.sup.AA is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R.sup.AA is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R.sup.AA is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0266] In some embodiments where ligand L.sub.A is selected from LIST 5, at least one R.sup.BB

and 291 to 296, R.sup.J is selected from R469 to R774 for L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L) (R.sup.M) and R.sup.J is selected from R1 to R774 for L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L) (R.sup.M) and each of R.sup.K, R.sup.L, and R.sup.M, is independently selected from R1 to R774, wherein each of L.sub.Aib(R469)(R1)(R1)(R1) to L.sub.A296(R774)(R774)(R774)(R744) is defined in the following LIST 7:

TABLE-US-00002 L.sub.A Structure of L.sub.A L.sub.A1(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A1(R469)(R1)(R1)(R1) to L.sub.A1(R774)(R774)(R774)(R774) have the structure [00145]  L.sub.A2(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A2(R469)(R1)(R1)(R1) to L.sub.A2(R774)(R774)(R774)(R774) have the structure [00146]  L.sub.A3(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A3(R469)(R1) (R1)(R1) to L.sub.A3(R774)(R774)(R774)(R774) have the structure [00147]  L.sub.A4(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A4(R469)(R1)(R1)(R1) to L.sub.A4(R774)(R774)(R774)(R774) have the structure [00148]  L.sub.A5(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A5(R469)(R1)(R1)(R1) to L.sub.A5(R774)(R774)(R774)(R774) have the structure [00149]  L.sub.A6(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A6(R469)(R1)(R1)(R1) to L.sub.A6(R774)(R774)(R774)(R774) have the structure [00150]  L.sub.A7(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A7(R469)(R1)(R1)(R1) to L.sub.A7(R774)(R774)(R774)(R774) have the structure [00151]  L.sub.A8(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A8(R469)(R1)(R1)(R1) to L.sub.A8(R774)(R774)(R774)(R774) have the structure [00152]  L.sub.A9(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A9(R469)(R1)(R1)(R1) to L.sub.A9(R774)(R774)(R774)(R774) have the structure [00153]  L.sub.A10(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A10(R469)(R1)(R1)(R1) to L.sub.A10(R774)(R774)(R774)(R774) have the structure [00154]  L.sub.A11(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A11(R469)(R1)(R1)(R1) to L.sub.A11(R774)(R774)(R774)(R774) have the structure [00155]  L.sub.A12(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A12(R469)(R1)(R1)(R1) to L.sub.A12(R774)(R774)(R774)(R774) have the structure [00156]  L.sub.A13(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A13(R469)(R1)(R1)(R1) to L.sub.A13(R774)(R774)(R774)(R774) have the structure [00157]  L.sub.A14(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A14(R469)(R1)(R1)(R1) to L.sub.A14(R774)(R774)(R774)(R774) have the structure [00158]  L.sub.A15(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A15(R469)(R1)(R1)(R1) to L.sub.A15(R774)(R774)(R774)(R774) have the structure [00159]  L.sub.A16(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A16(R469)(R1)(R1)(R1) to L.sub.A16(R774)(R774)(R774)(R774) have the structure [00160]  L.sub.A17(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A17(R469)(R1)(R1)(R1) to L.sub.A17(R774)(R774)(R774)(R774) have the structure [00161]  L.sub.A18(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A18(R469)(R1)(R1)(R1) to L.sub.A18(R774)(R774)(R774)(R774) have the structure [00162]  L.sub.A19(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A19(R469)(R1)(R1)(R1) to L.sub.A19(R774)(R774)(R774)(R774) have the structure [00163]  L.sub.A20(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A20(R469)(R1)(R1)(R1) to L.sub.A20(R774)(R774)(R774)(R774) have the structure [00164]  L.sub.A21(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A21(R469)(R1)(R1)(R1) to L.sub.A21(R774)(R774)(R774)(R774) have the structure [00165]  L.sub.A22(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A22(R469)(R1)(R1)(R1) to L.sub.A22(R774)(R774)(R774)(R774) have the structure [00166]  L.sub.A23(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A23(R469)(R1)(R1)(R1) to

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

L.sub.A273(R774)(R774)(R774)(R774) have the structure [00417] 
L.sub.A274(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A274(R469)(R1)(R1)(R1) to
L.sub.A274(R774)(R774)(R774)(R774) have the structure [00418] 
L.sub.A275(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A275(R469)(R1)(R1)(R1) to
L.sub.A275(R774)(R774)(R774)(R774) have the structure [00419] 
L.sub.A276(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A276(R469)(R1)(R1)(R1) to
L.sub.A276(R774)(R774)(R774)(R774) have the structure [00420] 
L.sub.A277(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A277(R469)(R1)(R1)(R1) to
L.sub.A277(R774)(R774)(R774)(R774) have the structure [00421] 
L.sub.A278(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A278(R469)(R1)(R1)(R1) to
L.sub.A278(R774)(R774)(R774)(R774) have the structure [00422] 
L.sub.A279(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A279(R469)(R1)(R1)(R1) to
L.sub.A279(R774)(R774)(R774)(R774) have the structure [00423] 
L.sub.A280(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A280(R469)(R1)(R1)(R1) to
L.sub.A280(R774)(R774)(R774)(R774) have the structure [00424] 
L.sub.A281(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A281(R469)(R1)(R1)(R1) to
L.sub.A281(R774)(R774)(R774)(R774) have the structure [00425] 
L.sub.A282(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A282(R469)(R1)(R1)(R1) to
L.sub.A282(R774)(R774)(R774)(R774) have the structure [00426] 
L.sub.A283(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A283(R469)(R1)(R1)(R1) to
L.sub.A283(R774)(R774)(R774)(R774) have the structure [00427] 
L.sub.A284(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A284(R469)(R1)(R1)(R1) to
L.sub.A284(R774)(R774)(R774)(R774) have the structure [00428] 
L.sub.A285(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A285(R469)(R1)(R1)(R1) to
L.sub.A285(R774)(R774)(R774)(R774) have the structure [00429] 
L.sub.A286(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A286(R469)(R1)(R1)(R1) to
L.sub.A286(R774)(R774)(R774)(R774) have the structure [00430] 
L.sub.A287(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A287(R469)(R1)(R1)(R1) to
L.sub.A287(R774)(R774)(R774)(R774) have the structure [00431] 
L.sub.A288(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A288(R469)(R1)(R1)(R1) to
L.sub.A288(R774)(R774)(R774)(R774) have the structure [00432] 
L.sub.A289(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A289(R469)(R1)(R1)(R1) to
L.sub.A289(R774)(R774)(R774)(R774) have the structure [00433] 
L.sub.A290(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A290(R469)(R1)(R1)(R1) to
L.sub.A290(R774)(R774)(R774)(R774) have the structure [00434] 
L.sub.A291(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A291(R1)(R1)(R1)(R1) to
L.sub.A291(R774)(R774)(R774)(R774) have the structure [00435] 
L.sub.A292(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A292(R1)(R1)(R1)(R1) to
L.sub.A292(R774)(R774)(R774)(R774) have the structure [00436] 
L.sub.A293(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A293(R1)(R1)(R1)(R1) to
L.sub.A293(R774)(R774)(R774)(R774) have the structure [00437] 
L.sub.A294(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A294(R1)(R1)(R1)(R1) to
L.sub.A294(R774)(R774)(R774)(R774) have the structure [00438] 
L.sub.A295(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A295(R1)(R1)(R1)(R1) to
L.sub.A295(R774)(R774)(R774)(R774) have the structure [00439] 
L.sub.A296(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A296(R1)(R1)(R1)(R1) to
L.sub.A296(R774)(R774)(R774)(R774) have the structure [00440]  [0273]
wherein R1 to R468 have the structures defined in the following LIST 8:

##STR00441## ##STR00442## ##STR00443## ##STR00444## ##STR00445## ##STR00446##
##STR00447## ##STR00448## ##STR00449## ##STR00450## ##STR00451## ##STR00452##

##STR00453## ##STR00454## ##STR00455## ##STR00456## ##STR00457## ##STR00458##
##STR00459##
##STR00460## ##STR00461## ##STR00462## ##STR00463## ##STR00464## ##STR00465##
##STR00466## ##STR00467## ##STR00468## ##STR00469## ##STR00470## ##STR00471##
##STR00472## ##STR00473## ##STR00474## ##STR00475## ##STR00476## ##STR00477##
##STR00478## ##STR00479## ##STR00480## ##STR00481## ##STR00482## ##STR00483##
##STR00484## ##STR00485## ##STR00486## ##STR00487## ##STR00488## ##STR00489##
##STR00490## ##STR00491## ##STR00492## ##STR00493## ##STR00494## ##STR00495##
##STR00496## ##STR00497##
##STR00498## ##STR00499## ##STR00500## ##STR00501## ##STR00502## ##STR00503##
##STR00504## ##STR00505## ##STR00506## ##STR00507## ##STR00508## ##STR00509##
##STR00510## ##STR00511## ##STR00512## [0274] wherein R469 to R774 have the structures
defined in the following LIST 9:
##STR00513## ##STR00514## ##STR00515## ##STR00516## ##STR00517## ##STR00518##
##STR00519## ##STR00520## ##STR00521## ##STR00522## ##STR00523## ##STR00524##
##STR00525## ##STR00526## ##STR00527## ##STR00528## ##STR00529## ##STR00530##
##STR00531## ##STR00532## ##STR00533## ##STR00534## ##STR00535## ##STR00536##
##STR00537## ##STR00538## ##STR00539## ##STR00540## ##STR00541##
##STR00542## ##STR00543## ##STR00544## ##STR00545## ##STR00546## ##STR00547##
##STR00548## ##STR00549## ##STR00550## ##STR00551## ##STR00552## ##STR00553##
##STR00554## ##STR00555## ##STR00556## ##STR00557## ##STR00558## ##STR00559##
##STR00560## ##STR00561## ##STR00562## ##STR00563## ##STR00564## ##STR00565##
##STR00566## ##STR00567## ##STR00568## ##STR00569## ##STR00570## ##STR00571##
[0275] 12. The compound of claim 1, wherein the compound has a formula of
 $M(L_{sub}A)_{sub}p(L_{sub}B)_{sub}q(L_{sub}C)_{sub}r$ wherein $L_{sub}B$ and $L_{sub}C$ are each a bidentate
ligand; and wherein p is 1, 2, or 3; q is 0, 1, or 2; r is 0, 1, or 2; and $p+q+r$ is the oxidation state of
the metal M . [0276] 13. The compound of claim 12, wherein the compound has a formula selected
from the group consisting of $Ir(L_{sub}A)_{sub}3$, $Ir(L_{sub}A)(L_{sub}B)_{sub}2$,
 $Ir(L_{sub}A)_{sub}2(L_{sub}B)$, $Ir(L_{sub}A)_{sub}2(L_{sub}C)$, and $Ir(L_{sub}A)(L_{sub}B)(L_{sub}C)$; and
wherein $L_{sub}A$, $L_{sub}B$, and $L_{sub}C$ are different from each other; or a formula of $Pt(L_{sub}A)$
 $(L_{sub}B)$; and wherein $L_{sub}A$ and $L_{sub}B$ can be same or different. [0277] 14. The compound of
claim 12, wherein the compound has formula $Ir(L_{sub}A)_{sub}3$, formula $Ir(L_{sub}A)$
 $(L_{sub}Bk)_{sub}2$, formula $Ir(L_{sub}A)_{sub}2(L_{sub}Bk)$, formula $Ir(L_{sub}A)_{sub}2(L_{sub}Cj-1)$, or
 $Ir(L_{sub}A)_{sub}2(L_{sub}Cj-II)$; [0278] wherein $L_{sub}A$ is according to Formula II; [0279] wherein
 k is an integer from 1 to 541, and each $L_{sub}Bk$ has a structure defined as follows:
##STR00572## ##STR00573## ##STR00574## ##STR00575## ##STR00576## ##STR00577##
##STR00578## ##STR00579## ##STR00580## ##STR00581## ##STR00582## ##STR00583##
##STR00584## ##STR00585## ##STR00586## ##STR00587## ##STR00588## ##STR00589##
##STR00590## ##STR00591## ##STR00592## ##STR00593## ##STR00594## ##STR00595##
##STR00596## ##STR00597## ##STR00598## ##STR00599## ##STR00600## ##STR00601##
##STR00602## ##STR00603## ##STR00604## ##STR00605## ##STR00606## ##STR00607##
##STR00608## ##STR00609## ##STR00610## ##STR00611## ##STR00612## ##STR00613##
##STR00614## ##STR00615## ##STR00616## ##STR00617## ##STR00618## ##STR00619##
##STR00620## ##STR00621## ##STR00622## ##STR00623## ##STR00624## ##STR00625##
##STR00626## ##STR00627## ##STR00628## ##STR00629## ##STR00630## ##STR00631##
##STR00632## ##STR00633## ##STR00634## ##STR00635## ##STR00636## ##STR00637##
##STR00638## ##STR00639## ##STR00640## ##STR00641## ##STR00642## ##STR00643##
##STR00644## ##STR00645## ##STR00646## ##STR00647## ##STR00648## ##STR00649##
##STR00650## ##STR00651## ##STR00652## ##STR00653## ##STR00654## ##STR00655##
##STR00656## ##STR00657## ##STR00658## ##STR00659## ##STR00660## ##STR00661##

##STR00662## ##STR00663## ##STR00664## ##STR00665## ##STR00666## ##STR00667##
##STR00668## ##STR00669## ##STR00670## ##STR00671## ##STR00672## ##STR00673##
##STR00674## ##STR00675## ##STR00676## ##STR00677## ##STR00678## ##STR00679##
##STR00680## ##STR00681## ##STR00682## ##STR00683## ##STR00684## ##STR00685##
##STR00686## ##STR00687## ##STR00688## ##STR00689## ##STR00690##
##STR00691##

wherein each L.sub.Cj-I has a structure based on formula

##STR00692##

and

each L.sub.Cj-II has a structure based on formula

##STR00693##

wherein for each L.sub.Cj in L.sub.Cj-I and L.sub.Cj-II, R.sup.201 and R.sup.202 are defined as follows:

TABLE-US-00003 L.sub.Cj R.sup.201 R.sup.202 L.sub.C1 R.sup.D1 R.sup.D1 L.sub.C2 R.sup.D2
R.sup.D2 L.sub.C3 R.sup.D3 R.sup.D3 L.sub.C4 R.sup.D4 R.sup.D4 L.sub.C5 R.sup.D5 R.sup.D5
L.sub.C6 R.sup.D6 R.sup.D6 L.sub.C7 R.sup.D7 R.sup.D7 L.sub.C8 R.sup.D8 R.sup.D8 L.sub.C9
R.sup.D9 R.sup.D9 L.sub.C10 R.sup.D10 R.sup.D10 L.sub.C11 R.sup.D11 R.sup.D11 L.sub.C12
R.sup.D12 R.sup.D12 L.sub.C13 R.sup.D13 R.sup.D13 L.sub.C14 R.sup.D14 R.sup.D14
L.sub.C15 R.sup.D15 R.sup.D15 L.sub.C16 R.sup.D16 R.sup.D16 L.sub.C17 R.sup.D17
R.sup.D17 L.sub.C18 R.sup.D18 R.sup.D18 L.sub.C19 R.sup.D19 R.sup.D19 L.sub.C20
R.sup.D20 R.sup.D20 L.sub.C21 R.sup.D21 R.sup.D21 L.sub.C22 R.sup.D22 R.sup.D22
L.sub.C23 R.sup.D23 R.sup.D23 L.sub.C24 R.sup.D24 R.sup.D24 L.sub.C25 R.sup.D25
R.sup.D25 L.sub.C26 R.sup.D26 R.sup.D26 L.sub.C27 R.sup.D27 R.sup.D27 L.sub.C28
R.sup.D28 R.sup.D28 L.sub.C29 R.sup.D29 R.sup.D29 L.sub.C30 R.sup.D30 R.sup.D30
L.sub.C31 R.sup.D31 R.sup.D31 L.sub.C32 R.sup.D32 R.sup.D32 L.sub.C33 R.sup.D33
R.sup.D33 L.sub.C34 R.sup.D34 R.sup.D34 L.sub.C35 R.sup.D35 R.sup.D35 L.sub.C36
R.sup.D36 R.sup.D36 L.sub.C37 R.sup.D37 R.sup.D37 L.sub.C38 R.sup.D38 R.sup.D38
L.sub.C39 R.sup.D39 R.sup.D39 L.sub.C40 R.sup.D40 R.sup.D40 L.sub.C41 R.sup.D41
R.sup.D41 L.sub.C42 R.sup.D42 R.sup.D42 L.sub.C43 R.sup.D43 R.sup.D43 L.sub.C44
R.sup.D44 R.sup.D44 L.sub.C45 R.sup.D45 R.sup.D45 L.sub.C46 R.sup.D46 R.sup.D46
L.sub.C47 R.sup.D47 R.sup.D47 L.sub.C48 R.sup.D48 R.sup.D48 L.sub.C49 R.sup.D49
R.sup.D49 L.sub.C50 R.sup.D50 R.sup.D50 L.sub.C51 R.sup.D51 R.sup.D51 L.sub.C52
R.sup.D52 R.sup.D52 L.sub.C53 R.sup.D53 R.sup.D53 L.sub.C54 R.sup.D54 R.sup.D54
L.sub.C55 R.sup.D55 R.sup.D55 L.sub.C56 R.sup.D56 R.sup.D56 L.sub.C57 R.sup.D57
R.sup.D57 L.sub.C58 R.sup.D58 R.sup.D58 L.sub.C59 R.sup.D59 R.sup.D59 L.sub.C60
R.sup.D60 R.sup.D60 L.sub.C61 R.sup.D61 R.sup.D61 L.sub.C62 R.sup.D62 R.sup.D62
L.sub.C63 R.sup.D63 R.sup.D63 L.sub.C64 R.sup.D64 R.sup.D64 L.sub.C65 R.sup.D65
R.sup.D65 L.sub.C66 R.sup.D66 R.sup.D66 L.sub.C67 R.sup.D67 R.sup.D67 L.sub.C68
R.sup.D68 R.sup.D68 L.sub.C69 R.sup.D69 R.sup.D69 L.sub.C70 R.sup.D70 R.sup.D70
L.sub.C71 R.sup.D71 R.sup.D71 L.sub.C72 R.sup.D72 R.sup.D72 L.sub.C73 R.sup.D73
R.sup.D73 L.sub.C74 R.sup.D74 R.sup.D74 L.sub.C75 R.sup.D75 R.sup.D75 L.sub.C76
R.sup.D76 R.sup.D76 L.sub.C77 R.sup.D77 R.sup.D77 L.sub.C78 R.sup.D78 R.sup.D78
L.sub.C79 R.sup.D79 R.sup.D79 L.sub.C80 R.sup.D80 R.sup.D80 L.sub.C81 R.sup.D81
R.sup.D81 L.sub.C82 R.sup.D82 R.sup.D82 L.sub.C83 R.sup.D83 R.sup.D83 L.sub.C84
R.sup.D84 R.sup.D84 L.sub.C85 R.sup.D85 R.sup.D85 L.sub.C86 R.sup.D86 R.sup.D86
L.sub.C87 R.sup.D87 R.sup.D87 L.sub.C88 R.sup.D88 R.sup.D88 L.sub.C89 R.sup.D89
R.sup.D89 L.sub.C90 R.sup.D90 R.sup.D90 L.sub.C91 R.sup.D91 R.sup.D91 L.sub.C92
R.sup.D92 R.sup.D92 L.sub.C93 R.sup.D93 R.sup.D93 L.sub.C94 R.sup.D94 R.sup.D94
L.sub.C95 R.sup.D95 R.sup.D95 L.sub.C96 R.sup.D96 R.sup.D96 L.sub.C97 R.sup.D97
R.sup.D97 L.sub.C98 R.sup.D98 R.sup.D98 L.sub.C99 R.sup.D99 R.sup.D99 L.sub.C100

R.sup.D100 R.sup.D100 L.sub.C101 R.sup.D101 L.sub.C102 R.sup.D102 R.sup.D102
L.sub.C103 R.sup.D103 R.sup.D103 L.sub.C104 R.sup.D104 R.sup.D104 L.sub.C105 R.sup.D105
R.sup.D105 L.sub.C106 R.sup.D106 R.sup.D106 L.sub.C107 R.sup.D107 R.sup.D107 L.sub.C108
R.sup.D108 R.sup.D108 L.sub.C109 R.sup.D109 R.sup.D109 L.sub.C110 R.sup.D110 R.sup.D110
L.sub.C111 R.sup.D111 R.sup.D111 L.sub.C112 R.sup.D112 R.sup.D112 L.sub.C113 R.sup.D113
R.sup.D113 L.sub.C114 R.sup.D114 R.sup.D114 L.sub.C115 R.sup.D115 R.sup.D115 L.sub.C116
R.sup.D116 R.sup.D116 L.sub.C117 R.sup.D117 R.sup.D117 L.sub.C118 R.sup.D118 R.sup.D118
L.sub.C119 R.sup.D119 R.sup.D119 L.sub.C120 R.sup.D120 R.sup.D120 L.sub.C121 R.sup.D121
R.sup.D121 L.sub.C122 R.sup.D122 R.sup.D122 L.sub.C123 R.sup.D123 R.sup.D123 L.sub.C124
R.sup.D124 R.sup.D124 L.sub.C125 R.sup.D125 R.sup.D125 L.sub.C126 R.sup.D126 R.sup.D126
L.sub.C127 R.sup.D127 R.sup.D127 L.sub.C128 R.sup.D128 R.sup.D128 L.sub.C129 R.sup.D129
R.sup.D129 L.sub.C130 R.sup.D130 R.sup.D130 L.sub.C131 R.sup.D131 R.sup.D131 L.sub.C132
R.sup.D132 R.sup.D132 L.sub.C133 R.sup.D133 R.sup.D133 L.sub.C134 R.sup.D134 R.sup.D134
L.sub.C135 R.sup.D135 R.sup.D135 L.sub.C136 R.sup.D136 R.sup.D136 L.sub.C137 R.sup.D137
R.sup.D137 L.sub.C138 R.sup.D138 R.sup.D138 L.sub.C139 R.sup.D139 R.sup.D139 L.sub.C140
R.sup.D140 R.sup.D140 L.sub.C141 R.sup.D141 R.sup.D141 L.sub.C142 R.sup.D142 R.sup.D142
L.sub.C143 R.sup.D143 R.sup.D143 L.sub.C144 R.sup.D144 R.sup.D144 L.sub.C145 R.sup.D145
R.sup.D145 L.sub.C146 R.sup.D146 R.sup.D146 L.sub.C147 R.sup.D147 R.sup.D147 L.sub.C148
R.sup.D148 R.sup.D148 L.sub.C149 R.sup.D149 R.sup.D149 L.sub.C150 R.sup.D150 R.sup.D150
L.sub.C151 R.sup.D151 R.sup.D151 L.sub.C152 R.sup.D152 R.sup.D152 L.sub.C153 R.sup.D153
R.sup.D153 L.sub.C154 R.sup.D154 R.sup.D154 L.sub.C155 R.sup.D155 R.sup.D155 L.sub.C156
R.sup.D156 R.sup.D156 L.sub.C157 R.sup.D157 R.sup.D157 L.sub.C158 R.sup.D158 R.sup.D158
L.sub.C159 R.sup.D159 R.sup.D159 L.sub.C160 R.sup.D160 R.sup.D160 L.sub.C161 R.sup.D161
R.sup.D161 L.sub.C162 R.sup.D162 R.sup.D162 L.sub.C163 R.sup.D163 R.sup.D163 L.sub.C164
R.sup.D164 R.sup.D164 L.sub.C165 R.sup.D165 R.sup.D165 L.sub.C166 R.sup.D166 R.sup.D166
L.sub.C167 R.sup.D167 R.sup.D167 L.sub.C168 R.sup.D168 R.sup.D168 L.sub.C169 R.sup.D169
R.sup.D169 L.sub.C170 R.sup.D170 R.sup.D170 L.sub.C171 R.sup.D171 R.sup.D171 L.sub.C172
R.sup.D172 R.sup.D172 L.sub.C173 R.sup.D173 R.sup.D173 L.sub.C174 R.sup.D174 R.sup.D174
L.sub.C175 R.sup.D175 R.sup.D175 L.sub.C176 R.sup.D176 R.sup.D176 L.sub.C177 R.sup.D177
R.sup.D177 L.sub.C178 R.sup.D178 R.sup.D178 L.sub.C179 R.sup.D179 R.sup.D179 L.sub.C180
R.sup.D180 R.sup.D180 L.sub.C181 R.sup.D181 R.sup.D181 L.sub.C182 R.sup.D182 R.sup.D182
L.sub.C183 R.sup.D183 R.sup.D183 L.sub.C184 R.sup.D184 R.sup.D184 L.sub.C185 R.sup.D185
R.sup.D185 L.sub.C186 R.sup.D186 R.sup.D186 L.sub.C187 R.sup.D187 R.sup.D187 L.sub.C188
R.sup.D188 R.sup.D188 L.sub.C189 R.sup.D189 R.sup.D189 L.sub.C190 R.sup.D190 R.sup.D190
L.sub.C191 R.sup.D191 R.sup.D191 L.sub.C192 R.sup.D192 R.sup.D192 L.sub.C193 R.sup.D1
R.sup.D3 L.sub.C194 R.sup.D1 R.sup.D4 L.sub.C195 R.sup.D1 R.sup.D5 L.sub.C196 R.sup.D1
R.sup.D9 L.sub.C197 R.sup.D1 R.sup.D10 L.sub.C198 R.sup.D1 R.sup.D17 L.sub.C199 R.sup.D1
R.sup.D18 L.sub.C200 R.sup.D1 R.sup.D20 L.sub.C201 R.sup.D1 R.sup.D22 L.sub.C202
R.sup.D1 R.sup.D37 L.sub.C203 R.sup.D1 R.sup.D40 L.sub.C204 R.sup.D1 R.sup.D41
L.sub.C205 R.sup.D1 R.sup.D42 L.sub.C206 R.sup.D1 R.sup.D43 L.sub.C207 R.sup.D1
R.sup.D48 L.sub.C208 R.sup.D1 R.sup.D49 L.sub.C209 R.sup.D1 R.sup.D50 L.sub.C210
R.sup.D1 R.sup.D54 L.sub.C211 R.sup.D1 R.sup.D55 L.sub.C212 R.sup.D1 R.sup.D58
L.sub.C213 R.sup.D1 R.sup.D59 L.sub.C214 R.sup.D1 R.sup.D78 L.sub.C215 R.sup.D1
R.sup.D79 L.sub.C216 R.sup.D1 R.sup.D81 L.sub.C217 R.sup.D1 R.sup.D87 L.sub.C218
R.sup.D1 R.sup.D88 L.sub.C219 R.sup.D1 R.sup.D89 L.sub.C220 R.sup.D1 R.sup.D93
L.sub.C221 R.sup.D1 R.sup.D116 L.sub.C222 R.sup.D1 R.sup.D117 L.sub.C223 R.sup.D1
R.sup.D118 L.sub.C224 R.sup.D1 R.sup.D119 L.sub.C225 R.sup.D1 R.sup.D120 L.sub.C226
R.sup.D1 R.sup.D133 L.sub.C227 R.sup.D1 R.sup.D134 L.sub.C228 R.sup.D1 R.sup.D135
L.sub.C229 R.sup.D1 R.sup.D136 L.sub.C230 R.sup.D1 R.sup.D143 L.sub.C231 R.sup.D1
R.sup.D144 L.sub.C232 R.sup.D1 R.sup.D145 L.sub.C233 R.sup.D1 R.sup.D146 L.sub.C234

R.sup.D1 R.sup.D149 L.sub.C235 R.sup.D1 R.sup.D149 L.sub.C236 R.sup.D1 R.sup.D151
L.sub.C237 R.sup.D1 R.sup.D154 L.sub.C238 R.sup.D1 R.sup.D155 L.sub.C239 R.sup.D1
R.sup.D161 L.sub.C240 R.sup.D1 R.sup.D175 L.sub.C241 R.sup.D4 R.sup.D3 L.sub.C242
R.sup.D4 R.sup.D5 L.sub.C243 R.sup.D4 R.sup.D9 L.sub.C244 R.sup.D4 R.sup.D10 L.sub.C245
R.sup.D4 R.sup.D17 L.sub.C246 R.sup.D4 R.sup.D18 L.sub.C247 R.sup.D4 R.sup.D20
L.sub.C248 R.sup.D4 R.sup.D22 L.sub.C249 R.sup.D4 R.sup.D37 L.sub.C250 R.sup.D4
R.sup.D40 L.sub.C251 R.sup.D4 R.sup.D41 L.sub.C252 R.sup.D4 R.sup.D42 L.sub.C253
R.sup.D4 R.sup.D43 L.sub.C254 R.sup.D4 R.sup.D48 L.sub.C255 R.sup.D4 R.sup.D49
L.sub.C256 R.sup.D4 R.sup.D50 L.sub.C257 R.sup.D4 R.sup.D54 L.sub.C258 R.sup.D4
R.sup.D55 L.sub.C259 R.sup.D4 R.sup.D58 L.sub.C260 R.sup.D4 R.sup.D59 L.sub.C261
R.sup.D4 R.sup.D78 L.sub.C262 R.sup.D4 R.sup.D79 L.sub.C263 R.sup.D4 R.sup.D81
L.sub.C264 R.sup.D4 R.sup.D87 L.sub.C265 R.sup.D4 R.sup.D88 L.sub.C266 R.sup.D4
R.sup.D89 L.sub.C267 R.sup.D4 R.sup.D93 L.sub.C268 R.sup.D4 R.sup.D116 L.sub.C269
R.sup.D4 R.sup.D117 L.sub.C270 R.sup.D4 R.sup.D118 L.sub.C271 R.sup.D4 R.sup.D119
L.sub.C272 R.sup.D4 R.sup.D120 L.sub.C273 R.sup.D4 R.sup.D133 L.sub.C274 R.sup.D4
R.sup.D134 L.sub.C275 R.sup.D4 R.sup.D135 L.sub.C276 R.sup.D4 R.sup.D136 L.sub.C277
R.sup.D4 R.sup.D143 L.sub.C278 R.sup.D4 R.sup.D144 L.sub.C279 R.sup.D4 R.sup.D145
L.sub.C280 R.sup.D4 R.sup.D146 L.sub.C281 R.sup.D4 R.sup.D147 L.sub.C282 R.sup.D4
R.sup.D149 L.sub.C283 R.sup.D4 R.sup.D151 L.sub.C284 R.sup.D4 R.sup.D154 L.sub.C285
R.sup.D4 R.sup.D155 L.sub.C286 R.sup.D4 R.sup.D161 L.sub.C287 R.sup.D4 R.sup.D175
L.sub.C288 R.sup.D9 R.sup.D3 L.sub.C289 R.sup.D9 R.sup.D5 L.sub.C290 R.sup.D9 R.sup.D10
L.sub.C291 R.sup.D9 R.sup.D17 L.sub.C292 R.sup.D9 R.sup.D18 L.sub.C293 R.sup.D9
R.sup.D20 L.sub.C294 R.sup.D9 R.sup.D22 L.sub.C295 R.sup.D9 R.sup.D37 L.sub.C296
R.sup.D9 R.sup.D40 L.sub.C297 R.sup.D9 R.sup.D41 L.sub.C298 R.sup.D9 R.sup.D42
L.sub.C299 R.sup.D9 R.sup.D43 L.sub.C300 R.sup.D9 R.sup.D48 L.sub.C301 R.sup.D9
R.sup.D49 L.sub.C302 R.sup.D9 R.sup.D50 L.sub.C303 R.sup.D9 R.sup.D54 L.sub.C304
R.sup.D9 R.sup.D55 L.sub.C305 R.sup.D9 R.sup.D58 L.sub.C306 R.sup.D9 R.sup.D59
L.sub.C307 R.sup.D9 R.sup.D78 L.sub.C308 R.sup.D9 R.sup.D79 L.sub.C309 R.sup.D9
R.sup.D81 L.sub.C310 R.sup.D9 R.sup.D87 L.sub.C311 R.sup.D9 R.sup.D88 L.sub.C312
R.sup.D9 R.sup.D89 L.sub.C313 R.sup.D9 R.sup.D93 L.sub.C314 R.sup.D9 R.sup.D116
L.sub.C315 R.sup.D9 R.sup.D117 L.sub.C316 R.sup.D9 R.sup.D118 L.sub.C317 R.sup.D9
R.sup.D119 L.sub.C318 R.sup.D9 R.sup.D120 L.sub.C319 R.sup.D9 R.sup.D133 L.sub.C320
R.sup.D9 R.sup.D134 L.sub.C321 R.sup.D9 R.sup.D135 L.sub.C322 R.sup.D9 R.sup.D136
L.sub.C323 R.sup.D9 R.sup.D143 L.sub.C324 R.sup.D9 R.sup.D144 L.sub.C325 R.sup.D9
R.sup.D145 L.sub.C326 R.sup.D9 R.sup.D146 L.sub.C327 R.sup.D9 R.sup.D147 L.sub.C328
R.sup.D9 R.sup.D149 L.sub.C329 R.sup.D9 R.sup.D151 L.sub.C330 R.sup.D9 R.sup.D154
L.sub.C331 R.sup.D9 R.sup.D155 L.sub.C332 R.sup.D9 R.sup.D161 L.sub.C333 R.sup.D9
R.sup.D175 L.sub.C334 R.sup.D10 R.sup.D3 L.sub.C335 R.sup.D10 R.sup.D5 L.sub.C336
R.sup.D10 R.sup.D17 L.sub.C337 R.sup.D10 R.sup.D18 L.sub.C338 R.sup.D10 R.sup.D20
L.sub.C339 R.sup.D10 R.sup.D22 L.sub.C340 R.sup.D10 R.sup.D37 L.sub.C341 R.sup.D10
R.sup.D40 L.sub.C342 R.sup.D10 R.sup.D41 L.sub.C343 R.sup.D10 R.sup.D42 L.sub.C344
R.sup.D10 R.sup.D43 L.sub.C345 R.sup.D10 R.sup.D48 L.sub.C346 R.sup.D10 R.sup.D49
L.sub.C347 R.sup.D10 R.sup.D50 L.sub.C348 R.sup.D10 R.sup.D54 L.sub.C349 R.sup.D10
R.sup.D55 L.sub.C350 R.sup.D10 R.sup.D58 L.sub.C351 R.sup.D10 R.sup.D59 L.sub.C352
R.sup.D10 R.sup.D78 L.sub.C353 R.sup.D10 R.sup.D79 L.sub.C354 R.sup.D10 R.sup.D81
L.sub.C355 R.sup.D10 R.sup.D87 L.sub.C356 R.sup.D10 R.sup.D88 L.sub.C357 R.sup.D10
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R.sup.D10 R.sup.D117 L.sub.C361 R.sup.D10 R.sup.D118 L.sub.C362 R.sup.D10 R.sup.D119
L.sub.C363 R.sup.D10 R.sup.D120 L.sub.C364 R.sup.D10 R.sup.D133 L.sub.C365 R.sup.D10
R.sup.D134 L.sub.C366 R.sup.D10 R.sup.D135 L.sub.C367 R.sup.D10 R.sup.D136 L.sub.C368

R.sup.D10 R.sup.D10 L.sub.C369 R.sup.D10 R.sup.D144 L.sub.C370 R.sup.D10 R.sup.D145
L.sub.C371 R.sup.D10 R.sup.D146 L.sub.C372 R.sup.D10 R.sup.D147 L.sub.C373 R.sup.D10
R.sup.D149 L.sub.C374 R.sup.D10 R.sup.D151 L.sub.C375 R.sup.D10 R.sup.D154 L.sub.C376
R.sup.D10 R.sup.D155 L.sub.C377 R.sup.D10 R.sup.D161 L.sub.C378 R.sup.D10 R.sup.D175
L.sub.C379 R.sup.D17 R.sup.D3 L.sub.C380 R.sup.D17 R.sup.D5 L.sub.C381 R.sup.D17
R.sup.D18 L.sub.C382 R.sup.D17 R.sup.D20 L.sub.C383 R.sup.D17 R.sup.D22 L.sub.C384
R.sup.D17 R.sup.D37 L.sub.C385 R.sup.D17 R.sup.D40 L.sub.C386 R.sup.D17 R.sup.D41
L.sub.C387 R.sup.D17 R.sup.D42 L.sub.C388 R.sup.D17 R.sup.D43 L.sub.C389 R.sup.D17
R.sup.D48 L.sub.C390 R.sup.D17 R.sup.D49 L.sub.C391 R.sup.D17 R.sup.D50 L.sub.C392
R.sup.D17 R.sup.D54 L.sub.C393 R.sup.D17 R.sup.D55 L.sub.C394 R.sup.D17 R.sup.D58
L.sub.C395 R.sup.D17 R.sup.D59 L.sub.C396 R.sup.D17 R.sup.D78 L.sub.C397 R.sup.D17
R.sup.D79 L.sub.C398 R.sup.D17 R.sup.D81 L.sub.C399 R.sup.D17 R.sup.D87 L.sub.C400
R.sup.D17 R.sup.D88 L.sub.C401 R.sup.D17 R.sup.D89 L.sub.C402 R.sup.D17 R.sup.D93
L.sub.C403 R.sup.D17 R.sup.D116 L.sub.C404 R.sup.D17 R.sup.D117 L.sub.C405 R.sup.D17
R.sup.D118 L.sub.C406 R.sup.D17 R.sup.D119 L.sub.C407 R.sup.D17 R.sup.D120 L.sub.C408
R.sup.D17 R.sup.D133 L.sub.C409 R.sup.D17 R.sup.D134 L.sub.C410 R.sup.D17 R.sup.D135
L.sub.C411 R.sup.D17 R.sup.D136 L.sub.C412 R.sup.D17 R.sup.D143 L.sub.C413 R.sup.D17
R.sup.D144 L.sub.C414 R.sup.D17 R.sup.D145 L.sub.C415 R.sup.D17 R.sup.D146 L.sub.C416
R.sup.D17 R.sup.D147 L.sub.C417 R.sup.D17 R.sup.D149 L.sub.C418 R.sup.D17 R.sup.D151
L.sub.C419 R.sup.D17 R.sup.D154 L.sub.C420 R.sup.D17 R.sup.D155 L.sub.C421 R.sup.D17
R.sup.D161 L.sub.C422 R.sup.D17 R.sup.D175 L.sub.C423 R.sup.D50 R.sup.D3 L.sub.C424
R.sup.D50 R.sup.D5 L.sub.C425 R.sup.D50 R.sup.D18 L.sub.C426 R.sup.D50 R.sup.D20
L.sub.C427 R.sup.D50 R.sup.D22 L.sub.C428 R.sup.D50 R.sup.D37 L.sub.C429 R.sup.D50
R.sup.D40 L.sub.C430 R.sup.D50 R.sup.D41 L.sub.C431 R.sup.D50 R.sup.D42 L.sub.C432
R.sup.D50 R.sup.D43 L.sub.C433 R.sup.D50 R.sup.D48 L.sub.C434 R.sup.D50 R.sup.D49
L.sub.C435 R.sup.D50 R.sup.D54 L.sub.C436 R.sup.D50 R.sup.D55 L.sub.C437 R.sup.D50
R.sup.D58 L.sub.C438 R.sup.D50 R.sup.D59 L.sub.C439 R.sup.D50 R.sup.D78 L.sub.C440
R.sup.D50 R.sup.D79 L.sub.C441 R.sup.D50 R.sup.D81 L.sub.C442 R.sup.D50 R.sup.D87
L.sub.C443 R.sup.D50 R.sup.D88 L.sub.C444 R.sup.D50 R.sup.D89 L.sub.C445 R.sup.D50
R.sup.D93 L.sub.C446 R.sup.D50 R.sup.D116 L.sub.C447 R.sup.D50 R.sup.D117 L.sub.C448
R.sup.D50 R.sup.D118 L.sub.C449 R.sup.D50 R.sup.D119 L.sub.C450 R.sup.D50 R.sup.D120
L.sub.C451 R.sup.D50 R.sup.D133 L.sub.C452 R.sup.D50 R.sup.D134 L.sub.C453 R.sup.D50
R.sup.D135 L.sub.C454 R.sup.D50 R.sup.D136 L.sub.C455 R.sup.D50 R.sup.D143 L.sub.C456
R.sup.D50 R.sup.D144 L.sub.C457 R.sup.D50 R.sup.D145 L.sub.C458 R.sup.D50 R.sup.D146
L.sub.C459 R.sup.D50 R.sup.D147 L.sub.C460 R.sup.D50 R.sup.D149 L.sub.C461 R.sup.D50
R.sup.D151 L.sub.C462 R.sup.D50 R.sup.D154 L.sub.C463 R.sup.D50 R.sup.D155 L.sub.C464
R.sup.D50 R.sup.D161 L.sub.C465 R.sup.D50 R.sup.D175 L.sub.C466 R.sup.D55 R.sup.D3
L.sub.C467 R.sup.D55 R.sup.D5 L.sub.C468 R.sup.D55 R.sup.D18 L.sub.C469 R.sup.D55
R.sup.D20 L.sub.C470 R.sup.D55 R.sup.D22 L.sub.C471 R.sup.D55 R.sup.D37 L.sub.C472
R.sup.D55 R.sup.D40 L.sub.C473 R.sup.D55 R.sup.D41 L.sub.C474 R.sup.D55 R.sup.D42
L.sub.C475 R.sup.D55 R.sup.D43 L.sub.C476 R.sup.D55 R.sup.D48 L.sub.C477 R.sup.D55
R.sup.D49 L.sub.C478 R.sup.D55 R.sup.D54 L.sub.C479 R.sup.D55 R.sup.D58 L.sub.C480
R.sup.D55 R.sup.D59 L.sub.C481 R.sup.D55 R.sup.D78 L.sub.C482 R.sup.D55 R.sup.D79
L.sub.C483 R.sup.D55 R.sup.D81 L.sub.C484 R.sup.D55 R.sup.D87 L.sub.C485 R.sup.D55
R.sup.D88 L.sub.C486 R.sup.D55 R.sup.D89 L.sub.C487 R.sup.D55 R.sup.D93 L.sub.C488
R.sup.D55 R.sup.D116 L.sub.C489 R.sup.D55 R.sup.D117 L.sub.C490 R.sup.D55 R.sup.D118
L.sub.C491 R.sup.D55 R.sup.D119 L.sub.C492 R.sup.D55 R.sup.D120 L.sub.C493 R.sup.D55
R.sup.D133 L.sub.C494 R.sup.D55 R.sup.D134 L.sub.C495 R.sup.D55 R.sup.D135 L.sub.C496
R.sup.D55 R.sup.D136 L.sub.C497 R.sup.D55 R.sup.D143 L.sub.C498 R.sup.D55 R.sup.D144
L.sub.C499 R.sup.D55 R.sup.D145 L.sub.C500 R.sup.D55 R.sup.D146 L.sub.C501 R.sup.D55

R.sup.D147 L.sub.C502 R.sup.D502 R.sup.D149 L.sub.C503 R.sup.D55 R.sup.D155 L.sub.C504
R.sup.D55 R.sup.D154 L.sub.C505 R.sup.D55 R.sup.D155 L.sub.C506 R.sup.D55 R.sup.D161
L.sub.C507 R.sup.D55 R.sup.D175 L.sub.C508 R.sup.D116 R.sup.D3 L.sub.C509 R.sup.D116
R.sup.D5 L.sub.C510 R.sup.D116 R.sup.D17 L.sub.C511 R.sup.D116 R.sup.D18 L.sub.C512
R.sup.D116 R.sup.D20 L.sub.C513 R.sup.D116 R.sup.D22 L.sub.C514 R.sup.D116 R.sup.D37
L.sub.C515 R.sup.D116 R.sup.D40 L.sub.C516 R.sup.D116 R.sup.D41 L.sub.C517 R.sup.D116
R.sup.D42 L.sub.C518 R.sup.D116 R.sup.D43 L.sub.C519 R.sup.D116 R.sup.D48 L.sub.C520
R.sup.D116 R.sup.D49 L.sub.C521 R.sup.D116 R.sup.D54 L.sub.C522 R.sup.D116 R.sup.D58
L.sub.C523 R.sup.D116 R.sup.D59 L.sub.C524 R.sup.D116 R.sup.D78 L.sub.C525 R.sup.D116
R.sup.D79 L.sub.C526 R.sup.D116 R.sup.D81 L.sub.C527 R.sup.D116 R.sup.D87 L.sub.C528
R.sup.D116 R.sup.D88 L.sub.C529 R.sup.D116 R.sup.D89 L.sub.C530 R.sup.D116 R.sup.D93
L.sub.C531 R.sup.D116 R.sup.D117 L.sub.C532 R.sup.D116 R.sup.D118 L.sub.C533 R.sup.D116
R.sup.D119 L.sub.C534 R.sup.D116 R.sup.D120 L.sub.C535 R.sup.D116 R.sup.D133 L.sub.C536
R.sup.D116 R.sup.D134 L.sub.C537 R.sup.D116 R.sup.D135 L.sub.C538 R.sup.D116 R.sup.D136
L.sub.C539 R.sup.D116 R.sup.D143 L.sub.C540 R.sup.D116 R.sup.D144 L.sub.C541 R.sup.D116
R.sup.D145 L.sub.C542 R.sup.D116 R.sup.D146 L.sub.C543 R.sup.D116 R.sup.D147 L.sub.C544
R.sup.D116 R.sup.D149 L.sub.C545 R.sup.D116 R.sup.D151 L.sub.C546 R.sup.D116 R.sup.D154
L.sub.C547 R.sup.D116 R.sup.D155 L.sub.C548 R.sup.D116 R.sup.D161 L.sub.C549 R.sup.D116
R.sup.D175 L.sub.C550 R.sup.D143 R.sup.D3 L.sub.C551 R.sup.D143 R.sup.D5 L.sub.C552
R.sup.D143 R.sup.D17 L.sub.C553 R.sup.D143 R.sup.D18 L.sub.C554 R.sup.D143 R.sup.D20
L.sub.C555 R.sup.D143 R.sup.D22 L.sub.C556 R.sup.D143 R.sup.D37 L.sub.C557 R.sup.D143
R.sup.D40 L.sub.C558 R.sup.D143 R.sup.D41 L.sub.C559 R.sup.D143 R.sup.D42 L.sub.C560
R.sup.D143 R.sup.D43 L.sub.C561 R.sup.D143 R.sup.D48 L.sub.C562 R.sup.D143 R.sup.D49
L.sub.C563 R.sup.D143 R.sup.D54 L.sub.C564 R.sup.D143 R.sup.D58 L.sub.C565 R.sup.D143
R.sup.D59 L.sub.C566 R.sup.D143 R.sup.D78 L.sub.C567 R.sup.D143 R.sup.D79 L.sub.C568
R.sup.D143 R.sup.D81 L.sub.C569 R.sup.D143 R.sup.D87 L.sub.C570 R.sup.D143 R.sup.D88
L.sub.C571 R.sup.D143 R.sup.D89 L.sub.C572 R.sup.D143 R.sup.D93 L.sub.C573 R.sup.D143
R.sup.D116 L.sub.C574 R.sup.D143 R.sup.D117 L.sub.C575 R.sup.D143 R.sup.D118 L.sub.C576
R.sup.D143 R.sup.D119 L.sub.C577 R.sup.D143 R.sup.D120 L.sub.C578 R.sup.D143 R.sup.D133
L.sub.C579 R.sup.D143 R.sup.D134 L.sub.C580 R.sup.D143 R.sup.D135 L.sub.C581 R.sup.D143
R.sup.D136 L.sub.C582 R.sup.D143 R.sup.D144 L.sub.C583 R.sup.D143 R.sup.D145 L.sub.C584
R.sup.D143 R.sup.D146 L.sub.C585 R.sup.D143 R.sup.D147 L.sub.C586 R.sup.D143 R.sup.D149
L.sub.C587 R.sup.D143 R.sup.D151 L.sub.C588 R.sup.D143 R.sup.D154 L.sub.C589 R.sup.D143
R.sup.D155 L.sub.C590 R.sup.D143 R.sup.D161 L.sub.C591 R.sup.D143 R.sup.D175 L.sub.C592
R.sup.D144 R.sup.D3 L.sub.C593 R.sup.D144 R.sup.D5 L.sub.C594 R.sup.D144 R.sup.D17
L.sub.C595 R.sup.D144 R.sup.D18 L.sub.C596 R.sup.D144 R.sup.D20 L.sub.C597 R.sup.D144
R.sup.D22 L.sub.C598 R.sup.D144 R.sup.D37 L.sub.C599 R.sup.D144 R.sup.D40 L.sub.C600
R.sup.D144 R.sup.D41 L.sub.C601 R.sup.D144 R.sup.D42 L.sub.C602 R.sup.D144 R.sup.D43
L.sub.C603 R.sup.D144 R.sup.D48 L.sub.C604 R.sup.D144 R.sup.D49 L.sub.C605 R.sup.D144
R.sup.D54 L.sub.C606 R.sup.D144 R.sup.D58 L.sub.C607 R.sup.D144 R.sup.D59 L.sub.C608
R.sup.D144 R.sup.D78 L.sub.C609 R.sup.D144 R.sup.D79 L.sub.C610 R.sup.D144 R.sup.D81
L.sub.C611 R.sup.D144 R.sup.D87 L.sub.C612 R.sup.D144 R.sup.D88 L.sub.C613 R.sup.D144
R.sup.D89 L.sub.C614 R.sup.D144 R.sup.D93 L.sub.C615 R.sup.D144 R.sup.D116 L.sub.C616
R.sup.D144 R.sup.D117 L.sub.C617 R.sup.D144 R.sup.D118 L.sub.C618 R.sup.D144 R.sup.D119
L.sub.C619 R.sup.D144 R.sup.D120 L.sub.C620 R.sup.D144 R.sup.D133 L.sub.C621 R.sup.D144
R.sup.D134 L.sub.C622 R.sup.D144 R.sup.D135 L.sub.C623 R.sup.D144 R.sup.D136 L.sub.C624
R.sup.D144 R.sup.D145 L.sub.C625 R.sup.D144 R.sup.D146 L.sub.C626 R.sup.D144 R.sup.D147
L.sub.C627 R.sup.D144 R.sup.D149 L.sub.C628 R.sup.D144 R.sup.D151 L.sub.C629 R.sup.D144
R.sup.D154 L.sub.C630 R.sup.D144 R.sup.D155 L.sub.C631 R.sup.D144 R.sup.D161 L.sub.C632
R.sup.D144 R.sup.D175 L.sub.C633 R.sup.D145 R.sup.D3 L.sub.C634 R.sup.D145 R.sup.D5

L.sub.C635 R.sup.D145 R.sup.D17 L.sub.C636 R.sup.D145 R.sup.D18 L.sub.C637 R.sup.D145
R.sup.D20 L.sub.C638 R.sup.D145 R.sup.D22 L.sub.C639 R.sup.D145 R.sup.D37 L.sub.C640
R.sup.D145 R.sup.D40 L.sub.C641 R.sup.D145 R.sup.D41 L.sub.C642 R.sup.D145 R.sup.D42
L.sub.C643 R.sup.D145 R.sup.D43 L.sub.C644 R.sup.D145 R.sup.D48 L.sub.C645 R.sup.D145
R.sup.D49 L.sub.C646 R.sup.D145 R.sup.D54 L.sub.C647 R.sup.D145 R.sup.D58 L.sub.C648
R.sup.D145 R.sup.D59 L.sub.C649 R.sup.D145 R.sup.D78 L.sub.C650 R.sup.D145 R.sup.D79
L.sub.C651 R.sup.D145 R.sup.D81 L.sub.C652 R.sup.D145 R.sup.D87 L.sub.C653 R.sup.D145
R.sup.D88 L.sub.C654 R.sup.D145 R.sup.D89 L.sub.C655 R.sup.D145 R.sup.D93 L.sub.C656
R.sup.D145 R.sup.D116 L.sub.C657 R.sup.D145 R.sup.D117 L.sub.C658 R.sup.D145 R.sup.D118
L.sub.C659 R.sup.D145 R.sup.D119 L.sub.C660 R.sup.D145 R.sup.D120 L.sub.C661 R.sup.D145
R.sup.D133 L.sub.C662 R.sup.D145 R.sup.D134 L.sub.C663 R.sup.D145 R.sup.D135 L.sub.C664
R.sup.D145 R.sup.D136 L.sub.C665 R.sup.D145 R.sup.D146 L.sub.C666 R.sup.D145 R.sup.D147
L.sub.C667 R.sup.D145 R.sup.D149 L.sub.C668 R.sup.D145 R.sup.D151 L.sub.C669 R.sup.D145
R.sup.D154 L.sub.C670 R.sup.D145 R.sup.D155 L.sub.C671 R.sup.D145 R.sup.D161 L.sub.C672
R.sup.D145 R.sup.D175 L.sub.C673 R.sup.D146 R.sup.D3 L.sub.C674 R.sup.D146 R.sup.D5
L.sub.C675 R.sup.D146 R.sup.D17 L.sub.C676 R.sup.D146 R.sup.D18 L.sub.C677 R.sup.D146
R.sup.D20 L.sub.C678 R.sup.D146 R.sup.D22 L.sub.C679 R.sup.D146 R.sup.D37 L.sub.C680
R.sup.D146 R.sup.D40 L.sub.C681 R.sup.D146 R.sup.D41 L.sub.C682 R.sup.D146 R.sup.D42
L.sub.C683 R.sup.D146 R.sup.D43 L.sub.C684 R.sup.D146 R.sup.D48 L.sub.C685 R.sup.D146
R.sup.D49 L.sub.C686 R.sup.D146 R.sup.D54 L.sub.C687 R.sup.D146 R.sup.D58 L.sub.C688
R.sup.D146 R.sup.D59 L.sub.C689 R.sup.D146 R.sup.D78 L.sub.C690 R.sup.D146 R.sup.D79
L.sub.C691 R.sup.D146 R.sup.D81 L.sub.C692 R.sup.D146 R.sup.D87 L.sub.C693 R.sup.D146
R.sup.D88 L.sub.C694 R.sup.D146 R.sup.D89 L.sub.C695 R.sup.D146 R.sup.D93 L.sub.C696
R.sup.D146 R.sup.D117 L.sub.C697 R.sup.D146 R.sup.D118 L.sub.C698 R.sup.D146 R.sup.D119
L.sub.C699 R.sup.D146 R.sup.D120 L.sub.C700 R.sup.D146 R.sup.D133 L.sub.C701 R.sup.D146
R.sup.D134 L.sub.C702 R.sup.D146 R.sup.D135 L.sub.C703 R.sup.D146 R.sup.D136 L.sub.C704
R.sup.D146 R.sup.D146 L.sub.C705 R.sup.D146 R.sup.D147 L.sub.C706 R.sup.D146 R.sup.D149
L.sub.C707 R.sup.D146 R.sup.D151 L.sub.C708 R.sup.D146 R.sup.D154 L.sub.C709 R.sup.D146
R.sup.D155 L.sub.C710 R.sup.D146 R.sup.D161 L.sub.C711 R.sup.D146 R.sup.D175 L.sub.C712
R.sup.D133 R.sup.D3 L.sub.C713 R.sup.D133 R.sup.D5 L.sub.C714 R.sup.D133 R.sup.D3
L.sub.C715 R.sup.D133 R.sup.D18 L.sub.C716 R.sup.D133 R.sup.D20 L.sub.C717 R.sup.D133
R.sup.D22 L.sub.C718 R.sup.D133 R.sup.D37 L.sub.C719 R.sup.D133 R.sup.D40 L.sub.C720
R.sup.D133 R.sup.D41 L.sub.C721 R.sup.D133 R.sup.D42 L.sub.C722 R.sup.D133 R.sup.D43
L.sub.C723 R.sup.D133 R.sup.D48 L.sub.C724 R.sup.D133 R.sup.D49 L.sub.C725 R.sup.D133
R.sup.D54 L.sub.C726 R.sup.D133 R.sup.D58 L.sub.C727 R.sup.D133 R.sup.D59 L.sub.C728
R.sup.D133 R.sup.D78 L.sub.C729 R.sup.D133 R.sup.D79 L.sub.C730 R.sup.D133 R.sup.D81
L.sub.C731 R.sup.D133 R.sup.D87 L.sub.C732 R.sup.D133 R.sup.D88 L.sub.C733 R.sup.D133
R.sup.D89 L.sub.C734 R.sup.D133 R.sup.D93 L.sub.C735 R.sup.D133 R.sup.D117 L.sub.C736
R.sup.D133 R.sup.D118 L.sub.C737 R.sup.D133 R.sup.D119 L.sub.C738 R.sup.D133 R.sup.D120
L.sub.C739 R.sup.D133 R.sup.D133 L.sub.C740 R.sup.D133 R.sup.D134 L.sub.C741 R.sup.D133
R.sup.D135 L.sub.C742 R.sup.D133 R.sup.D136 L.sub.C743 R.sup.D133 R.sup.D146 L.sub.C744
R.sup.D133 R.sup.D147 L.sub.C745 R.sup.D133 R.sup.D149 L.sub.C746 R.sup.D133 R.sup.D151
L.sub.C747 R.sup.D133 R.sup.D154 L.sub.C748 R.sup.D133 R.sup.D155 L.sub.C749 R.sup.D133
R.sup.D161 L.sub.C750 R.sup.D133 R.sup.D175 L.sub.C751 R.sup.D175 R.sup.D3 L.sub.C752
R.sup.D175 R.sup.D5 L.sub.C753 R.sup.D175 R.sup.D18 L.sub.C754 R.sup.D175 R.sup.D20
L.sub.C755 R.sup.D175 R.sup.D22 L.sub.C756 R.sup.D175 R.sup.D37 L.sub.C757 R.sup.D175
R.sup.D40 L.sub.C758 R.sup.D175 R.sup.D41 L.sub.C759 R.sup.D175 R.sup.D42 L.sub.C760
R.sup.D175 R.sup.D43 L.sub.C761 R.sup.D175 R.sup.D48 L.sub.C762 R.sup.D175 R.sup.D49
L.sub.C763 R.sup.D175 R.sup.D54 L.sub.C764 R.sup.D175 R.sup.D58 L.sub.C765 R.sup.D175
R.sup.D59 L.sub.C766 R.sup.D175 R.sup.D78 L.sub.C767 R.sup.D175 R.sup.D79 L.sub.C768

R.sup.D175 R.sup.D181 L.sub.C769 R.sup.D193 R.sup.D193 L.sub.C770 R.sup.D194 R.sup.D194
L.sub.C771 R.sup.D195 R.sup.D195 L.sub.C772 R.sup.D196 R.sup.D196 L.sub.C773 R.sup.D197
R.sup.D197 L.sub.C774 R.sup.D198 R.sup.D198 L.sub.C775 R.sup.D199 R.sup.D199 L.sub.C776
R.sup.D200 R.sup.D200 L.sub.C777 R.sup.D201 R.sup.D201 L.sub.C778 R.sup.D202 R.sup.D202
L.sub.C779 R.sup.D203 R.sup.D203 L.sub.C780 R.sup.D204 R.sup.D204 L.sub.C781 R.sup.D205
R.sup.D205 L.sub.C782 R.sup.D206 R.sup.D206 L.sub.C783 R.sup.D207 R.sup.D207 L.sub.C784
R.sup.D208 R.sup.D208 L.sub.C785 R.sup.D209 R.sup.D209 L.sub.C786 R.sup.D210 R.sup.D210
L.sub.C787 R.sup.D211 R.sup.D211 L.sub.C788 R.sup.D212 R.sup.D212 L.sub.C789 R.sup.D213
R.sup.D213 L.sub.C790 R.sup.D214 R.sup.D214 L.sub.C791 R.sup.D215 R.sup.D215 L.sub.C792
R.sup.D216 R.sup.D216 L.sub.C793 R.sup.D217 R.sup.D217 L.sub.C794 R.sup.D218 R.sup.D218
L.sub.C795 R.sup.D219 R.sup.D219 L.sub.C796 R.sup.D220 R.sup.D220 L.sub.C797 R.sup.D221
R.sup.D221 L.sub.C798 R.sup.D222 R.sup.D222 L.sub.C799 R.sup.D223 R.sup.D223 L.sub.C800
R.sup.D224 R.sup.D224 L.sub.C801 R.sup.D225 R.sup.D225 L.sub.C802 R.sup.D226 R.sup.D226
L.sub.C803 R.sup.D227 R.sup.D227 L.sub.C804 R.sup.D228 R.sup.D228 L.sub.C805 R.sup.D229
R.sup.D229 L.sub.C806 R.sup.D230 R.sup.D230 L.sub.C807 R.sup.D231 R.sup.D231 L.sub.C808
R.sup.D232 R.sup.D232 L.sub.C809 R.sup.D233 R.sup.D233 L.sub.C810 R.sup.D234 R.sup.D234
L.sub.C811 R.sup.D235 R.sup.D235 L.sub.C812 R.sup.D236 R.sup.D236 L.sub.C813 R.sup.D237
R.sup.D237 L.sub.C814 R.sup.D238 R.sup.D238 L.sub.C815 R.sup.D239 R.sup.D239 L.sub.C816
R.sup.D240 R.sup.D240 L.sub.C817 R.sup.D241 R.sup.D241 L.sub.C818 R.sup.D242 R.sup.D242
L.sub.C819 R.sup.D243 R.sup.D243 L.sub.C820 R.sup.D244 R.sup.D244 L.sub.C821 R.sup.D245
R.sup.D245 L.sub.C822 R.sup.D246 R.sup.D246 L.sub.C823 R.sup.D17 R.sup.D193 L.sub.C824
R.sup.D17 R.sup.D194 L.sub.C825 R.sup.D17 R.sup.D195 L.sub.C826 R.sup.D17 R.sup.D196
L.sub.C827 R.sup.D17 R.sup.D197 L.sub.C828 R.sup.D17 R.sup.D198 L.sub.C829 R.sup.D17
R.sup.D199 L.sub.C830 R.sup.D17 R.sup.D200 L.sub.C831 R.sup.D17 R.sup.D201 L.sub.C832
R.sup.D17 R.sup.D202 L.sub.C833 R.sup.D17 R.sup.D203 L.sub.C834 R.sup.D17 R.sup.D204
L.sub.C835 R.sup.D17 R.sup.D205 L.sub.C836 R.sup.D17 R.sup.D206 L.sub.C837 R.sup.D17
R.sup.D207 L.sub.C838 R.sup.D17 R.sup.D208 L.sub.C839 R.sup.D17 R.sup.D209 L.sub.C840
R.sup.D17 R.sup.D210 L.sub.C841 R.sup.D17 R.sup.D211 L.sub.C842 R.sup.D17 R.sup.D212
L.sub.C843 R.sup.D17 R.sup.D213 L.sub.C844 R.sup.D17 R.sup.D214 L.sub.C845 R.sup.D17
R.sup.D215 L.sub.C846 R.sup.D17 R.sup.D216 L.sub.C847 R.sup.D17 R.sup.D217 L.sub.C848
R.sup.D17 R.sup.D218 L.sub.C849 R.sup.D17 R.sup.D219 L.sub.C850 R.sup.D17 R.sup.D220
L.sub.C851 R.sup.D17 R.sup.D221 L.sub.C852 R.sup.D17 R.sup.D222 L.sub.C853 R.sup.D17
R.sup.D223 L.sub.C854 R.sup.D17 R.sup.D224 L.sub.C855 R.sup.D17 R.sup.D225 L.sub.C856
R.sup.D17 R.sup.D226 L.sub.C857 R.sup.D17 R.sup.D227 L.sub.C858 R.sup.D17 R.sup.D228
L.sub.C859 R.sup.D17 R.sup.D229 L.sub.C860 R.sup.D17 R.sup.D230 L.sub.C861 R.sup.D17
R.sup.D231 L.sub.C862 R.sup.D17 R.sup.D232 L.sub.C863 R.sup.D17 R.sup.D233 L.sub.C864
R.sup.D17 R.sup.D234 L.sub.C865 R.sup.D17 R.sup.D235 L.sub.C866 R.sup.D17 R.sup.D236
L.sub.C867 R.sup.D17 R.sup.D237 L.sub.C868 R.sup.D17 R.sup.D238 L.sub.C869 R.sup.D17
R.sup.D239 L.sub.C870 R.sup.D17 R.sup.D240 L.sub.C871 R.sup.D17 R.sup.D241 L.sub.C872
R.sup.D17 R.sup.D242 L.sub.C873 R.sup.D17 R.sup.D243 L.sub.C874 R.sup.D17 R.sup.D244
L.sub.C875 R.sup.D17 R.sup.D245 L.sub.C876 R.sup.D17 R.sup.D246 L.sub.C877 R.sup.D1
R.sup.D193 L.sub.C878 R.sup.D1 R.sup.D194 L.sub.C879 R.sup.D1 R.sup.D195 L.sub.C880
R.sup.D1 R.sup.D196 L.sub.C881 R.sup.D1 R.sup.D197 L.sub.C882 R.sup.D1 R.sup.D198
L.sub.C883 R.sup.D1 R.sup.D199 L.sub.C884 R.sup.D1 R.sup.D200 L.sub.C885 R.sup.D1
R.sup.D201 L.sub.C886 R.sup.D1 R.sup.D202 L.sub.C887 R.sup.D1 R.sup.D203 L.sub.C888
R.sup.D1 R.sup.D204 L.sub.C889 R.sup.D1 R.sup.D205 L.sub.C890 R.sup.D1 R.sup.D206
L.sub.C891 R.sup.D1 R.sup.D207 L.sub.C892 R.sup.D1 R.sup.D208 L.sub.C893 R.sup.D1
R.sup.D209 L.sub.C894 R.sup.D1 R.sup.D210 L.sub.C895 R.sup.D1 R.sup.D211 L.sub.C896
R.sup.D1 R.sup.D212 L.sub.C897 R.sup.D1 R.sup.D213 L.sub.C898 R.sup.D1 R.sup.D214
L.sub.C899 R.sup.D1 R.sup.D215 L.sub.C900 R.sup.D1 R.sup.D216 L.sub.C901 R.sup.D1

R.sup.D217 L.sub.C902 R.sup.D1 R.sup.D218 L.sub.C903 R.sup.D1 R.sup.D219 L.sub.C904
R.sup.D1 R.sup.D220 L.sub.C905 R.sup.D1 R.sup.D221 L.sub.C906 R.sup.D1 R.sup.D222
L.sub.C907 R.sup.D1 R.sup.D223 L.sub.C908 R.sup.D1 R.sup.D224 L.sub.C909 R.sup.D1
R.sup.D225 L.sub.C910 R.sup.D1 R.sup.D226 L.sub.C911 R.sup.D1 R.sup.D227 L.sub.C912
R.sup.D1 R.sup.D228 L.sub.C913 R.sup.D1 R.sup.D229 L.sub.C914 R.sup.D1 R.sup.D230
L.sub.C915 R.sup.D1 R.sup.D231 L.sub.C916 R.sup.D1 R.sup.D232 L.sub.C917 R.sup.D1
R.sup.D233 L.sub.C918 R.sup.D1 R.sup.D234 L.sub.C919 R.sup.D1 R.sup.D235 L.sub.C920
R.sup.D1 R.sup.D236 L.sub.C921 R.sup.D1 R.sup.D237 L.sub.C922 R.sup.D1 R.sup.D238
L.sub.C923 R.sup.D1 R.sup.D239 L.sub.C924 R.sup.D1 R.sup.D240 L.sub.C925 R.sup.D1
R.sup.D241 L.sub.C926 R.sup.D1 R.sup.D242 L.sub.C927 R.sup.D1 R.sup.D243 L.sub.C928
R.sup.D1 R.sup.D244 L.sub.C929 R.sup.D1 R.sup.D245 L.sub.C930 R.sup.D1 R.sup.D246
L.sub.C931 R.sup.D50 R.sup.D193 L.sub.C932 R.sup.D50 R.sup.D194 L.sub.C933 R.sup.D50
R.sup.D195 L.sub.C934 R.sup.D50 R.sup.D196 L.sub.C935 R.sup.D50 R.sup.D197 L.sub.C936
R.sup.D50 R.sup.D198 L.sub.C937 R.sup.D50 R.sup.D199 L.sub.C938 R.sup.D50 R.sup.D200
L.sub.C939 R.sup.D50 R.sup.D201 L.sub.C940 R.sup.D50 R.sup.D202 L.sub.C941 R.sup.D50
R.sup.D203 L.sub.C942 R.sup.D50 R.sup.D204 L.sub.C943 R.sup.D50 R.sup.D205 L.sub.C944
R.sup.D50 R.sup.D206 L.sub.C945 R.sup.D50 R.sup.D207 L.sub.C946 R.sup.D50 R.sup.D208
L.sub.C947 R.sup.D50 R.sup.D209 L.sub.C948 R.sup.D50 R.sup.D210 L.sub.C949 R.sup.D50
R.sup.D211 L.sub.C950 R.sup.D50 R.sup.D212 L.sub.C951 R.sup.D50 R.sup.D213 L.sub.C952
R.sup.D50 R.sup.D214 L.sub.C953 R.sup.D50 R.sup.D215 L.sub.C954 R.sup.D50 R.sup.D216
L.sub.C955 R.sup.D50 R.sup.D217 L.sub.C956 R.sup.D50 R.sup.D218 L.sub.C957 R.sup.D50
R.sup.D219 L.sub.C958 R.sup.D50 R.sup.D220 L.sub.C959 R.sup.D50 R.sup.D221 L.sub.C960
R.sup.D50 R.sup.D222 L.sub.C961 R.sup.D50 R.sup.D223 L.sub.C962 R.sup.D50 R.sup.D224
L.sub.C963 R.sup.D50 R.sup.D225 L.sub.C964 R.sup.D50 R.sup.D226 L.sub.C965 R.sup.D50
R.sup.D227 L.sub.C966 R.sup.D50 R.sup.D228 L.sub.C967 R.sup.D50 R.sup.D229 L.sub.C968
R.sup.D50 R.sup.D230 L.sub.C969 R.sup.D50 R.sup.D231 L.sub.C970 R.sup.D50 R.sup.D232
L.sub.C971 R.sup.D50 R.sup.D233 L.sub.C972 R.sup.D50 R.sup.D234 L.sub.C973 R.sup.D50
R.sup.D235 L.sub.C974 R.sup.D50 R.sup.D236 L.sub.C975 R.sup.D50 R.sup.D237 L.sub.C976
R.sup.D50 R.sup.D238 L.sub.C977 R.sup.D50 R.sup.D239 L.sub.C978 R.sup.D50 R.sup.D240
L.sub.C979 R.sup.D50 R.sup.D241 L.sub.C980 R.sup.D50 R.sup.D242 L.sub.C981 R.sup.D50
R.sup.D243 L.sub.C982 R.sup.D50 R.sup.D244 L.sub.C983 R.sup.D50 R.sup.D245 L.sub.C984
R.sup.D50 R.sup.D246 L.sub.C985 R.sup.D4 R.sup.D193 L.sub.C986 R.sup.D4 R.sup.D194
L.sub.C987 R.sup.D4 R.sup.D195 L.sub.C988 R.sup.D4 R.sup.D196 L.sub.C989 R.sup.D4
R.sup.D197 L.sub.C990 R.sup.D4 R.sup.D198 L.sub.C991 R.sup.D4 R.sup.D199 L.sub.C992
R.sup.D4 R.sup.D200 L.sub.C993 R.sup.D4 R.sup.D201 L.sub.C994 R.sup.D4 R.sup.D202
L.sub.C995 R.sup.D4 R.sup.D203 L.sub.C996 R.sup.D4 R.sup.D204 L.sub.C997 R.sup.D4
R.sup.D205 L.sub.C998 R.sup.D4 R.sup.D206 L.sub.C999 R.sup.D4 R.sup.D207 L.sub.C1000
R.sup.D4 R.sup.D208 L.sub.C1001 R.sup.D4 R.sup.D209 L.sub.C1002 R.sup.D4 R.sup.D210
L.sub.C1003 R.sup.D4 R.sup.D211 L.sub.C1004 R.sup.D4 R.sup.D212 L.sub.C1005 R.sup.D4
R.sup.D213 L.sub.C1006 R.sup.D4 R.sup.D214 L.sub.C1007 R.sup.D4 R.sup.D215 L.sub.C1008
R.sup.D4 R.sup.D216 L.sub.C1009 R.sup.D4 R.sup.D217 L.sub.C1010 R.sup.D4 R.sup.D218
L.sub.C1011 R.sup.D4 R.sup.D219 L.sub.C1012 R.sup.D4 R.sup.D220 L.sub.C1013 R.sup.D4
R.sup.D221 L.sub.C1014 R.sup.D4 R.sup.D222 L.sub.C1015 R.sup.D4 R.sup.D223 L.sub.C1016
R.sup.D4 R.sup.D224 L.sub.C1017 R.sup.D4 R.sup.D225 L.sub.C1018 R.sup.D4 R.sup.D226
L.sub.C1019 R.sup.D4 R.sup.D227 L.sub.C1020 R.sup.D4 R.sup.D228 L.sub.C1021 R.sup.D4
R.sup.D229 L.sub.C1022 R.sup.D4 R.sup.D230 L.sub.C1023 R.sup.D4 R.sup.D231 L.sub.C1024
R.sup.D4 R.sup.D232 L.sub.C1025 R.sup.D4 R.sup.D233 L.sub.C1026 R.sup.D4 R.sup.D234
L.sub.C1027 R.sup.D4 R.sup.D235 L.sub.C1028 R.sup.D4 R.sup.D236 L.sub.C1029 R.sup.D4
R.sup.D237 L.sub.C1030 R.sup.D4 R.sup.D238 L.sub.C1031 R.sup.D4 R.sup.D239 L.sub.C1032
R.sup.D4 R.sup.D240 L.sub.C1033 R.sup.D4 R.sup.D241 L.sub.C1034 R.sup.D4 R.sup.D242

[illegible]

L.sub.C1159 R.sup.D168 R.sup.D205 L.sub.C1160 R.sup.D168 R.sup.D206 L.sub.C1161 R.sup.D168 R.sup.D207 L.sub.C1162 R.sup.D168 R.sup.D208 L.sub.C1163 R.sup.D168 R.sup.D209 L.sub.C1164 R.sup.D168 R.sup.D210 L.sub.C1165 R.sup.D168 R.sup.D211 L.sub.C1166 R.sup.D168 R.sup.D212 L.sub.C1167 R.sup.D168 R.sup.D213 L.sub.C1168 R.sup.D168 R.sup.D214 L.sub.C1169 R.sup.D168 R.sup.D215 L.sub.C1170 R.sup.D168 R.sup.D216 L.sub.C1171 R.sup.D168 R.sup.D217 L.sub.C1172 R.sup.D168 R.sup.D218 L.sub.C1173 R.sup.D168 R.sup.D219 L.sub.C1174 R.sup.D168 R.sup.D220 L.sub.C1175 R.sup.D168 R.sup.D221 L.sub.C1176 R.sup.D168 R.sup.D222 L.sub.C1177 R.sup.D168 R.sup.D223 L.sub.C1178 R.sup.D168 R.sup.D224 L.sub.C1179 R.sup.D168 R.sup.D225 L.sub.C1180 R.sup.D168 R.sup.D226 L.sub.C1181 R.sup.D168 R.sup.D227 L.sub.C1182 R.sup.D168 R.sup.D228 L.sub.C1183 R.sup.D168 R.sup.D229 L.sub.C1184 R.sup.D168 R.sup.D230 L.sub.C1185 R.sup.D168 R.sup.D231 L.sub.C1186 R.sup.D168 R.sup.D232 L.sub.C1187 R.sup.D168 R.sup.D233 L.sub.C1188 R.sup.D168 R.sup.D234 L.sub.C1189 R.sup.D168 R.sup.D235 L.sub.C1190 R.sup.D168 R.sup.D236 L.sub.C1191 R.sup.D168 R.sup.D237 L.sub.C1192 R.sup.D168 R.sup.D238 L.sub.C1193 R.sup.D168 R.sup.D239 L.sub.C1194 R.sup.D168 R.sup.D240 L.sub.C1195 R.sup.D168 R.sup.D241 L.sub.C1196 R.sup.D168 R.sup.D242 L.sub.C1197 R.sup.D168 R.sup.D243 L.sub.C1198 R.sup.D168 R.sup.D244 L.sub.C1199 R.sup.D168 R.sup.D245 L.sub.C1200 R.sup.D168 R.sup.D246 L.sub.C1201 R.sup.D10 R.sup.D193 L.sub.C1202 R.sup.D10 R.sup.D194 L.sub.C1203 R.sup.D10 R.sup.D195 L.sub.C1204 R.sup.D10 R.sup.D196 L.sub.C1205 R.sup.D10 R.sup.D197 L.sub.C1206 R.sup.D10 R.sup.D198 L.sub.C1207 R.sup.D10 R.sup.D199 L.sub.C1208 R.sup.D10 R.sup.D200 L.sub.C1209 R.sup.D10 R.sup.D201 L.sub.C1210 R.sup.D10 R.sup.D202 L.sub.C1211 R.sup.D10 R.sup.D203 L.sub.C1212 R.sup.D10 R.sup.D204 L.sub.C1213 R.sup.D10 R.sup.D205 L.sub.C1214 R.sup.D10 R.sup.D206 L.sub.C1215 R.sup.D10 R.sup.D207 L.sub.C1216 R.sup.D10 R.sup.D208 L.sub.C1217 R.sup.D10 R.sup.D209 L.sub.C1218 R.sup.D10 R.sup.D210 L.sub.C1219 R.sup.D10 R.sup.D211 L.sub.C1220 R.sup.D10 R.sup.D212 L.sub.C1221 R.sup.D10 R.sup.D213 L.sub.C1222 R.sup.D10 R.sup.D214 L.sub.C1223 R.sup.D10 R.sup.D215 L.sub.C1224 R.sup.D10 R.sup.D216 L.sub.C1225 R.sup.D10 R.sup.D217 L.sub.C1226 R.sup.D10 R.sup.D218 L.sub.C1227 R.sup.D10 R.sup.D219 L.sub.C1228 R.sup.D10 R.sup.D220 L.sub.C1229 R.sup.D10 R.sup.D221 L.sub.C1230 R.sup.D10 R.sup.D222 L.sub.C1231 R.sup.D10 R.sup.D223 L.sub.C1232 R.sup.D10 R.sup.D224 L.sub.C1233 R.sup.D10 R.sup.D225 L.sub.C1234 R.sup.D10 R.sup.D226 L.sub.C1235 R.sup.D10 R.sup.D227 L.sub.C1236 R.sup.D10 R.sup.D228 L.sub.C1237 R.sup.D10 R.sup.D229 L.sub.C1238 R.sup.D10 R.sup.D230 L.sub.C1239 R.sup.D10 R.sup.D231 L.sub.C1240 R.sup.D10 R.sup.D232 L.sub.C1241 R.sup.D10 R.sup.D233 L.sub.C1242 R.sup.D10 R.sup.D234 L.sub.C1243 R.sup.D10 R.sup.D235 L.sub.C1244 R.sup.D10 R.sup.D236 L.sub.C1245 R.sup.D10 R.sup.D237 L.sub.C1246 R.sup.D10 R.sup.D238 L.sub.C1247 R.sup.D10 R.sup.D239 L.sub.C1248 R.sup.D10 R.sup.D240 L.sub.C1249 R.sup.D10 R.sup.D241 L.sub.C1250 R.sup.D10 R.sup.D242 L.sub.C1251 R.sup.D10 R.sup.D243 L.sub.C1252 R.sup.D10 R.sup.D244 L.sub.C1253 R.sup.D10 R.sup.D245 L.sub.C1254 R.sup.D10 R.sup.D246 L.sub.C1255 R.sup.D55 R.sup.D193 L.sub.C1256 R.sup.D55 R.sup.D194 L.sub.C1257 R.sup.D55 R.sup.D195 L.sub.C1258 R.sup.D55 R.sup.D196 L.sub.C1259 R.sup.D55 R.sup.D197 L.sub.C1260 R.sup.D55 R.sup.D198 L.sub.C1261 R.sup.D55 R.sup.D199 L.sub.C1262 R.sup.D55 R.sup.D200 L.sub.C1263 R.sup.D55 R.sup.D201 L.sub.C1264 R.sup.D55 R.sup.D202 L.sub.C1265 R.sup.D55 R.sup.D203 L.sub.C1266 R.sup.D55 R.sup.D204 L.sub.C1267 R.sup.D55 R.sup.D205 L.sub.C1268 R.sup.D55 R.sup.D206 L.sub.C1269 R.sup.D55 R.sup.D207 L.sub.C1270 R.sup.D55 R.sup.D208 L.sub.C1271 R.sup.D55 R.sup.D209 L.sub.C1272 R.sup.D55 R.sup.D210 L.sub.C1273 R.sup.D55 R.sup.D211 L.sub.C1274 R.sup.D55 R.sup.D212 L.sub.C1275 R.sup.D55 R.sup.D213 L.sub.C1276 R.sup.D55 R.sup.D214 L.sub.C1277 R.sup.D55 R.sup.D215 L.sub.C1278 R.sup.D55 R.sup.D216 L.sub.C1279 R.sup.D55 R.sup.D217 L.sub.C1280 R.sup.D55 R.sup.D218

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L.sub.C1403 R.sup.D143 R.sup.D233 L.sub.C1404 R.sup.D143 R.sup.D234 L.sub.C1405 R.sup.D143 R.sup.D235 L.sub.C1406 R.sup.D143 R.sup.D236 L.sub.C1407 R.sup.D143 R.sup.D237 L.sub.C1408 R.sup.D143 R.sup.D238 L.sub.C1409 R.sup.D143 R.sup.D239 L.sub.C1410 R.sup.D143 R.sup.D240 L.sub.C1411 R.sup.D143 R.sup.D241 L.sub.C1412 R.sup.D143 R.sup.D242 L.sub.C1413 R.sup.D143 R.sup.D243 L.sub.C1414 R.sup.D143 R.sup.D244 L.sub.C1415 R.sup.D143 R.sup.D245 L.sub.C1416 R.sup.D143 R.sup.D246 wherein R.sup.D1 to R.sup.D246 have the structures defined as follows:

##STR00694## ##STR00695## ##STR00696## ##STR00697## ##STR00698## ##STR00699##
##STR00700## ##STR00701## ##STR00702## ##STR00703## ##STR00704## ##STR00705##
##STR00706## ##STR00707## ##STR00708## ##STR00709## ##STR00710## ##STR00711##
##STR00712## ##STR00713## ##STR00714## ##STR00715## ##STR00716## [0280] 15. The compound of claim 1, wherein the compound is selected from the group consisting of:

##STR00717## ##STR00718## ##STR00719## ##STR00720## ##STR00721## ##STR00722##
##STR00723## ##STR00724## ##STR00725## [0281] 16. The compound of claim 12, wherein the compound comprises a structure of Formula III:

##STR00726## wherein: [0282] M.sup.1 is Pd or Pt; [0283] moieties E and F are each independently monocyclic or polycyclic ring structure, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring; [0284] Z.sup.3 and Z.sup.4 are each independently C or N; [0285] K.sup.1, K.sup.2, K.sup.3, and K.sup.4 are each independently selected from the group consisting of a direct bond, O, and S, wherein at least two of them are direct bonds; [0286] L.sup.1, L.sup.2, and L.sup.3 are each independently absent or selected from the group consisting of a direct bond, BR, BRR', NR, PR.sub.eP(O)R, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO.sub.2, CR, CRR', SiRR', GeRR', alkylene, cycloalkyl, aryl, cycloalkylene, arylene, heteroarylene, and combinations thereof, wherein at least one of L.sup.1 and L.sup.2 is present; [0287] R.sup.E and R.sup.F each independently represent zero, mono, or up to a maximum allowed number of substitutions; [0288] each of R, R', R.sup.E, and R.sup.F is independently a hydrogen, or a substituent selected from the group consisting of deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acid, ester, nitrile, isonitrile, sulfanyl, selenyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and [0289] two adjacent R.sup.A, R.sup.B, R.sup.C, R.sup.E, and RF can be joined or fused together to form a ring. [0290] 17. An organic light emitting device (OLED) comprising: [0291] an anode; [0292] a cathode; and [0293] an organic layer disposed between the anode and the cathode, wherein the organic layer comprises a compound according to claim 1.

[0294] 18. The OLED of claim 17, wherein the host is selected from the group consisting of:

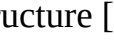
##STR00727## ##STR00728## ##STR00729## ##STR00730## ##STR00731## ##STR00732##
##STR00733## ##STR00734## ##STR00735## ##STR00736## ##STR00737## ##STR00738##
##STR00739## ##STR00740## ##STR00741## ##STR00742## ##STR00743##
##STR00744##

wherein: [0295] each of J.sub.1 to J.sub.6 is independently C or N; [0296] L' is a direct bond or an organic linker; [0297] each Y.sup.AA, Y.sup.BB, Y.sup.CC, and Y.sup.DD is independently selected from the group consisting of absent a bond, direct bond, O, S, Se, CRR', SiRR', GeRR', NR, BR, BRR'; [0298] each of R.sup.A1, R.sup.B', R.sup.C, R.sup.D', R.sup.E', R.sup.F', and R.sup.G' independently represents mono, up to the maximum substitutions, or no substitutions; [0299] each R, R', R.sup.A1, R.sup.B', R.sup.C', R.sup.D', R.sup.E', R.sup.F', and R.sup.G' is independently a hydrogen or a substituent selected from the group consisting of the General Substituents as defined herein; any two substituents can be joined or fused to form a ring; and where possible, each unsubstituted aromatic carbon atom can be replaced with N to form an aza-substituted ring. [0300] 19. The OLED of claim 17, wherein the compound is a sensitizer, and the OLED further comprises an acceptor selected from the group consisting of a fluorescent emitter, a delayed fluorescence

emitter, and combination thereof. [0301] 20. A consumer product comprising an organic light-emitting device comprising: [0302] an anode; [0303] a cathode; and [0304] an organic layer disposed between the anode and the cathode, wherein the organic layer comprises a compound according to claim 1.

[0305] In some embodiments of Formula III, the compound is selected from the group consisting of the compounds having the formula of $\text{Pt}(\text{L.sub.A}')(\text{Ly})$:

##STR00745## [0306] wherein L.sub.A' is selected from the group consisting of the structures defined in LIST 20 or LIST 21, wherein for LIST 20 each of Ri, Rj, Rk, and Rl is independently selected from R1 to R468, and Rq is selected from R469 to R774, wherein each of L.sub.A'1-(R1)(R1)(R469)(R1) to L.sub.A'42-(R468)(R774)(R468) is defined in the following LIST 20:

TABLE-US-00004 L.sub.A' Structure of L.sub.A' L.sub.A'1-(Ri)(Rj)(Rq)(Rl), wherein L.sub.A'1-(R1)(R1)(R469)(R1) to L.sub.A'1- (R468)(R468)(R774)(R468), having the structure [00746]  L.sub.A'2-(Ri)(Rj)(Rk)(Rq), wherein L.sub.A'2- (R1)(R1)(R1)(R469) to L.sub.A'2- (R468)(R468)(R468)(R774), having the structure [00747]  L.sub.A'3-(Ri)(Rj)(Rk)(Rq), wherein L.sub.A'3- (R1)(R1)(R1)(R469) to L.sub.A'3- (R468)(R468)(R468)(R774), having the structure [00748]  L.sub.A'4-(Ri)(Rj)(Rq)(Rl), wherein L.sub.A'4- (R1)(R1)(R469)(R1) to L.sub.A'4- (R468)(R468)(R774)(R468), having the structure [00749]  L.sub.A'5-(Ri)(Rj)(Rq)(Rl), wherein L.sub.A'5- (R1)(R1)(R469)(R1) to L.sub.A'5- (R468)(R468)(R774)(R468), having the structure [00750]  L.sub.A'6-(Ri)(Rj)(Rk)(Rq), wherein L.sub.A'6- (R1)(R1)(R1)(R469) to L.sub.A'6- (R468)(R468)(R468)(R774), having the structure [00751]  L.sub.A'7-(Ri)(Rq)(Rk)(Rl), wherein L.sub.A'7- (R1)(R469)(R469)(R1) to L.sub.A'7- (R468)(R774)(R468)(R468), having the structure [00752]  L.sub.A'8-(Ri)(Rj)(Rq)(Rl), wherein L.sub.A'8- (R1)(R1)(R469)(R1) to L.sub.A'8- (R468)(R468)(R774)(R468), having the structure [00753]  L.sub.A'9-(Rq)(Rj)(Rk)(Rl), wherein L.sub.A'9- (R469)(R1)(R1)(R1) to L.sub.A'9- (R774)(R468)(R468)(R468), having the structure [00754]  L.sub.A'10-(Rq)(Rj)(Rk)(Rl), wherein L.sub.A'10- (R469)(R1)(R1)(R1) to L.sub.A'10- (R774)(R468)(R468)(R468), having the structure [00755]  L.sub.A'11-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'11- (R469)(R1)(R1)(R1) to L.sub.A'11- (R774)(R468)(R468)(R468), having the structure [00756]  L.sub.A'12-(Rq)(Rj)(Rk)(Rl), wherein L.sub.A'12- (R469)(R1)(R1)(R1) to L.sub.A'12- (R774)(R468)(R468)(R468), having the structure [00757]  L.sub.A'13-(Rq)(Rj)(Rk)(Rl), wherein L.sub.A'13- (R469)(R1)(R1)(R1) to L.sub.A'13- (R774)(R468)(R468)(R468), having the structure [00758]  L.sub.A'14-(Ri)(Rj)(Rk)(Rq), wherein L.sub.A'14- (R1)(R1)(R1)(R469) to L.sub.A'14- (R468)(R468)(R468)(R774), having the structure [00759]  L.sub.A'15-(Ri)(Rq)(Rk)(Rl), wherein L.sub.A'15- (R1)(R469)(R1)(R1) to L.sub.A'15- (R468)(R774)(R468)(R468), having the structure [00760]  L.sub.A'16-(Rq)(Rj)(Rk)(Rl), wherein L.sub.A'16- (R469)(R1)(R1)(R1) to L.sub.A'16- (R774)(R468)(R468)(R468), having the structure [00761]  L.sub.A'17-(Rq)(Rj)(Rk)(Rl), wherein L.sub.A'17- (R469)(R1)(R1)(R1) to L.sub.A'17- (R774)(R468)(R468)(R468), having the structure [00762]  L.sub.A'18-(Ri)(Rj)(Rq)(Rl), wherein L.sub.A'18- (R1)(R1)(R469)(R1) to L.sub.A'18- (R468)(R468)(R774)(R468), having the structure [00763]  L.sub.A'19-(Ri)(Rq)(Rk)(Rl), wherein L.sub.A'19- (R1)(R469)(R1)(R1) to L.sub.A'19- (R468)(R774)(R468)(R468), having the structure [00764]  L.sub.A'20-(Ri)(Rj)(Rq)(Rl), wherein L.sub.A'20- (R1)(R1)(R469)(R1) to L.sub.A'20- (R468)(R774)(R468)(R468), having the structure [00765]  L.sub.A'21-(Ri)(Rq)(Rk)(Rl), wherein L.sub.A'21- (R1)(R469)(R1)(R1) to L.sub.A'21- (R468)(R774)(R468)(R468), having the structure [00766]  L.sub.A'22-(Ri)(Rj)(Rk)(Rq), wherein L.sub.A'22- (R1)(R1)(R469)(R1) to L.sub.A'22- (R468)(R468)(R468)(R774), having the structure [00767]  L.sub.A'23-(Ri)(Rq)(Rk)(Rl), wherein L.sub.A'23- (R1)(R469)(R1)(R1) to L.sub.A'23- (R468)(R774)(R468)(R468), having the structure [00768] 

L.sub.A'24-(Ri)(Rj)(Rq)(Rl), wherein L.sub.A'24- (R1)(R1)(R469)(R1) to L.sub.A'24- (R468)(R774)(R468), having the structure [00769]  embedded image L.sub.A'25-(Rq)(Rj)(Rk)(Rl), wherein L.sub.A'25- (R469)(R1)(R1)(R1) to L.sub.A'25- (R774)(R468)(R468)(R468), having the structure [00770]  embedded image L.sub.A'26-(Ri)(Rq)(Rk)(Rl), wherein L.sub.A'26- (R1)(R469)(R1)(R1) to L.sub.A'26- (R468)(R774)(R468)(R468), having the structure [00771]  embedded image L.sub.A'27-(Ri)(Rq)(Rk)(Rl), wherein L.sub.A'27- (R1)(R469)(R1)(R1) to L.sub.A'27- (R468)(R774)(R468)(R468), having the structure [00772]  embedded image L.sub.A'28-(Ri)(Rj)(Rq)(Rl), wherein L.sub.A'28- (R1)(R1)(R469)(R1) to L.sub.A'28- (R468)(R774)(R468), having the structure [00773]  embedded image L.sub.A'29-(Ri)(Rj)(Rq)(Rl), wherein L.sub.A'29- (R1)(R1)(R469)(R1) to L.sub.A'29- (R468)(R468)(R774)(R468), having the structure [00774]  embedded image L.sub.A'30-(Ri)(Rq)(Rk)(Rl), wherein L.sub.A'30- (R1)(R469)(R1)(R1) to L.sub.A'30- (R468)(R774)(R468)(R468), having the structure [00775]  embedded image L.sub.A'31-(Ri)(Rq)(Rk)(Rl), wherein L.sub.A'31- (R1)(R469)(R1)(R1) to L.sub.A'31- (R468)(R774)(R468)(R468), having the structure [00776]  embedded image L.sub.A'32-(Ri)(Rq)(Rk), wherein L.sub.A'32-(R1)(R469)(R1) to L.sub.A'32-(R468)(R774)(R468), having the structure [00777]  embedded image L.sub.A'33-(Ri)(Rj)(Rq), wherein L.sub.A'33-(R1)(R1)(R469) to L.sub.A'33-(R468)(R468)(R774), having the structure [00778]  embedded image L.sub.A'34-(Ri)(Rj)(Rq), wherein L.sub.A'34-(R1)(R1)(R469) to L.sub.A'34-(R468)(R468)(R774), having the structure [00779]  embedded image L.sub.A'35-(Ri)(Rq)(Rk), wherein L.sub.A'35-(R1)(R469)(R1) to L.sub.A'35-(R468)(R774)(R468), having the structure [00780]  embedded image L.sub.A'36-(Ri)(Rj)(Rq), wherein L.sub.A'36-(R1)(R1)(R469) to L.sub.A'36-(R468)(R468)(R774), having the structure [00781]  embedded image L.sub.A'37-(Ri)(Rj)(Rq), wherein L.sub.A'37-(R1)(R1)(R469) to L.sub.A'37-(R468)(R468)(R774), having the structure [00782]  embedded image L.sub.A'38-(Ri)(Rq)(Rk), wherein L.sub.A'38-(R1)(R469)(R1) to L.sub.A'38-(R468)(R774)(R468), having the structure [00783]  embedded image L.sub.A'39-(Ri)(Rq)(Rk), wherein L.sub.A'39-(R1)(R469)(R1) to L.sub.A'39-(R468)(R774)(R468), having the structure [00784]  embedded image L.sub.A'40-(Ri)(Rj)(Rq), wherein L.sub.A'40-(R1)(R1)(R469) to L.sub.A'40-(R468)(R468)(R774), having the structure [00785]  embedded image L.sub.A'41-(Ri)(Rq)(Rk), wherein L.sub.A'41-(R1)(R469)(R1) to L.sub.A'41-(R468)(R774)(R468), having the structure [00786]  embedded image L.sub.A'42-(Ri)(Rq)(Rk), wherein L.sub.A'42-(R1)(R469)(R1) to L.sub.A'42-(R468)(R774)(R468), having the structure [00787]  embedded image [0307] wherein for LIST 21 each of Ri, Rj, Rk, and Rl is independently selected from R1 to R468, and Rq is selected from R469 to R774, wherein each of L.sub.A'43-(R1)(R1)(R1) to L.sub.A'87-(R468)(R468)(F468)(R468) is defined in the following LIST 21:

TABLE-US-00005 L.sub.A' Structure of L.sub.A' L.sub.A'43-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'43- (R1)(R1)(R1)(R1) to L.sub.A'43- (R468)(R468)(R468)(R468), having the structure [00788]  embedded image L.sub.A'44-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'44- (R1)(R1)(R1)(R1) to L.sub.A'44- (R468)(R468)(R468)(R468), having the structure [00789]  embedded image L.sub.A'45-(Ri)(Rj)(Rk)(Rq), wherein L.sub.A'45- (R1)(R1)(R1)(R469) to L.sub.A'45- (R468)(R468)(R774), having the structure [00790]  embedded image L.sub.A'46-(Ri)(Rj)(Rq)(Rl), wherein L.sub.A'46- (R1)(R1)(R469)(R1) to L.sub.A'46- (R468)(R468)(R774)(R468), having the structure [00791]  embedded image L.sub.A'47-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'47- (R1)(R1)(R1) to L.sub.A'47- (R468)(R468)(R468)(R468), having the structure [00792]  embedded image L.sub.A'48-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'48- (R1)(R1)(R1)(R1) to L.sub.A'48- (R468)(R468)(R468)(R468), having the structure [00793]  embedded image L.sub.A'49-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'49- (R1)(R1)(R1)(R1) to L.sub.A'49- (R468)(R468)(R468)(R468), having the structure [00794] embedded image L.sub.A'50-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'50- (R1)(R1)(R1)(R1) to L.sub.A'50- (R468)(R468)(R468)(R468), having the structure [00795] embedded image L.sub.A'51-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'51- (R1)(R1)(R1)(R1) to L.sub.A'51- (R468)(R468)(R468)(R468), having the structure [00796]

[illegible]

L.sub.A'82-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'82- (R1)(R1)(R1)(R1) to L.sub.A'82- (R468)(R468)(R468)(R468), having the structure [00827]  L.sub.A'83-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'83- (R1)(R1)(R1)(R1) to L.sub.A'83- (R468)(R468)(R468)(R468), having the structure [00828]  L.sub.A'84-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'84- (R1)(R1)(R1)(R1) to L.sub.A'84- (R468)(R468)(R468)(R468), having the structure [00829]  L.sub.A'85-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'85- (R1)(R1)(R1)(R1) to L.sub.A'85- (R468)(R468)(R468)(R468), having the structure [00830]  L.sub.A'86-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'86- (R1)(R1)(R1)(R1) to L.sub.A'86- (R468)(R468)(R468)(R468), having the structure [00831]  L.sub.A'87-(Ri)(Rj)(Rk)(Rl), wherein L.sub.A'87- (R1)(R1)(R1)(R1) to L.sub.A'87- (R468)(R468)(R468)(R468), having the structure [00832]  L.sub.y' is selected from the group consisting of the structures defined in LIST 22 or LIST 23, wherein for LIST 22 each of Rs, Rt, and Ru is independently selected from R1 to R468, and Rq is selected from R469 to R774, wherein each of L.sub.y1-(R469)(R1)(R1) to L.sub.y47-(R774)(R468)(R468) is defined in the following LIST 22: TABLE-US-00006 L.sub.y Structure of L.sub.y L.sub.y1-(Rq)(Rt)(Ru), wherein L.sub.y1-(R469)(R1)(R1) to L.sub.y1- (R774)(R468)(R468), having the structure [00833]  L.sub.y2-(Rq)(Rt)(Ru), wherein L.sub.y2-(R469)(R1)(R1) to L.sub.y2- (R774)(R468)(R468), having the structure [00834]  L.sub.y3-(Rq)(Rt)(Ru), wherein L.sub.y3-(R469)(R1)(R1) to L.sub.y3- (R774)(R468)(R468), having the structure [00835]  L.sub.y4-(Rq)(Rt)(Ru), wherein L.sub.y4-(R469)(R1)(R1) to L.sub.y4- (R774)(R468)(R468), having the structure [00836]  L.sub.y5-(Rs)(Rq)(Ru), wherein L.sub.y5-(R1)(R469)(R1) to L.sub.y5- (R468)(R774)(R468), having the structure [00837]  L.sub.y6-(Rs)(Rq)(Ru), wherein L.sub.y6-(R1)(R469)(R1) to L.sub.y6- (R468)(R774)(R468), having the structure [00838]  L.sub.y7-(Rq)(Rt)(Ru), wherein L.sub.y7-(R469)(R1)(R1) to L.sub.y7- (R774)(R468)(R468), having the structure [00839]  L.sub.y8-(Rq)(Rt)(Ru), wherein L.sub.y8-(R469)(R1)(R1) to L.sub.y8- (R774)(R468)(R468), having the structure [00840]  L.sub.y9-(Rs)(Rt)(Rq), wherein L.sub.y9-(R1)(R1)(R469) to L.sub.y9- (R468)(R468)(R774), having the structure [00841]  L.sub.y10-(Rs)(Rt)(Rq), wherein L.sub.y10-(R1)(R1)(R469) to L.sub.y10-(R468)(R468)(R774), having the structure [00842]  L.sub.y11-(Rs)(Rt)(Rq), wherein L.sub.y11-(R1)(R1)(R469) to L.sub.y11-(R468)(R468)(R774), having the structure [00843]  L.sub.y12-(Rs)(Rt)(Rq), wherein L.sub.y12-(R1)(R1)(R469) to L.sub.y12-(R468)(R468)(R774), having the structure [00844]  L.sub.y13-(Rs)(Rt)(Rq), wherein L.sub.y13-(R1)(R1)(R469) to L.sub.y13-(R468)(R468)(R774), having the structure [00845]  L.sub.y14-(Rs)(Rt)(Rq), wherein L.sub.y14-(R1)(R1)(R469) to L.sub.y14-(R468)(R468)(R774), having the structure [00846]  L.sub.y15-(Rq)(Rt)(Ru), wherein L.sub.y15-(R469)(R1)(R1) to L.sub.y15-(R774)(R468)(R468), having the structure [00847]  L.sub.y16-(Rq)(Rt)(Ru), wherein L.sub.y16-(R469)(R1)(R1) to L.sub.y16- (R774)(R468)(R468), having the structure [00848]  L.sub.y17-(Rs)(Rq)(Ru), wherein L.sub.y17-(R1)(R469)(R1) to L.sub.y17-(R468)(R774)(R468), having the structure [00849]  L.sub.y18-(Rq)(Rt)(Ru), wherein L.sub.y18-(R469)(R1)(R1) to L.sub.y18-(R774)(R468)(R468), having the structure [00850]  L.sub.y19-(Rq)(Rt)(Ru), wherein L.sub.y19-(R469)(R1)(R1) to L.sub.y19-(R774)(R468)(R468), having the structure [00851]  L.sub.y20-(Rq)(Rt)(Ru), wherein L.sub.y20-(R469)(R1)(R1) to L.sub.y20-(R774)(R468)(R468), having the structure [00852]  L.sub.y21-(Rq)(Rt)(Ru), wherein L.sub.y21-(R469)(R1)(R1) to L.sub.y21-(R774)(R468)(R468), having the structure [00853]  L.sub.y22-(Rq)(Rt)(Ru), wherein L.sub.y22-(R469)(R1)(R1) to L.sub.y22-(R774)(R468)(R468), having the structure [00854]  L.sub.y23-(Rq)(Rt)(Ru), wherein L.sub.y23-(R469)(R1)(R1) to L.sub.y23-(R774)(R468)(R468), having the structure [00855]  L.sub.y24-(Rs)(Rt)(Rq), wherein L.sub.y24-(R1)(R1)(R469)

to L.sub.y24-(R468)(R468)(R774), having the structure [00856]  embedded image L.sub.y25-(Rs)(Rt)(Rq), wherein L.sub.y25-(R1)(R1)(R469) to L.sub.y25- (R468)(R468)(R774), having the structure [00857]  embedded image L.sub.y26-(Rs)(Rt)(Rq), wherein L.sub.y26-(R1)(R1)(R469) to L.sub.y26- (R468)(R468)(R774), having the structure [00858]  embedded image L.sub.y27-(Rs)(Rt)(Rq), wherein L.sub.y27-(R1)(R1)(R469) to L.sub.y27- (R468)(R468)(R774), having the structure [00859]  embedded image L.sub.y28-(Rs)(Rq)(Ru), wherein L.sub.y28-(R1)(R469)(R1) to L.sub.y28- (R468)(R774)(R468), having the structure [00860]  embedded image L.sub.y29-(Rq)(Rt)(Ru), wherein L.sub.y29-(R469)(R1)(R1) to L.sub.y29- (R774)(R468)(R468), having the structure [00861]  embedded image L.sub.y30-(Rs)(Rq)(Ru), wherein L.sub.y30-(R1)(R469)(R1) to L.sub.y30- (R468)(R774)(R468), having the structure [00862]  embedded image L.sub.y31-(Rs)(Rt)(Rq), wherein L.sub.y31-(R1)(R1)(R469) to L.sub.y31- (R468)(R468)(R774), having the structure [00863]  embedded image L.sub.y32-(Rq)(Rt)(Ru), wherein L.sub.y32-(R469)(R1)(R1) to L.sub.y32- (R774)(R468)(R468), having the structure [00864]  embedded image L.sub.y33-(Rs)(Rq)(Ru), wherein L.sub.y33-(R1)(R469)(R1) to L.sub.y33- (R468)(R774)(R468), having the structure [00865]  embedded image L.sub.y34-(Rs)(Rq)(Ru), wherein L.sub.y34-(R1)(R469)(R1) to L.sub.y34- (R468)(R774)(R468), having the structure [00866]  embedded image L.sub.y35-(Rs)(Rt)(Rq), wherein L.sub.y35-(R1)(R1)(R469) to L.sub.y35- (R468)(R468)(R774), having the structure [00867]  embedded image L.sub.y36-(Rs)(Rt)(Rq), wherein L.sub.y36-(R1)(R1)(R469) to L.sub.y36- (R468)(R468)(R774), having the structure [00868]  embedded image L.sub.y37-(Rs)(Rq)(Ru), wherein L.sub.y37-(R1)(R469)(R1) to L.sub.y37- (R468)(R774)(R468), having the structure [00869]  embedded image L.sub.y38-(Rs)(Rt)(Rq), wherein L.sub.y38-(R1)(R1)(R469) to L.sub.y38- (R468)(R468)(R774), having the structure [00870]  embedded image L.sub.y39-(Rs)(Rt)(Rq), wherein L.sub.y39-(R1)(R1)(R469) to L.sub.y39- (R468)(R468)(R774), having the structure [00871]  embedded image L.sub.y40-(Rs)(Rq)(Ru), wherein L.sub.y40-(R1)(R469)(R1) to L.sub.y40- (R468)(R774)(R468), having the structure [00872]  embedded image L.sub.y41-(Rs)(Rq)(Ru), wherein L.sub.y41-(R1)(R469)(R1) to L.sub.y41- (R468)(R774)(R468), having the structure [00873]  embedded image L.sub.y42-(Rs)(Rt)(Rq), wherein L.sub.y42-(R1)(R1)(R469) to L.sub.y42- (R468)(R468)(R774), having the structure [00874]  embedded image L.sub.y43-(Rs)(Rq)(Ru), wherein L.sub.y43-(R1)(R469)(R1) to L.sub.y43- (R468)(R774)(R468), having the structure [00875]  embedded image L.sub.y44-(Rs)(Rq)(Ru), wherein L.sub.y44-(R1)(R469)(R1) to L.sub.y44- (R468)(R774)(R468), having the structure [00876]  embedded image L.sub.y45-(Rq)(Rt)(Ru), wherein L.sub.y45-(R469)(R1)(R1) to L.sub.y45- (R774)(R468)(R468), having the structure [00877]  embedded image L.sub.y46-(Rq)(Rt)(Ru), wherein L.sub.y46-(R469)(R1)(R1) to L.sub.y46- (R774)(R468)(R468), having the structure [00878]  embedded image L.sub.y47-(Rq)(Rt)(Ru), wherein L.sub.y47-(R469)(R1)(R1) to L.sub.y47- (R774)(R468)(R468), having the structure [00879]  embedded image [0309] wherein for LIST 23 each of Rs, Rt, and Ru is independently selected from R1 to R468, wherein each of L.sub.y48-(R1)(R1)(R1) to L.sub.y94-(R468)(R468)(R468) is defined in the following LIST 23:

TABLE-US-00007 L.sub.y Structure of L.sub.y L.sub.y48-(Rs)(Rt)(Ru), wherein L.sub.y48-(R1)(R1)(R1) to L.sub.y48- (R468)(R468)(R468), having the structure [00880]  embedded image L.sub.y49-(Rs)(Rt)(Ru), wherein L.sub.y49-(R1)(R1)(R1) to L.sub.y49- (R468)(R468)(R468), having the structure [00881]  embedded image L.sub.y50-(Rs)(Rt)(Ru), wherein L.sub.y50-(R1)(R1)(R1) to L.sub.y50- (R468)(R468)(R468), having the structure [00882]  embedded image L.sub.y51-(Rs)(Rt)(Ru), wherein L.sub.y51-(R1)(R1)(R1) to L.sub.y51- (R468)(R468)(R468), having the structure [00883]  embedded image L.sub.y52-(Rs)(Rt)(Ru), wherein L.sub.y52-(R1)(R1)(R1) to L.sub.y52- (R468)(R468)(R468), having the structure [00884]  embedded image L.sub.y53-(Rs)(Rt)(Ru), wherein L.sub.y53-(R1)(R1)(R1) to L.sub.y53- (R468)(R468)(R468), having the structure [00885]  embedded image L.sub.y54-(Rs)(Rt)(Ru), wherein L.sub.y54-(R1)(R1)(R1) to L.sub.y54- (R468)(R468)(R468), having the structure [00886]  embedded image L.sub.y55-(Rs)(Rt)(Ru), wherein L.sub.y55-(R1)(R1)(R1) to L.sub.y55- (R468)(R468)(R468),

having the structure [00887]  L.sub.y56-(Rs)(Rt)(Ru), wherein L.sub.y56-(R1)(R1) to L.sub.y56- (R468)(R468)(R468), having the structure [00888]  L.sub.y57-(Rs)(Rt)(Ru), wherein L.sub.y57-(R1)(R1)(R1) to L.sub.y57- (R468)(R468)(R468), having the structure [00889]  L.sub.y58-(Rs)(Rt)(Ru), wherein L.sub.y58-(R1)(R1) to L.sub.y58- (R468)(R468)(R468), having the structure [00890]  L.sub.y59-(Rs)(Rt)(Ru), wherein L.sub.y59-(R1)(R1)(R1) to L.sub.y59- (R468)(R468)(R468), having the structure [00891]  L.sub.y60-(Rs)(Rt)(Ru), wherein L.sub.y60-(R1)(R1) to L.sub.y60- (R468)(R468) (R468), having the structure [00892]  L.sub.y61-(Rs)(Rt) Ru), wherein L.sub.y61-(R1)(R1)(R1) to L.sub.y61- (R468)(R468)(R468), having the structure [00893]  L.sub.y62-(Rs)(Rt)(Ru), wherein L.sub.y62-(R1)(R1) to L.sub.y62- (R468)(R468)(R468), having the structure [00894]  L.sub.y63-(Rs)(Rt)(Ru), wherein L.sub.y63-(R1)(R1)(R1) to L.sub.y63- (R468)(R468)(R468), having the structure [00895]  L.sub.y64-(Rs)(Rt)(Ru), wherein L.sub.y64-(R1)(R1) to L.sub.y64- (R468)(R468)(R468), having the structure [00896]  L.sub.y65-(Rs)(Rt)(Ru), wherein L.sub.y65-(R1)(R1)(R1) to L.sub.y65- (R468)(R468)(R468), having the structure [00897]  L.sub.y66-(Rs)(Rt)(Ru), wherein L.sub.y66-(R1)(R1) to L.sub.y66- (R468)(R468)(R468), having the structure [00898]  L.sub.y67-(Rs)(Rt)(Ru), wherein L.sub.y67-(R1)(R1)(R1) to L.sub.y67- (R468)(R468)(R468), having the structure [00899]  L.sub.y68-(Rs)(Rt)(Ru), wherein L.sub.y68-(R1)(R1) to L.sub.y68- (R468)(R468)(R468), having the structure [00900]  L.sub.y69-(Rs)(Rt)(Ru), wherein L.sub.y69-(R1)(R1)(R1) to L.sub.y69- (R468)(R468)(R468), having the structure [00901]  L.sub.y70-(Rs)(Rt)(Ru), wherein L.sub.y70-(R1)(R1) to L.sub.y70- (R468)(R468)(R468), having the structure [00902]  L.sub.y71-(Rs)(Rt)(Ru), wherein L.sub.y71-(R1)(R1)(R1) to L.sub.y71- (R468)(R468)(R468), having the structure [00903]  L.sub.y72-(Rs)(Rt)(Ru), wherein L.sub.y72-(R1)(R1) to L.sub.y72- (R468)(R468)(R468), having the structure [00904]  L.sub.y73-(Rs)(Rt)(Ru), wherein L.sub.y73-(R1)(R1)(R1) to L.sub.y73- (R468)(R468)(R468), having the structure [00905]  L.sub.y74-(Rs)(Rt)(Ru), wherein L.sub.y74-(R1)(R1) to L.sub.y74- (R468)(R468)(R468), having the structure [00906]  L.sub.y75-(Rs)(Rt)(Ru), wherein L.sub.y75-(R1)(R1)(R1) to L.sub.y75- (R468)(R468)(R468), having the structure [00907]  L.sub.y76-(Rs)(Rt)(Ru), wherein L.sub.y76-(R1)(R1) to L.sub.y76- (R468)(R468)(R468), having the structure [00908]  L.sub.y77-(Rs)(Rt)(Ru), wherein L.sub.y77-(R1)(R1)(R1) to L.sub.y77- (R468)(R468)(R468), having the structure [00909]  L.sub.y78-(Rs)(Rt)(Ru), wherein L.sub.y78-(R1)(R1) to L.sub.y78- (R468) R468)(R468), having the structure [00910]  L.sub.y79-(Rs)(Rt)(Ru), wherein L.sub.y79-(R1)(R1)(R1) to L.sub.y79- (R468)(R468)(R468), having the structure [00911]  L.sub.y80-(Rs)(Rt)(Ru), wherein L.sub.y80-(R1)(R1) to L.sub.y80- (R468)(R468)(R468), having the structure [00912]  L.sub.y81-(Rs)(Rt)(Ru), wherein L.sub.y81-(R1)(R1)(R1) to L.sub.y81- (R468)(R468)(R468), having the structure [00913]  L.sub.y82-(Rs)(Rt)(Ru), wherein L.sub.y82-(R1)(R1) to L.sub.y82- (R468)(R468)(R468), having the structure [00914]  L.sub.y83-(Rs)(Rt)(Ru), wherein L.sub.y83-(R1)(R1)(R1) to L.sub.y83- (R468)(R468)(R468), having the structure [00915]  L.sub.y84-(Rs)(Rt)(Ru), wherein L.sub.y84-(R1)(R1) to L.sub.y84- (R468)(R468)(R468), having the structure [00916]  L.sub.y85-(Rs)(Rt)(Ru), wherein L.sub.y85-(R1)(R1)(R1) to L.sub.y85- (R468)(R468)(R468), having the structure [00917]  L.sub.y86-(Rs)(Rt)(Ru), wherein L.sub.y86-(R1)(R1) to L.sub.y86- (R468)(R468)(R468), having the structure [00918]  L.sub.y87-(Rs)(Rt)(Ru), wherein L.sub.y87-(R1)(R1)(R1) to L.sub.y87- (R468)(R468)(R468), having the structure [00919]  L.sub.y88-(Rs)(Rt)(Ru), wherein L.sub.y88-(R1)(R1) to L.sub.y88- (R468)(R468)(R468), having the structure [00920] 

L.sub.y89-(Rs)(Rt)(Ru), wherein L.sub.y89-(R1)(R1) to L.sub.y89- (R468)(R468)(R468), having the structure [00921]  L.sub.y90-(Rs)(Rt)(Ru), wherein L.sub.y90-(R1)(R1) to L.sub.y90- (R468)(R468)(R468), having the structure [00922]  L.sub.y91-(Rs)(Rt)(Ru), wherein L.sub.y91-(R1)(R1)(R1) to L.sub.y91- (R468)(R468)(R468), having the structure [00923]  L.sub.y92-(Rs)(Rt)(Ru), wherein L.sub.y92-(R1)(R1) to L.sub.y92- (R468)(R468)(R468), having the structure [00924]  L.sub.y93-(Rs)(Rt)(Ru), wherein L.sub.y93-(R1)(R1)(R1) to L.sub.y93- (R468)(R468)(R468), having the structure [00925]  L.sub.y94-(Rs)(Rt)(Ru), wherein L.sub.y94-(R1)(R1) to L.sub.y94- (R468)(R468)(R468), having the structure [00926]  [0310] wherein, when L.sub.A' is selected from the structures of LIST 20, Ly is selected from the structures of LIST 22 or LIST 23; [0311] wherein, when L.sub.A' is selected from the structures of LIST 21, Ly is selected from the structures of LIST 22; [0312] wherein R1 to R468 have the structures defined in LIST 8 defined herein; and [0313] wherein R469 to R774 have the structures defined in LIST 8 defined herein.

[0314] In the above embodiments for combinations of LIST 20 with LIST 22 or LIST P23, Rq is selected from R469 to R774, and the rest of Ri, Rj, Rk, and Rl is each independently an integer from R1 to R774. However, it should be understood that the same is intended for each of Ri, Rj, Rk, and Ri. For example, only one of Ri, Rj, Rk, Ri, or Rq must be selected from R469 to R774, and the rest can each independently be selected from R1 to R774. In other embodiments, only one of Ri, Rj, Rk, Ri, or Rq can be selected from R469 to R774. In still other embodiments, two of Ri, Rj, Rk, Ri, or Rq can be independently selected from R469 to R774. As a result, all the above identified embodiments/combinations are intended to be specifically covered. In other words, in alternate embodiments, the structure of Formula I (e.g., R469 to R774) can be required in one of positions Ri, Rj, Rk, or Rl, while Rq can be selected from R1 to R468.

[0315] By the same token, even though Rq is selected from R469 to R774 for LIST 22, the same is intended for each of Rs, Rt, and Ru. For example, only one of Rs, Rt, Ru, or Rq must be selected from R469 to R774, and the rest can each independently be selected from R1 to R774. In other embodiments, only one of Rs, Rt, Ru, or Rq can be selected from R469 to R774. In still other embodiments, two of Rs, Rt, Ru, or Rq can be independently selected from R469 to R774. As a result, all the above identified embodiments/combinations are intended to be specifically covered. In other words, in alternate embodiments, the structure of Formula I (e.g., R469 to R774) can be required in one of positions Rs, Rt, or Ru, while Rq can be selected from R1 to R468.

[0316] In some embodiments, the compound is selected from the group consisting of the structures of the following LIST 24:

##STR00927## ##STR00928## ##STR00929## ##STR00930## ##STR00931##

[0317] In some embodiments, the compound having a first ligand L.sub.A of Formula I described herein can be at least 30% deuterated, at least 40% deuterated, at least 50% deuterated, at least 60% deuterated, at least 70% deuterated, at least 80% deuterated, at least 90% deuterated, at least 95% deuterated, at least 99% deuterated, or 100% deuterated. As used herein, percent deuteration has its ordinary meaning and includes the percent of all possible hydrogen atoms in the compound (e.g., positions that are hydrogen or deuterium) that are occupied by deuterium atoms. In some embodiments, carbon atoms comprised the ring coordinated to the metal M are fully or partially deuterated. In some embodiments, carbon atoms comprised by a polycyclic ring system coordinated to the metal M are fully or partially deuterated. In some embodiments, a substituent attached to a monocyclic or fused polycyclic ring system coordinated to the metal M is fully or partially deuterated.

[0318] In some embodiments, the compound of formula I has an emission at room temperature with a full width at half maximum (FWHM) of equal to or less than 50, 45, 40, 35, 30, 25, 20, 15, 10, or 5 nm. Narrower FWHM means better color purity for the OLED display application.

[0319] In some embodiments of heteroleptic compound having the formula of

M(L.sub.A).sub.p(L.sub.B).sub.q(L.sub.C).sub.r as defined above, the ligand L.sub.A has a first substituent R.sup.I, where the first substituent R.sup.I has a first atom a-I that is the farthest away from the metal M among all atoms in the ligand L.sub.A. Additionally, the ligand L.sub.B, if present, has a second substituent R.sup.II, where the second substituent R.sup.II has a first atom a-II that is the farthest away from the metal M among all atoms in the ligand L.sub.B. Furthermore, the ligand L.sub.C, if present, has a third substituent R.sup.III, where the third substituent R.sup.III has a first atom a-III that is the farthest away from the metal M among all atoms in the ligand L.sub.C.

[0320] In such heteroleptic compounds, vectors V.sub.D1, V.sub.D2, and V.sub.D3 can be defined as follows. V.sub.D1 represents the direction from the metal M to the first atom a-I and the vector V.sub.D1 has a value D.sup.1 that represents the straight line distance between the metal M and the first atom a-I in the first substituent R.sup.I. V.sub.D2 represents the direction from the metal M to the first atom a-II and the vector V.sub.D2 has a value D.sup.2 that represents the straight line distance between the metal M and the first atom a-II in the second substituent R.sup.II. V.sub.D3 represents the direction from the metal M to the first atom a-III and the vector V.sub.D3 has a value D.sup.3 that represents the straight line distance between the metal M and the first atom a-III in the third substituent R.sup.III.

[0321] In such heteroleptic compounds, a sphere having a radius r is defined whose center is the metal M and the radius r is the smallest radius that will allow the sphere to enclose all atoms in the compound that are not part of the substituents R.sup.I, R.sup.II and R.sup.III; and where at least one of D.sup.1, D.sup.2, and D.sup.3 is greater than the radius r by at least 1.5 Å. In some embodiments, at least one of D.sup.1, D.sup.2, and D.sup.3 is greater than the radius r by at least 2.9, 3.0, 4.3, 4.4, 5.2, 5.9, 7.3, 8.8, 10.3, 13.1, 17.6, or 19.1 Å. In some embodiments, at least two of D.sup.1, D.sup.2, and D.sup.3 is greater than the radius r by at least 1.5, 2.9, 3.0, 4.3, 4.4, 5.2, 5.9, 7.3, 8.8, 10.3, 13.1, 17.6, or 19.1 Å.

[0322] In some embodiments of such heteroleptic compound, the compound has a transition dipole moment axis and angles are defined between the transition dipole moment axis and the vectors V.sub.D1, V.sub.D2, and V.sub.D3, where at least one of the angles between the transition dipole moment axis and the vectors V.sub.D1, V.sub.D2, and V.sub.D3 is less than 40°. In some embodiments, at least one of the angles between the transition dipole moment axis and the vectors V.sub.D1, V.sub.D2, and V.sub.D3 is less than 30°, 20°, 15°, or 10°. In some embodiments, at least two of the angles between the transition dipole moment axis and the vectors V.sub.D1, V.sub.D2, and V.sub.D3 are less than 20°. In some embodiments, at least two of the angles between the transition dipole moment axis and the vectors V.sub.D1, V.sub.D2, and V.sub.D3 are less than 15° or 10°.

[0323] In some embodiments, all three angles between the transition dipole moment axis and the vectors V.sub.D1, V.sub.D2, and V.sub.D3 are less than 20°. In some embodiments, all three angles between the transition dipole moment axis and the vectors V.sub.D1, V.sub.D2, and V.sub.D3 are less than 15° or 10°.

[0324] In some embodiments of such heteroleptic compounds, the compound has a vertical dipole ratio (VDR) of 0.33 or less. In some embodiments of such heteroleptic compounds, the compound has a VDR of 0.30, 0.25, 0.20, or 0.15 or less.

[0325] One of ordinary skill in the art would readily understand the meaning of the terms transition dipole moment axis of a compound and vertical dipole ratio of a compound. Nevertheless, the meaning of these terms can be found in U.S. Pat. No. 10,672,997 whose disclosure is incorporated herein by reference in its entirety. In U.S. Pat. No. 10,672,997, horizontal dipole ratio (HDR) of a compound, rather than VDR, is discussed. However, one skilled in the art readily understands that $VDR=1-HDR$.

[0326] In some embodiments, the compound can be an emissive dopant. In some embodiments, the compound can produce emissions via phosphorescence, fluorescence, thermally activated delayed

fluorescence, triplet-triplet annihilation, or combinations of these processes. In some embodiments, the emissive dopant can be a racemic mixture, or can be enriched in one enantiomer. In some embodiments, the present compounds can have different stereoisomers, such as fac and mer. The current compound relates both to individual isomers and to mixtures of various isomers in any mixing ratio. In some embodiments, the compound can be homoleptic (each ligand is the same). In some embodiments, the compound can be heteroleptic (at least one ligand is different from others). When there are more than one ligand coordinated to a metal, the ligands can all be the same in some embodiments. In some other embodiments, at least one ligand is different from the other ligands. In some embodiments, every ligand can be different from every other ligand. This is also true in embodiments where a ligand being coordinated to a metal can be linked with other ligands being coordinated to that metal to form a tridentate, tetradentate, pentadentate, or hexadentate ligands. Thus, where the coordinating ligands are being linked together, all of the ligands can be the same in some embodiments, and at least one of the ligands being linked can be different from the other ligand(s) in some other embodiments.

[0327] In yet another aspect of the present disclosure, a formulation that comprises the novel compound disclosed herein is described. The formulation can include one or more components selected from the group consisting of a solvent, an emitter, a host, a hole injection material, hole transport material, electron blocking material, hole blocking material, and an electron transport material, disclosed herein.

[0328] The present disclosure encompasses any chemical structure comprising the novel compound of the present disclosure, or a monovalent or polyvalent variant thereof. In other words, the inventive compound, or a monovalent or polyvalent variant thereof, can be a part of a larger chemical structure. Such chemical structure can be selected from the group consisting of a monomer, a polymer, a macromolecule, and a supramolecule (also known as supermolecule). As used herein, a “monovalent variant of a compound” refers to a moiety that is identical to the compound except that one hydrogen has been removed and replaced with a bond to the rest of the chemical structure. As used herein, a “polyvalent variant of a compound” refers to a moiety that is identical to the compound except that more than one hydrogen has been removed and replaced with a bond or bonds to the rest of the chemical structure. In the instance of a supramolecule, the inventive compound can also be incorporated into the supramolecule complex without covalent bonds. As used in this context, the description that a structure A comprises a moiety B means that the structure A includes the structure of moiety B not including the H or D atoms that can be attached to the moiety B. This is because at least one H or D on a given moiety structure has to be replaced to become a substituent so that the moiety B can be part of the structure A, and one or more of the H or D on a given moiety B structure can be further substituted once it becomes a part of structure A.

C. The OLEDs and the Devices of the Present Disclosure

[0329] In another aspect, the present disclosure also provides an OLED device comprising a first organic layer that contains a compound as disclosed in the above compounds section of the present disclosure.

[0330] In some embodiments, the OLED comprises: an anode; a cathode; and an organic layer disposed between the anode and the cathode, where the organic layer comprises a compound having a first ligand L.sub.A comprising a structure of Formula I as described herein.

[0331] In some embodiments, the organic layer is selected from the group consisting of HIL, HTL, EBL, EML, HBL, ETL, and EIL. In some embodiments, the organic layer may be an emissive layer and the compound as described herein may be an emissive dopant or a non-emissive dopant.

[0332] In some embodiments, the organic layer may further comprise a host, wherein host comprises at least one chemical group selected from the group consisting of triphenylene, carbazole, indolocarbazole, dibenzothiophene, dibenzofuran, dibenzoselenophene, 5λ.sup.2-benzo[d]benzo[4,5]imidazo[3,2-a]imidazole, 5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene,

azaborinine, oxaborinine, dihydroacridine, xanthene, dihydrobenzoazasiline, dibenzooxasiline, phenoxazine, phenoxathiine, phenothiazine, dihydrophenazine, fluorene, naphthalene, anthracene, phenanthrene, phenanthroline, benzoquinoline, quinoline, isoquinoline, quinazoline, pyrimidine, pyrazine, pyridine, triazine, boryl, silyl, aza-triphenylene, aza-carbazole, aza-indolocarbazole, aza-dibenzothiophene, aza-dibenzofuran, aza-dibenzoselenophene, aza-5 λ .sup.2-benzo[d]benzo[4,5]imidazo[3,2-a]imidazole, and aza-(5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene).

[0333] In some embodiments, the host can be selected from the group consisting of the structures of the following HOST Group 1:

##STR00932## ##STR00933## ##STR00934## ##STR00935## ##STR00936## ##STR00937##
##STR00938## ##STR00939## ##STR00940## ##STR00941## ##STR00942## ##STR00943##
##STR00944## ##STR00945## ##STR00946## ##STR00947## ##STR00948##

wherein: [0334] each of J.sub.1 to J.sub.6 is independently C or N; [0335] L' is a direct bond or an organic linker; [0336] each Y.sup.AA, Y.sup.BB, Y.sup.CC, and Y.sup.DD is independently selected from the group consisting of absent a bond, direct bond, O, S, Se, CRR', SiRR', GeRR', NR, BR, BRR'; [0337] each of R.sup.A1, R.sup.B', R.sup.C', R.sup.D', R.sup.E', R.sup.F', and R.sup.G' independently represents mono, up to the maximum substitutions, or no substitutions; [0338] each R, R', R.sup.A', R.sup.B', R.sup.C', R.sup.D', R.sup.E', R.sup.F', and R.sup.G' is independently a hydrogen or a substituent selected from the group consisting of the General Substituents as defined herein; any two substituents can be joined or fused to form a ring; [0339] and where possible, each unsubstituted aromatic carbon atom can be replaced with N to form an aza-substituted ring.

[0340] In some embodiments at least one of J.sub.1 to J.sub.3 is N. In some embodiments at least two of J.sub.1 to J.sub.3 are N. In some embodiments, all three of J.sub.1 to J.sub.3 are N. In some embodiments, each Y.sup.CC and Y.sup.DD is independently O, S, or SiRR', or more preferably O or S. In some embodiments, at least one unsubstituted aromatic carbon atom is replaced with N to form an aza-ring.

[0341] In some embodiments, the host is selected from the group consisting of EG1-MG1-EG1 to EG53-MG27-EG53 with a formula of EGa-MGb-EGc, or EG1-EG1 to EG53-EG53 with a formula of EGa-EGc when MGb is absent, wherein a is an integer from 1 to 53, b is an integer from 1 to 27, c is an integer from 1 to 53. The structure of EG1 to EG53 is shown below:

##STR00949## ##STR00950##

The structure of MG1 to MG27 is shown below:

##STR00951##

In the MGb structures shown above, the two bonding positions in the asymmetric structures MG10, MG11, MG12, MG13, MG14, MG17, MG24, and MG25 are labeled with numbers for identification purposes.

[0342] In some embodiments, the host can be any of the aza-substituted variants thereof, fully or partially deuterated variants thereof, and combinations thereof. In some embodiments, the host has formula EGa-MGb-Egc and is selected from the group consisting of h1 to h112 defined in the following HOST Group 2 list, where each of MGb, EGa, and EGc are defined as follows:

TABLE-US-00008 h MGb EGa EGC h1 MG1 EG3 EG36 h2 MG1 EG8 EG12 h3 MG1 EG13 EG14 h4 MG1 EG13 EG18 h5 MG1 EG13 EG25 h6 MG1 EG13 EG36 h7 MG1 EG22 EG36 h8 MG1 EG25 EG46 h9 MG1 EG27 EG46 h10 MG1 EG27 EG48 h11 MG1 EG32 EG50 h12 MG1 EG35 EG46 h13 MG1 EG36 EG45 h14 MG1 EG36 EG49 h15 MG1 EG40 EG45 h16 MG2 EG3 EG36 h17 MG2 EG25 EG31 h18 MG2 EG31 EG33 h19 MG2 EG36 EG45 h20 MG2 EG36 EG46 h21 MG3 EG4 EG36 h22 MG3 EG34 EG45 h23 MG4 EG13 EG17 h24 MG5 EG13 EG45 h25 MG5 EG17 EG36 h26 MG5 EG18 EG36 h27 MG6 EG17 EG17 h28 MG7 EG43 EG45 h29 MG8 EG1 EG28 h30 MG8 EG6 EG7 h31 MG8 EG7 EG7 h32 MG8 EG7 EG11 h33 MG9 EG1 EG43 h34 MG10 4-EG1 2-EG37 h35 MG10 4-EG1 2-EG38 h36 MG10 EG1 EG42 h37 MG11 4-EG1 2-EG39 h38 MG12 1-EG17 9-EG31 h39 MG13 3-EG17 9-EG4 h40 MG13 3-EG17 9-EG13 h41

MG13 3-EG17 9-EG31 h42 MG13 3-EG17 9-EG45 h43 MG13 3-EG17 9-EG46 h44 MG13 3-EG17 9-EG48 h45 MG13 3-EG17 9-EG49 h46 MG13 3-EG32 9-EG31 h47 MG13 3-EG44 9-EG3 h48 MG14 3-EG13 5-EG45 h49 MG14 3-EG23 5-EG45 h50 MG15 EG3 EG48 h51 MG15 EG17 EG31 h52 MG15 EG31 EG36 h53 MG16 EG17 EG17 h54 MG17 EG17 EG17 h55 MG18 EG16 EG24 h56 MG18 EG16 EG30 h57 MG18 EG20 EG41 h58 MG19 EG16 EG29 h59 MG20 EG1 EG31 h60 MG20 EG17 EG18 h61 MG21 EG23 EG23 h62 MG22 EG1 EG45 h63 MG22 EG1 EG46 h64 MG22 EG3 EG46 h65 MG22 EG4 EG46 h66 MG22 EG4 EG47 h67 MG22 EG9 EG45 h68 MG23 EG1 EG3 h69 MG23 EG1 EG6 h70 MG23 EG1 EG14 h71 MG23 EG1 EG18 h72 MG23 EG1 EG19 h73 MG23 EG1 EG23 h74 MG23 EG1 EG51 h75 MG23 EG2 EG18 h76 MG23 EG3 EG3 h77 MG23 EG3 EG4 h78 MG23 EG3 EG5 h79 MG23 EG4 EG4 h80 MG23 EG4 EG5 h81 MG24 2-EG1 10-EG33 h82 MG24 2-EG4 10-EG36 h83 MG24 2-EG21 10-EG36 h84 MG24 2-EG23 10-EG36 h85 MG25 2-EG1 9-EG33 h86 MG25 2-EG3 9-EG36 h87 MG25 2-EG4 9-EG36 h88 MG25 2-EG17 9-EG27 h89 MG25 2-EG17 9-EG36 h90 MG25 2-EG21 9-EG36 h91 MG25 2-EG23 9-EG27 h92 MG25 2-EG23 9-EG36 h93 MG26 EG1 EG9 h94 MG26 EG1 EG10 h95 MG26 EG1 EG21 h96 MG26 EG1 EG23 h97 MG26 EG1 EG26 h98 MG26 EG3 EG3 h99 MG26 EG3 EG9 h100 MG26 EG3 EG23 h101 MG26 EG3 EG26 h102 MG26 EG4 EG10 h103 MG26 EG5 EG10 h104 MG26 EG6 EG10 h105 MG26 EG10 EG10 h106 MG26 EG10 EG14 h107 MG26 EG10 EG15 h108 MG27 EG52 EG53 h109 — EG13 EG18 h110 — EG17 EG31 h111 — EG17 EG50 h112 — EG40 EG45

In the table above, the EGa and EGc structures that are bonded to one of the asymmetric structures MG10, MG11, MG12, MG13, MG14, MG17, MG24, and MG25, are noted with a numeric prefix identifying their bonding position in the MGb structure.

[0343] In some embodiments, the organic layer may further comprise a host, wherein the host comprises a metal complex.

[0344] In some embodiments, the emissive layer can comprise two hosts, a first host and a second host. In some embodiments, the first host is a hole transporting host, and the second host is an electron transporting host. In some embodiments, the first host is a hole transporting host, and the second host is a bipolar host. In some embodiments, the first host is an electron transporting host, and the second host is a bipolar host. In some embodiments, the first host and the second host can form an exciplex. In some embodiments, the emissive layer can comprise a third host. In some embodiments, the third host is selected from the group consisting of an insulating host (wide band gap host), a hole transporting host, and an electron transporting host. In some embodiments, the third host forms an exciplex with one of the first host and the second host, or with both the first host and the second host. In some embodiments, the emissive layer can comprise a fourth host. In some embodiments, the fourth host is selected from the group consisting of an insulating host (wide band gap host), a hole transporting host, and an electron transporting host. In some embodiments, the fourth host forms an exciplex with one of the first host, the second host, and the third host, with two of the first host, the second host, and the third host, or with each of the first host, the second host, and the third host. In some embodiments, the electron transporting host has a LUMO less than -2.4 eV, less than -2.5 eV, less than -2.6 eV, or less than -2.7 eV. In some embodiments, the hole transporting host has a HOMO higher than -5.6 eV, higher than -5.5 eV, higher than -5.4 eV, or higher than -5.35 eV. The HOMO and LUMO values can be determined using solution electrochemistry. Solution cyclic voltammetry and differential pulsed voltammetry can be performed using a CH Instruments model 6201B potentiostat using anhydrous dimethylformamide (DMF) solvent and tetrabutylammonium hexafluorophosphate as the supporting electrolyte. Glassy carbon, platinum wire, and silver wire were used as the working, counter and reference electrodes, respectively. Electrochemical potentials can be referenced to an internal ferrocene-ferroconium redox couple (Fc/Fc⁺) by measuring the peak potential differences from differential pulsed voltammetry. The corresponding highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) energies can be determined by

referencing the cationic and anionic redox potentials to ferrocene (4.8 eV vs. vacuum) according to literature ((a) Fink, R.; Heischkel, Y.; Thelakkat, M.; Schmidt, H.-W. *Chem. Mater.* 1998, 10, 3620-3625. (b) Pommerehne, J.; Vestweber, H.; Guss, W; Mahrt, R. F.; Bassler, H.; Porsch, M.; Daub, J. *Adv. Mater* 1995, 7, 551).

[0345] In some embodiments, the compound as described herein may be a sensitizer or a component of a sensitizer; wherein the device may further comprise an acceptor that receives the energy from the sensitizer. In some embodiments, the acceptor is an emitter in the device. In some embodiments, the acceptor may be a fluorescent material. In some embodiments, the compound described herein can be used as a phosphorescent sensitizer in an OLED where one or multiple layers in the OLED contain an acceptor in the form of one or more non-delayed fluorescent and/or delayed fluorescence material. In some embodiments, the compound described herein can be used as one component of an exciplex to be used as a sensitizer. As a phosphorescent sensitizer, the compound must be capable of energy transfer to the acceptor and the acceptor will emit the energy or further transfer energy to a final emitter. The acceptor concentrations can range from 0.001% to 99.9%. The acceptor could be in either the same layer as the phosphorescent sensitizer or in one or more different layers. In some embodiments, the acceptor is a thermally activated delayed fluorescence (TADF) material. In some embodiments, the acceptor is a non-delayed fluorescent material. In some embodiments, the emission can arise from any or all of the sensitizer, acceptor, and final emitter. In some embodiments, the acceptor has an emission at room temperature with a full width at half maximum (FWHM) of equal to or less than 50, 45, 40, 35, 30, 25, 20, 15, 10, or 5 nm. Narrower FWHM means better color purity for the OLED display application.

[0346] As used herein, phosphorescence generally refers to emission of a photon with a change in electron spin quantum number, i.e., the initial and final states of the emission have different electron spin quantum numbers, such as from T1 to S0 state. Most of the Ir and Pt complexes currently used in OLED are phosphorescent emitters. In some embodiments, if an exciplex formation involves a triplet emitter, such exciplex can also emit phosphorescent light. On the other hand, fluorescent emitters generally refer to emission of a photon without a change in electron spin quantum number, such as from S1 to S0 state, or from D1 to D0 state. Fluorescent emitters can be delayed fluorescent or non-delayed fluorescent emitters. Depending on the spin state, fluorescent emitter can be a singlet emitter or a doublet emitter, or other multiplet emitter. It is believed that the internal quantum efficiency (IQE) of fluorescent OLEDs can exceed the 25% spin statistics limit through delayed fluorescence. There are two types of delayed fluorescence, i.e. P-type and E-type delayed fluorescence. P-type delayed fluorescence is generated from triplet-triplet annihilation (TTA). On the other hand, E-type delayed fluorescence does not rely on the collision of two triplets, but rather on the thermal population between the triplet states and the singlet excited states. Thermal energy can activate the transition from the triplet state back to the singlet state. This type of delayed fluorescence is also known as TADF. E-type delayed fluorescence characteristics can be found in an exciplex system or in a single compound. Without being bound by theory, it is believed that TADF emissions require a compound or an exciplex having a small singlet-triplet energy gap ($\Delta E_{\text{sub.S-T}}$) less than or equal to 400, 350, 300, 250, 200, 150, 100, or 50 meV. There are two major types of TADF emitters, one is called donor-acceptor type TADF, the other one is called multiple resonance (MR) TADF. Often, single compound donor-acceptor TADF compounds are constructed by connecting an electron donor moiety such as amino- or carbazole-derivatives and an electron acceptor moiety such as N-containing six-membered aromatic rings or cyano-substituted aromatic rings. Donor-acceptor exciplexes can be formed between a hole transporting compound and an electron transporting compound. Examples of MR-TADF materials include highly conjugated fused ring systems. In some embodiments, MR-TADF materials comprises boron, carbon, and nitrogen atoms. Such materials may comprise other atoms, such as oxygen, as well. In some embodiments, the reverse intersystem crossing time from T1 to S1 of the delayed fluorescent emission at 293K is less than or equal to 10 microseconds. In some embodiments, such time can be

greater than 10 microseconds and less than 100 microseconds.

[0347] In some embodiments, the OLED may comprise an additional compound selected from the group consisting of a non-delayed fluorescence material, a delayed fluorescence material, a phosphorescent material, and combination thereof.

[0348] In some embodiments, the inventive compound described herein is a phosphorescent material.

[0349] In some embodiments, the phosphorescent material is an emitter which emits light within the OLED. In some embodiments, the phosphorescent material does not emit light within the OLED. In some embodiments, the phosphorescent material energy transfers its excited state to another material within the OLED. In some embodiments, the phosphorescent material participates in charge transport within the OLED. In some embodiments, the phosphorescent material is a sensitizer or a component of a sensitizer, and the OLED further comprises an acceptor. In some embodiments, the phosphorescent material forms an exciplex with another material within the OLED, for example a host material, an emitter material.

[0350] In some embodiments, the non-delayed fluorescence material or the delayed fluorescence material is an emitter which emits light within the OLED. In some embodiments, the non-delayed fluorescence material or the delayed fluorescence material does not emit light within the OLED. In some embodiments, the non-delayed fluorescence material or the delayed fluorescence material energy transfers its excited state to another material within the OLED. In some embodiments, the non-delayed fluorescence material or the delayed fluorescence material participates in charge transport within the OLED. In some embodiments, the non-delayed fluorescence material or the delayed fluorescence material is an acceptor, and the OLED further comprises a sensitizer.

[0351] In some embodiments of the OLED, the delayed fluorescence material comprises at least one donor group and at least one acceptor group. In some embodiments, the delayed fluorescence material is a metal complex. In some embodiments, the delayed fluorescence material is a non-metal complex. In some embodiments, the delayed fluorescence material is a Pt, Pd, Zn, Cu, Ag, or Au complex (some of them are also called metal-assisted (MA) TADF). In some embodiments, the metal-assisted delayed fluorescence material comprises a metal-carbene bond. In some embodiments, the non-delayed fluorescence material or delayed fluorescence material comprises at least one chemical group selected from the group consisting of aryl-amine, aryloxy, arylthio, triphenylene, carbazole, indolocarbazole, dibenzothiophene, dibenzofuran, dibenzoselenophene, 5λ^{sup}.2-benzo[d]benzo[4,5]imidazo[3,2-a]imidazole, 5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene, 5λ^{sup}.2,9λ^{sup}.2-diaza-13b-boranaphtho[2,3,4-de]anthracene, 5-oxa-9%^{sup}.2-aza-13b-boranaphtho[3,2,1-de]anthracene, azaborinine, oxaborinine, dihydroacridine, xanthene, dihydrobenzoazasiline, dibenzooxasiline, phenoxazine, phenoxathiine, phenothiazine, dihydrophenazine, fluorene, naphthalene, anthracene, phenanthrene, phenanthroline, benzoquinoline, quinoline, isoquinoline, quinazoline, pyrimidine, pyrazine, pyridine, triazine, boryl, amino, silyl, aza-variants thereof, and combinations thereof. In some embodiments, non-delayed the fluorescence material or delayed fluorescence material comprises a tri(aryl/heteroaryl)borane with one or more pairs of the substituents from the aryl/heteroaryl being joined to form a ring. In some embodiments, the fluorescence material comprises at least one chemical group selected from the group consisting of naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, and azulene.

[0352] In yet another aspect, the OLED of the present disclosure may also comprise an emissive region containing a compound or a formulation of the compound as disclosed in the above compounds section of the present disclosure. In some embodiments, the emissive region can comprise a compound or a formulation of the compound having a first ligand L_{sub}.A comprising a structure of Formula I as described herein. In some embodiments, the emissive region consists of one or more organic layers, wherein at least one of the one or more organic layers has a minimum thickness selected from the group consisting of 350, 400, 450, 500, 550, 600, 650 and 700 Å. In

some embodiments, the at least one of the one or more organic layers are formed from an Emissive System that has a figure of merit (FOM) value equal to or larger than the number selected from the group consisting of 2.50, 2.55, 2.60, 2.65, 2.70, 2.75, 2.80, 2.85, 2.90, 2.95, 3.00, 5.00, 10.0, 15.0, and 20.0. The definition of FOM is available in U.S. patent Application Publication No. 2023/0292605, and its entire contents are incorporated herein by reference. In some embodiments, the at least one of the one or more organic layers comprises a compound or a formulation of the compound as disclosed in Sections A and D of the present disclosure.

[0353] In some embodiments, the OLED or the emissive region comprising the inventive compound disclosed herein can be incorporated into a full-color pixel arrangement of a device. The full-color pixel arrangement of such a device comprises at least one pixel, wherein the at least one pixel comprises a first subpixel and a second subpixel. The first subpixel includes a first OLED comprising a first emissive region. The second subpixel includes a second OLED comprising a second emissive region. In some embodiments, the first and/or second OLED, the first and/or second emissive region can be the same or different and each can independently have the various device characteristics and the various embodiments of the inventive compounds included therein, and various combinations and subcombinations of the various device characteristics and the various embodiments of the inventive compounds included therein, as disclosed herein.

[0354] In some embodiments, the first emissive region is configured to emit a light having a peak wavelength $\lambda_{\text{sub.max1}}$; the second emissive region is configured to emit a light having a peak wavelength $\lambda_{\text{sub.max2}}$. In some embodiments, the difference between the peak wavelengths $\lambda_{\text{sub.max1}}$ and $\lambda_{\text{sub.max2}}$ is at least 4 nm but within the same color. For example, a light blue and a deep blue light as described above. In some embodiments, a first emissive region is configured to emit a light having a peak wavelength $\lambda_{\text{sub.max1}}$ in one region of the visible spectrum of 400-500 nm, 500-600 nm, 600-700 nm; and a second emissive region is configured to emit light having a peak wavelength $\lambda_{\text{sub.max2}}$ in one of the remaining regions of the visible spectrum of 400-500 nm, 500-600 nm, 600-700 nm. In some embodiments, the first emissive region comprises a first number of emissive layers that are deposited one over the other if more than one; and the second emissive region comprises a second number of emissive layers that is deposited one over the other if more than one; and the first number is different from the second number. In some embodiments, both the first emissive region and the second emissive region comprise a phosphorescent materials, which may be the same or different. In some embodiments, the first emissive region comprises a phosphorescent material, while the second emissive region comprises a fluorescent material. In some embodiments, both the first emissive region and the second emissive region comprise a fluorescent materials, which may be the same or different.

[0355] In some embodiments, the at least one pixel of the OLED or emissive regions includes a total of N subpixels; wherein the N subpixels comprises the first subpixel and the second subpixel; wherein each of the N subpixels comprises an emissive region; wherein the total number of the emissive regions within the at least one pixel is equal to or less than N-1. In some embodiments, the second emissive region is exactly the same as the first emissive region; and each subpixel of the at least one pixel comprises the same one emissive region as the first emissive region. In some embodiments, the full-color pixel arrangements can have a plurality of pixels comprising a first pixel region and a second pixel region; wherein at least one display characteristic in the first pixel region is different from the corresponding display characteristic of the second pixel region, and wherein the at least one display characteristic is selected from the group consisting of resolution, cavity mode, color, outcoupling, and color filter.

[0356] In some embodiments, the OLED is a stacked OLED comprising one or more charge generation layers (CGLs). In some embodiments, the OLED comprises a first electrode, a first emissive region disposed over the first electrode, a first CGL disposed over the first emissive region, a second emissive region disposed over the first CGL, and a second electrode disposed over the second emissive region. In some embodiments, the first and/or the second emissive regions can

have the various device characteristics as described above for the pixelated device. In some embodiments, the stacked OLED is configured to emit white color. In some embodiments, one or more of the emissive regions in a pixelated or in a stacked OLED comprises a sensitizer and an acceptor with the various sensitizing device characteristics and the various embodiments of the inventive compounds disclosed herein. For example, the first emissive region is comprised in a sensitizing device, while the second emissive region is not comprised in a sensitizing device; in some instances, both the first and the second emissive regions are comprised in sensitizing devices. [0357] In some embodiments, the OLED can emit light having at least 1%, 5%, 10, 30%, 50%, 70%, 80%, 90%, 95%, 99%, or 100% from the plasmonic mode. In some embodiments, at least one of the anode, the cathode, or a new layer disposed over the organic emissive layer functions as an enhancement layer. The enhancement layer comprises a plasmonic material exhibiting surface plasmon resonance that non-radiatively couples to the emitter material and transfers excited state energy from the emitter material to non-radiative mode of surface plasmon polariton. In some embodiments, the enhancement layer is provided no more than a threshold distance away from the organic emissive layer, wherein the emitter material has a total non-radiative decay rate constant and a total radiative decay rate constant due to the presence of the enhancement layer. A threshold distance is where the total non-radiative decay rate constant is equal to the total radiative decay rate constant. Another threshold distance is the distance at which the total radiative decay rate constant divided by the sum of the total non-radiative decay rate constant and total radiative decay rate constant is equal to the photoluminescent yield of the emissive material without the enhancement layer present.

[0358] In some embodiments, the OLED further comprises an outcoupling layer. In some embodiments, the outcoupling layer is disposed over the enhancement layer on a side opposite the organic emissive layer. The outcoupling layer scatters the energy from the surface plasmon polaritons. In some embodiments this energy is scattered as photons to free space. In other embodiments, the energy is scattered from the surface plasmon mode into other modes of the device such as but not limited to the organic waveguide mode, the substrate mode, or another waveguiding mode. In some embodiments, one or more intervening layer can be disposed between the enhancement layer and the outcoupling layer. The examples for intervening layer(s) can be dielectric materials, including organic, inorganic, perovskites, oxides, and may include stacks and/or mixtures of these materials.

[0359] The enhancement layer modifies the effective properties of the medium in which the emitter material resides resulting in any or all of the following: a decreased rate of emission, a modification of emission line-shape, a change in emission intensity with angle, a change in the stability of the emitter material, a change in the efficiency of the OLED, and a reduced efficiency roll-off of the OLED device. Placement of the enhancement layer on the cathode side, anode side, or on both sides, or the enhancement layer itself being as the CGL, results in OLED devices which take advantage of any of the above-mentioned effects. In addition to the specific functional layers mentioned herein and illustrated in the various OLED examples shown in the figures, the OLEDs according to the present disclosure may include any of the other functional layers often found in OLEDs.

[0360] In some embodiments, the enhancement layer can be comprised of plasmonic materials, optically active metamaterials, or hyperbolic metamaterials. In some embodiments, the plasmonic material includes at least one metal. In such embodiments the metal may include at least one of Ag, Al, Au, Ir, Pt, Ni, Cu, W, Ta, Fe, Cr, Mg, Ga, Rh, Ti, Ru, Pd, In, Bi, or Ca, alloys or mixtures of these materials, and stacks of these materials. In some embodiments, the enhancement layer is provided as a planar layer. In other embodiments, the enhancement layer has wavelength-sized features that are arranged periodically, quasi-periodically, or randomly, or sub-wavelength-sized features that are arranged periodically, quasi-periodically, or randomly.

[0361] In some embodiments, the outcoupling layer has wavelength-sized or sub-wavelength sized

features that are arranged periodically, quasi-periodically, or randomly. In some embodiments, the outcoupling layer may be composed of a plurality of nanoparticles. In some embodiments, the outcoupling layer is composed of a plurality of nanoparticles disposed over a material. In these embodiments the outcoupling layer may be tunable by at least one of: varying a size of the plurality of nanoparticles, varying a shape of the plurality of nanoparticles, changing a material of the plurality of nanoparticles, adjusting a thickness of the material, changing the refractive index of the material, adding an additional layer disposed on the plurality of nanoparticles, varying a thickness of the enhancement layer, or varying the material of the enhancement layer. The plurality of nanoparticles of the device may be formed from at least one of metal, dielectric material, semiconductor materials, an alloy of metal, a mixture of dielectric materials, a stack or layering of one or more materials, and/or a core of one type of material and that is coated with a shell of a different type of material. In some embodiments, the outcoupling layer is composed of at least metal nanoparticles wherein the metal is selected from the group consisting of Ag, Al, Au, Ir, Pt, Ni, Cu, W, Ta, Fe, Cr, Mg, Ga, Rh, Ti, Ru, Pd, In, Bi, and Ca, alloys or mixtures of these materials, and stacks of these materials. In some embodiments the outcoupling layer is formed by lithography. [0362] In some embodiments of a plasmonic device, the emitter, and/or host compounds used in the emissive layer has a vertical dipole ratio (VDR) of 0.33 or more. In some such embodiments, the emitter, and/or host compounds have a VDR of 0.40, 0.50, 0.60, 0.70, or more.

[0363] In yet another aspect, the present disclosure also provides a consumer product comprising an organic light-emitting device (OLED) having an anode; a cathode; and an organic layer disposed between the anode and the cathode, wherein the organic layer may comprise a compound or a formulation of the compound as disclosed in the above compounds section of the present disclosure.

[0364] In some embodiments, the consumer product comprises an OLED having an anode; a cathode; and an organic layer disposed between the anode and the cathode, wherein the organic layer may comprise a compound having a first ligand L_{sub}.A comprising a structure of Formula I as described herein.

[0365] Generally, an OLED comprises at least one organic layer disposed between and electrically connected to an anode and a cathode. When a current is applied, the anode injects holes and the cathode injects electrons into the organic layer(s). The injected holes and electrons each migrate toward the oppositely charged electrode. When an electron and hole localize on the same molecule, and an “exciton,” which is a localized electron-hole pair having an excited energy state, is formed. Light is emitted when the exciton relaxes via a photoemissive mechanism. In some cases, the exciton may be localized as an excimer or an exciplex. Non-radiative mechanisms, such as thermal relaxation, may also occur, but are generally considered undesirable.

[0366] FIG. 1 shows an organic light emitting device **100**. The figures are not necessarily drawn to scale. Device **100** may include a substrate **110**, an anode **115**, a hole injection layer (HIL) **120**, a hole transport layer (HTL) **125**, an electron blocking layer (EBL) **130**, an emissive layer (EML) **135**, a hole blocking layer (HBL) **140**, an electron transport layer (ETL) **145**, an electron injection layer (EIL) **150**, a protective layer **155**, a cathode **160**, and a barrier layer **170**. Cathode **160** is a compound cathode having a first conductive layer **162** and a second conductive layer **164**. Device **100** may be fabricated by depositing the layers described, in order. The properties and functions of these various layers, as well as example materials, are described in more detail in U.S. Pat. No. 7,279,704 at cols. 6-10, which are incorporated by reference.

[0367] More examples for each of these layers are available. For example, a flexible and transparent substrate-anode combination is disclosed in U.S. Pat. No. 5,844,363, which is incorporated by reference in its entirety. An example of a p-doped hole transport layer is m-MTDATA doped with F_{sub}.4-TCNQ at a molar ratio of 50:1, as disclosed in U.S. Patent Application Publication No. 2003/0230980, which is incorporated by reference in its entirety. Examples of emissive and host materials are disclosed in U.S. Pat. No. 6,303,238 to Thompson et

al., which is incorporated by reference in its entirety. An example of an n-doped electron transport layer is BPhen doped with Li at a molar ratio of 1:1, as disclosed in U.S. Patent Application Publication No. 2003/0230980, which is incorporated by reference in its entirety. U.S. Pat. Nos. 5,703,436 and 5,707,745, which are incorporated by reference in their entireties, disclose examples of cathodes including compound cathodes having a thin layer of metal such as Mg:Ag with an overlying transparent, electrically-conductive, sputter-deposited ITO layer. The theory and use of blocking layers is described in more detail in U.S. Pat. No. 6,097,147 and U.S. Patent Application Publication No. 2003/0230980, which are incorporated by reference in their entireties. Examples of injection layers are provided in U.S. Patent Application Publication No. 2004/0174116, which is incorporated by reference in its entirety. A description of protective layers may be found in U.S. Patent Application Publication No. 2004/0174116, which is incorporated by reference in its entirety.

[0368] FIG. 2 shows an inverted OLED **200**. The device includes a substrate **210**, a cathode **215**, an emissive layer **220**, a hole transport layer **225**, and an anode **230**. Device **200** may be fabricated by depositing the layers described, in order. Because the most common OLED configuration has a cathode disposed over the anode, and device **200** has cathode **215** disposed under anode **230**, device **200** may be referred to as an “inverted” OLED. Materials similar to those described with respect to device **100** may be used in the corresponding layers of device **200**. FIG. 2 provides one example of how some layers may be omitted from the structure of device **100**.

[0369] The simple layered structure illustrated in FIGS. 1 and 2 is provided by way of non-limiting example, and it is understood that embodiments of the present disclosure may be used in connection with a wide variety of other structures. The specific materials and structures described are exemplary in nature, and other materials and structures may be used. Functional OLEDs may be achieved by combining the various layers described in different ways, or layers may be omitted entirely, based on design, performance, and cost factors. Other layers not specifically described may also be included. Materials other than those specifically described may be used. Although many of the examples provided herein describe various layers as comprising a single material, it is understood that combinations of materials, such as a mixture of host and dopant, or more generally a mixture, may be used. Also, the layers may have various sublayers. The names given to the various layers herein are not intended to be strictly limiting. For example, in device **200**, hole transport layer **225** transports holes and injects holes into emissive layer **220**, and may be described as a hole transport layer or a hole injection layer. In one embodiment, an OLED may be described as having an “organic layer” disposed between a cathode and an anode. This organic layer may comprise a single layer, or may further comprise multiple layers of different organic materials as described, for example, with respect to FIGS. 1 and 2.

[0370] Structures and materials not specifically described may also be used, such as OLEDs comprised of polymeric materials (PLEDs) such as disclosed in U.S. Pat. No. 5,247,190 to Friend et al., which is incorporated by reference in its entirety. By way of further example, OLEDs having a single organic layer may be used. OLEDs may be stacked, for example as described in U.S. Pat. No. 5,707,745 to Forrest et al., which is incorporated by reference in its entirety. The OLED structure may deviate from the simple layered structure illustrated in FIGS. 1 and 2. For example, the substrate may include an angled reflective surface to improve out-coupling, such as a mesa structure as described in U.S. Pat. No. 6,091,195 to Forrest et al., and/or a pit structure as described in U.S. Pat. No. 5,834,893 to Bulovic et al., which are incorporated by reference in their entireties.

[0371] Unless otherwise specified, any of the layers of the various embodiments may be deposited by any suitable method. For the organic layers, preferred methods include thermal evaporation, ink-jet, such as described in U.S. Pat. Nos. 6,013,982 and 6,087,196, which are incorporated by reference in their entireties, organic vapor phase deposition (OVPD), such as described in U.S. Pat. No. 6,337,102 to Forrest et al., which is incorporated by reference in its entirety, and deposition by organic vapor jet printing (OVJP, also referred to as organic vapor jet deposition (OVJD)), such as

described in U.S. Pat. No. 7,431,968, which is incorporated by reference in its entirety. Other suitable deposition methods include spin coating and other solution based processes. Solution based processes are preferably carried out in nitrogen or an inert atmosphere. For the other layers, preferred methods include thermal evaporation, sputtering, chemical vapor deposition, atomic layer deposition, and electron beam deposition. Preferred patterning methods include deposition through a mask, photolithography, and cold welding such as described in U.S. Pat. Nos. 6,294,398 and 6,468,819, which are incorporated by reference in their entireties, and patterning associated with some of the deposition methods such as ink-jet and organic vapor jet printing (OVJP). Other methods may also be used. The materials to be deposited may be modified to make them compatible with a particular deposition method. For example, substituents such as alkyl and aryl groups, branched or unbranched, and preferably containing at least 3 carbons, may be used in small molecules to enhance their ability to undergo solution processing. Substituents having 20 carbons or more may be used, and 3-20 carbons are a preferred range. Materials with asymmetric structures may have better solution processability than those having symmetric structures, because asymmetric materials may have a lower tendency to recrystallize. Dendrimer substituents may be used to enhance the ability of small molecules to undergo solution processing.

[0372] Devices fabricated in accordance with embodiments of the present disclosure may further optionally comprise a barrier layer. One purpose of the barrier layer is to protect the electrodes and organic layers from damaging exposure to harmful species in the environment including moisture, vapor and/or gases, etc. The barrier layer may be deposited over, under or next to a substrate, an electrode, or over any other parts of a device including an edge. The barrier layer may comprise a single layer, or multiple layers. The barrier layer may be formed by various known chemical vapor deposition techniques and may include compositions having a single phase as well as compositions having multiple phases. Any suitable material or combination of materials may be used for the barrier layer. The barrier layer may incorporate an inorganic or an organic compound or both. The preferred barrier layer comprises a plurality of alternative layers of polymeric material and non-polymeric material; organic material and inorganic material; or a mixture of a polymeric material and a non-polymeric material as one example described in U.S. Pat. No. 7,968,146, PCT Pat. Application Nos. PCT/US2007/023098 and PCT/US2009/042829, which are herein incorporated by reference in their entireties.

[0373] Devices fabricated in accordance with embodiments of the present disclosure can be incorporated into a wide variety of electronic component modules (or units) that can be incorporated into a variety of electronic products or intermediate components. Examples of such electronic products or intermediate components include display screens, lighting devices such as discrete light source devices or lighting panels, etc. that can be utilized by the end-user product manufacturers. Such electronic component modules can optionally include the driving electronics and/or power source(s). Devices fabricated in accordance with embodiments of the present disclosure can be incorporated into a wide variety of consumer products that have one or more of the electronic component modules (or units) incorporated therein. A consumer product comprising an OLED that includes the compound of the present disclosure in the organic layer in the OLED is disclosed. Such consumer products would include any kind of products that include one or more light source(s) and/or one or more of some type of visual displays. Some examples of such consumer products include flat panel displays, curved displays, computer monitors, medical monitors, televisions, billboards, lights for interior or exterior illumination and/or signaling, heads-up displays, fully or partially transparent displays, flexible displays, rollable displays, foldable displays, stretchable displays, laser printers, telephones, mobile phones, tablets, phablets, personal digital assistants (PDAs), wearable devices, laptop computers, digital cameras, camcorders, viewfinders, micro-displays (displays that are less than 2 inches diagonal), 3-D displays, virtual reality or augmented reality displays, vehicles, video walls comprising multiple displays tiled together, theater or stadium screen, a light therapy device, and a sign. Various control mechanisms

may be used to control devices fabricated in accordance with the present disclosure, including passive matrix and active matrix. Many of the devices are intended for use in a temperature range comfortable to humans, such as 18 degrees C. to 30 degrees C., and more preferably at room temperature (20-25° C.), but could be used outside this temperature range, for example, from -40 degree C. to +80° C.

[0374] More details on OLEDs, and the definitions described above, can be found in U.S. Pat. No. 7,279,704, which is incorporated herein by reference in its entirety.

[0375] The materials and structures described herein may have applications in devices other than OLEDs. For example, other optoelectronic devices such as organic solar cells and organic photodetectors may employ the materials and structures. More generally, organic devices, such as organic transistors, may employ the materials and structures.

[0376] In some embodiments, the OLED has one or more characteristics selected from the group consisting of being flexible, being rollable, being foldable, being stretchable, and being curved. In some embodiments, the OLED is transparent or semi-transparent. In some embodiments, the OLED further comprises a layer comprising carbon nanotubes. In some embodiments, the OLED further comprises one or more quantum dots. Such quantum dots can be in the emissive layer, or in other functional layers, such as a down conversion layer.

[0377] In some embodiments, the OLED comprises a RGB pixel arrangement or white plus color filter pixel arrangement. In some embodiments, the OLED is a mobile device, a handheld device, or a wearable device. In some embodiments, the OLED is a display panel having less than 10 inch diagonal or 50 square inch area. In some embodiments, the OLED is a display panel having at least 10 inch diagonal or 50 square inch area. In some embodiments, the OLED is a lighting panel.

D. Other Materials Used in the OLED

[0378] The materials described herein are as various examples useful for a particular layer in an OLED. They may also be used in combination with a wide variety of other materials present in the device. For example, emissive dopants disclosed herein may be used by themselves in the EML, or in conjunction with a wide variety of other emitters, hosts, transport layers, blocking layers, injection layers, electrodes and other layers that may be present. The materials described or referred to below are non-limiting examples of materials that may be useful in combination with the compounds and the devices disclosed herein, and one of skill in the art can readily consult the literature to identify other materials that may be useful in combination.

a) Conductivity Dopants:

[0379] A charge transport layer can be doped with conductivity dopants to substantially alter its density of charge carriers, which will in turn alter its conductivity. The conductivity is increased by generating charge carriers in the matrix material, and depending on the type of dopant, a change in the Fermi level of the semiconductor may also be achieved. Hole-transporting layer can be doped by p-type conductivity dopants and n-type conductivity dopants are used in the electron-transporting layer. In some embodiments, conductivity dopants comprises at least one chemical moiety selected from the group consisting of cyano, fluorinated aryl or heteroaryl, fluorinated alkyl or cycloalkyl, alkylene, heteroaryl, amide, benzodithiophene, and highly conjugated heteroaryl groups extended by non-ring double bonds.

b) HIL/HTL:

[0380] A hole injecting/transporting material to be used in the present disclosure is not particularly limited, and any compound may be used as long as the compound is typically used as a hole injecting/transporting material. Examples of the material include, but are not limited to: a phthalocyanine or porphyrin derivative; an aromatic amine derivative; an indolocarbazole derivative; a polymer containing fluorohydrocarbon; a polymer with conductivity dopants; a conducting polymer, such as PEDOT/PSS; a self-assembly monomer derived from compounds such as phosphonic acid and silane derivatives; a metal oxide derivative, such as MoO₃; a p-type semiconducting organic compound, such as 1,4,5,8,9,12-

Hexaazatriphenylenehexacarbonitrile; a metal complex, and a cross-linkable compounds.

[0381] Examples of aromatic amine derivatives used in HIL or HTL include, but not limit to the following general structures:

##STR00952##

[0382] Each of Ar.sup.1 to Ar.sup.9 is selected from the group consisting of aromatic hydrocarbon cyclic compounds such as benzene, biphenyl, triphenyl, triphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, and azulene; the group consisting of aromatic heterocyclic compounds such as dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, benzofuropyridine, furodipyridine, benzothienopyridine, thienodipyridine, benzoselenophenopyridine, and selenophenodipyridine; and the group consisting of 2 to 10 cyclic structural units which are groups of the same type or different types selected from the aromatic hydrocarbon cyclic group and the aromatic heterocyclic group and are bonded to each other directly or via at least one of oxygen atom, nitrogen atom, sulfur atom, silicon atom, phosphorus atom, boron atom, chain structural unit and the aliphatic cyclic group. Each of Ar.sup.1 to Ar.sup.9 may be unsubstituted or may be substituted by a general substituent as described above, any two substituents can be joined or fused into a ring.

[0383] In some embodiments, each Ar.sup.1 to Ar.sup.9 independently comprises a moiety selected from the group consisting of:

##STR00953##

wherein k is an integer from 1 to 20; X.sup.101 to X.sup.108 is C or N; Z.sup.101 is C, N, O, or S.

[0384] Examples of metal complexes used in HIL or HTL include, but are not limited to the following general formula:

##STR00954##

wherein Met is a metal, which can have an atomic weight greater than 40; (Y.sup.101-Y.sup.102) is a bidentate ligand, the coordinating atoms of Y.sup.101 and Y.sup.102 are independently selected from C, N, O, P, and S; L.sup.101 is an another ligand; k' is an integer value from 1 to the maximum number of ligands that may be attached to the metal; and k'+k'' is the maximum number of ligands that may be attached to the metal.

[0385] In some embodiments, (Y.sup.101-Y.sup.102) is a 2-phenylpyridine or 2-phenylimidazole derivative. In some embodiments, (Y.sup.101-Y.sup.102) is a carbene ligand. In some embodiments, Met is selected from Ir, Pt, Pd, Os, Cu, and Zn. In some embodiments, the metal complex has a smallest oxidation potential in solution vs. Fc.sup.+ / Fc couple less than about 0.6 V.

[0386] In some embodiments, the HIL/HTL material is selected from the group consisting of phthalocyanine and porphyrin compounds, starburst triarylamine, CF.sub.x fluorohydrocarbon polymer, conducting polymers (e.g., PEDOT:PSS, polyaniline, polythiophene), phosphonic acid and silane SAMs, triarylamine or polythiophene polymers with conductivity dopants, Organic compounds with conductive inorganic compounds (such as molybdenum and tungsten oxides), n-type semiconducting organic complexes, metal organometallic complexes, cross-linkable compounds, polythiophene based polymers and copolymers, triarylamine, triarylamine with spirofluorene core, arylamine carbazole compounds, triarylamine with (di)benzothiophene/(di)benzofuran, indolocarbazoles, isoindole compounds, and metal carbene complexes.

c) EBL:

[0387] An electron blocking layer (EBL) may be used to reduce the number of electrons and/or

excitons that leave the emissive layer. The presence of such a blocking layer in a device may result in substantially higher efficiencies, and/or longer lifetime, as compared to a similar device lacking a blocking layer. Also, a blocking layer may be used to confine emission to a desired region of an OLED. In some embodiments, the EBL material has a higher LUMO (closer to the vacuum level) and/or higher triplet energy than one or more emitters closest to the EBL interface. In some embodiments, the compound used in EBL contains at least one carbazole group and/or at least one arylamine group. In some embodiments the HOMO level of the compound used in the EBL is shallower than the HOMO level of one or more of the hosts in the EML. In some embodiments, the compound used in EBL contains the same molecule or the same functional groups used as one of the hosts described herein.

d) Hosts:

[0388] The light emitting layer of the organic EL device of the present disclosure preferably contains at least a light emitting material as the dopant, and a host material. Examples of the host material are not particularly limited, and any metal complexes or organic compounds may be used as long as the host won't fully quench the emission of the dopant.

[0389] Examples of metal complexes used as host are preferred to have the following general formula:

##STR00955##

wherein Met is a metal; (Y.sup.103-Y.sup.104) is a bidentate ligand, the coordinating atoms of Y.sup.103 and Y.sup.104 are independently selected from C, N, O, P, and S; L.sup.101 is an another ligand; k' is an integer value from 1 to the maximum number of ligands that may be attached to the metal; and k'+k'' is the maximum number of ligands that may be attached to the metal.

[0390] In some embodiments the metal complexes are:

##STR00956##

wherein (O—N) is a bidentate ligand, having metal coordinated to atoms O and N.

[0391] In some embodiments, Met is selected from Ir and Pt. In a further embodiments, (Y.sup.103-Y.sup.104) is a carbene ligand.

[0392] In some embodiments, the host compound contains at least one of the following groups selected from the group consisting of aromatic hydrocarbon cyclic compounds such as benzene, biphenyl, triphenyl, triphenylene, tetraphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, and azulene; the group consisting of aromatic heterocyclic compounds such as dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, aza-dibenzothiophene, aza-dibenzofuran, aza-dibenzoselenophene, aza-carbazole, aza-indolocarbazole, aza-triphenylene, aza-tetraphenylene, 5λ.sup.2-benzo[d]benzo[4,5]imidazo[3,2-a]imidazole, 5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene; and the group consisting of 2 to 10 cyclic structural units which are groups of the same type or different types selected from the aromatic hydrocarbon cyclic group and the aromatic heterocyclic group and are bonded to each other directly or via at least one of oxygen atom, nitrogen atom, sulfur atom, silicon atom, phosphorus atom, boron atom, chain structural unit and the aliphatic cyclic group. Each option within each group may be unsubstituted or may be substituted by the general substituents as described herein or may be further fused.

[0393] In some embodiments, the host compound comprises at least one of the moieties selected from the group consisting of:

##STR00957##

wherein k is an integer from 0 to 20 or 1 to 20. X.sup.101 to X.sup.108 are independently selected from C or N. Z.sup.101 and Z.sup.102 are independently selected from C, N, O, or S.

[0394] In some embodiments, the host material is selected from the group consisting of arylcarbazoles, metal 8-hydroxyquinolates, (e.g., alq3, balq), metal phenoxybenzothiazole compounds, conjugated oligomers and polymers (e.g., polyfluorene), aromatic fused rings, zinc complexes, chrysene based compounds, aryltriphenylene compounds, poly-fused heteroaryl compounds, donor acceptor type molecules, dibenzofuran/dibenzothiophene compounds, polymers (e.g., pvk), spirofluorene compounds, spirofluorene-carbazole compounds, indolocabazoles, 5-member ring electron deficient heterocycles (e.g., triazole, oxadiazole), tetraphenylene complexes, metal phenoxypyridine compounds, metal coordination complexes (e.g., Zn, Al with N{circumflex over ()}N ligands), dibenzothiophene/dibenzofuran-carbazole compounds, silicon/germanium aryl compounds, aryl benzoyl esters, carbazole linked by non-conjugated groups, aza-carbazole/dibenzofuran/dibenzothiophene compounds, and high triplet metal organometallic complexes (e.g., metal-carbene complexes).

e) Emitter Materials in EML:

[0395] One or more emitter materials may be used in conjunction with the compound or device of the present disclosure. The emitter material can be emissive or non-emissive in the current device as described herein. Examples of the emitter materials are not particularly limited, and any compounds may be used as long as the compounds are capable of producing emissions in a regular OLED device. Examples of suitable emitter materials include, but are not limited to, compounds which are capable of producing emissions via phosphorescence, non-delayed fluorescence, delayed fluorescence, especially the thermally activated delayed fluorescence, i.e., TADF (also referred to as E-type delayed fluorescence), triplet-triplet annihilation, or combinations of these processes.

[0396] In some embodiments, the emitter material has the formula of

M(L.sup.1).sub.x(L.sup.2).sub.y(L.sup.3).sub.z; [0397] wherein L.sup.1, L.sup.2, and L.sup.3 can be the same or different; [0398] wherein x is 1, 2, or 3; [0399] wherein y is 0, 1, or 2; [0400] wherein z is 0, 1, or 2; [0401] wherein x+y+z is the oxidation state of the metal M; [0402] wherein L.sup.1 is selected from the group consisting of the structures of LIGAND LIST:

##STR00958## ##STR00959## ##STR00960## ##STR00961## ##STR00962## ##STR00963##
wherein each L.sup.2 and L.sup.3 are independently selected from the group consisting of
##STR00964##

and the structures of LIGAND LIST; wherein: [0403] M is selected from the group consisting of Ir, Rh, Re, Ru, Os, Pt, Pd, Zn, Au, Ag, and Cu; [0404] T is selected from the group consisting of B, Al, Ga, and In; [0405] K.sup.1' is a direct bond or is selected from the group consisting of NR.sub.e, PR.sub.e, O, S, and Se; [0406] each Y.sup.1 to Y.sup.15 are independently selected from the group consisting of carbon and nitrogen; [0407] Y' is selected from the group consisting of BR.sub.e, NR.sub.e, PR.sub.e, O, S, Se, C=O, S=O, SO.sub.2, CR.sub.eR.sub.f, SiR.sub.eR.sub.f, and GeR.sub.eR.sub.f; [0408] each R.sub.a, R.sub.b, R.sub.c, and R.sub.d can independently represent from mono to the maximum possible number of substitutions, or no substitution; [0409] each R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1, R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sub.e, and R.sub.f is independently a hydrogen or a substituent selected from the group consisting of the general substituents as defined herein; and

wherein any two substituents can be fused or joined to form a ring or form a multidentate ligand.

[0410] In some embodiments, the emitter material is selected from the group consisting of the following Dopant Group 1:

##STR00965## ##STR00966## ##STR00967## ##STR00968## ##STR00969## ##STR00970##
##STR00971## ##STR00972## ##STR00973##

wherein [0411] each of X.sup.96 to X.sup.99 is independently C or N; [0412] each Y.sup.100 is independently selected from the group consisting of a NR", O, S, and Se; [0413] each of R.sup.10a, R.sup.20a, R.sup.30a, R.sup.40a, and R.sup.50a independently represents mono substitution, up to

the maximum substitutions, or no substitution; [0414] each of R, R', R'', R.sup.10a, R.sup.11a, R.sup.12a, R.sup.13a, R.sup.20a, R.sup.30a, R.sup.40a, R.sup.50a, R.sup.60, R.sup.70, R.sup.97, R.sup.98, and R.sup.99 is independently a hydrogen or a substituent selected from the group consisting of the general substituents as defined herein; any two substituents can be joined or fused to form a ring.

[0415] In some embodiments, the emitter material is selected from the group consisting of the following Dopant Group 2:

##STR00974## ##STR00975## ##STR00976## ##STR00977## ##STR00978## ##STR00979##
##STR00980## ##STR00981## ##STR00982## ##STR00983## ##STR00984## ##STR00985##
##STR00986## ##STR00987## ##STR00988## ##STR00989## ##STR00990## ##STR00991##

wherein: [0416] each Y.sup.100 is independently selected from the group consisting of a NR'', O, S, and Se; [0417] L is independently selected from the group consisting of a direct bond, BR'', BR''R', NR'', PR'', O, S, Se, C=O, C=S, C=Se, C=NR'', C=CR''R'', S=O, SO.sub.2, CR'', CR''R'', SiR''R'', GeR''R'', alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof; [0418] X.sup.100 and X.sup.200 for each occurrence is selected from the group consisting of O, S, Se, NR'', and CR''R''; [0419] each R.sup.A'', R.sup.B'', R.sup.C'', R.sup.D'', R.sup.E'', and R.sup.F''

independently represents mono-, up to the maximum substitutions, or no substitutions; each of R, R', R'', R''', R.sup.A1', R.sup.A2', R.sup.A'', R.sup.B'', R.sup.C'', R.sup.D'', R.sup.E'', R.sup.F'', R.sup.G'', R.sup.H'', R.sup.I'', R.sup.J'', R.sup.K'', R.sup.L'', R.sup.M'', and R.sup.N'' is independently a hydrogen or a substituent selected from the group consisting of the general substituents as defined herein; any two substituents can be joined or fused to form a ring;

[0420] In some embodiments of the above Dopant Groups 1 and 2, each unsubstituted aromatic carbon atom can be replaced with N to form an aza-ring. In some embodiments, the maximum number of N atom in one ring is 1 or 2. In some embodiments of the above Dopant Groups 2, Pt atom in each formula can be replaced by Pd atom.

[0421] In some embodiments of the OLED, the delayed fluorescence material comprises at least one donor group and at least one acceptor group. In some embodiments, the delayed fluorescence material is a metal complex. In some embodiments, the delayed fluorescence material is a non-metal complex. In some embodiments, the delayed fluorescence material is a Zn, Cu, Ag, or Au complex.

[0422] In some embodiments of the OLED, the delayed fluorescence material has the formula of M(L.sup.5)(L.sup.6), wherein M is Cu, Ag, or Au, L.sup.5 and L.sup.6 are different, and L.sup.5 and L.sup.6 are independently selected from the group consisting of:

##STR00992## ##STR00993##

wherein A.sup.1-A.sup.9 are each independently selected from C or N;

each R.sup.P, R.sup.Q, and R.sup.U independently represents mono-, up to the maximum substitutions, or no substitutions;

wherein each R.sup.P, R.sup.P, R.sup.U, R.sup.SA, R.sup.SB, R.sup.RA, R.sup.RB, R.sup.RC, R.sup.RD, R.sup.RE, and R.sup.RF is independently a hydrogen or a substituent selected from the group consisting of the general substituents as defined herein; any two substituents can be joined or fused to form a ring.

[0423] In some embodiments of the OLED, the delayed fluorescence material comprises at least one of the donor moieties selected from the group consisting of:

##STR00994## ##STR00995## ##STR00996##

wherein Y.sup.T, Y.sup.U, Y.sup.V, and Y.sup.W are each independently selected from the group consisting of B, C, Si, Ge, N, P, O, S, Se, C=O, S=O, and SO.sub.2.

[0424] In some of the above embodiments, any carbon ring atoms up to maximum of a total number of three, together with their substituents, in each phenyl ring of any of above structures can be replaced with N.

[0425] In some embodiments, the delayed fluorescence material comprises at least one of the

acceptor moieties selected from the group consisting of nitrile, isonitrile, borane, fluoride, pyridine, pyrimidine, pyrazine, triazine, aza-carbazole, aza-dibenzothiophene, aza-dibenzofuran, aza-dibenzoselenophene, aza-triphenylene, imidazole, pyrazole, oxazole, thiazole, isoxazole, isothiazole, triazole, thiadiazole, and oxadiazole. In some embodiments, the acceptor moieties and the donor moieties as described herein can be connected directly, through a conjugated linker, or a non-conjugated linker, such as a sp^3 carbon or silicon atom.

[0426] In some embodiments, the fluorescent material comprises at least one of the chemical moieties selected from the group consisting of:

##STR00997## ##STR00998## ##STR00999## ##STR01000## ##STR01001## ##STR01002##
##STR01003## ##STR01004## ##STR01005##

wherein Y^F, Y^G, Y^H, and Y^I are each independently selected from the group consisting of B, C, Si, Ge, N, P, O, S, Se, C=O, S=O, and SO₂;

wherein X^F and X^G are each independently selected from the group consisting of C and N.

[0427] In some of the above embodiments, any carbon ring atoms up to maximum of a total number of three, together with their substituents, in each phenyl ring of any of above structures can be replaced with N.

f) HBL:

[0428] A hole blocking layer (HBL) may be used to reduce the number of holes and/or excitons that leave the emissive layer. The presence of such a blocking layer in a device may result in substantially higher efficiencies and/or longer lifetime as compared to a similar device lacking a blocking layer. Also, a blocking layer may be used to confine emission to a desired region of an OLED. In some embodiments, the HBL material has a lower HOMO (further away from the vacuum level) and/or higher triplet energy than one or more of the emitters closest to the HBL interface.

[0429] In some embodiments, a compound used in the HBL contains the same molecule or the same functional groups used as host described above.

[0430] In some embodiments, a compound used in the HBL comprises at least one of the following moieties selected from the group consisting of:

##STR01006##

wherein k is an integer from 1 to 20; L¹⁰¹ is another ligand, k' is an integer from 1 to 3.

g) ETL:

[0431] Electron transport layer (ETL) may include a material capable of transporting electrons. Electron transport layer may be intrinsic (undoped), or doped. Doping may be used to enhance conductivity. Examples of the ETL material are not particularly limited, and any metal complexes or organic compounds may be used as long as they are typically used to transport electrons.

[0432] In some embodiments, compound used in ETL comprises at least one of the following moieties in the molecule:

##STR01007##

and fullerenes; wherein k is an integer from 1 to 20, X¹⁰¹ to X¹⁰⁸ is selected from C or N; Z¹⁰¹ is selected from the group consisting of C, N, O, and S.

[0433] In some embodiments, the metal complexes used in ETL contains, but not limit to the following general formula:

##STR01008##

wherein (O—N) or (N—N) is a bidentate ligand, having metal coordinated to atoms O, N or N, N; L¹⁰¹ is another ligand; k' is an integer value from 1 to the maximum number of ligands that may be attached to the metal.

[0434] In some embodiments, the ETL material is selected from the group consisting of anthracene-benzimidazole compounds, aza triphenylene derivatives, anthracene-benzothiazole compounds, metal 8-hydroxyquinolates, metal hydroxybenzoquinolates, bathocuprine compounds,

5-member ring electron deficient heterocycles (e.g., triazole, oxadiazole, imidazole, benzoimidazole), silole compounds, arylborane compounds, fluorinated aromatic compounds, fullerene (e.g., C60), triazine complexes, and Zn (N{circumflex over ()})N complexes.

h) Charge Generation Layer (CGL)

[0435] In tandem or stacked OLEDs, the CGL plays an essential role in the performance, which is composed of an n-doped layer and a p-doped layer for injection of electrons and holes, respectively. Electrons and holes are supplied from the CGL and electrodes. The consumed electrons and holes in the CGL are refilled by the electrons and holes injected from the cathode and anode, respectively; then, the bipolar currents reach a steady state gradually. Typical CGL materials include n and p conductivity dopants used in the transport layers.

[0436] In any compounds disclosed herein, the hydrogen atoms can be partially or fully deuterated. The minimum amount of hydrogen of the compound being deuterated is selected from the group consisting of 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 99%, and 100%. As used herein, percent deuteration has its ordinary meaning and includes the percent of all possible hydrogen and deuterium atoms that are replaced by deuterium atoms. In some embodiments, the deuterium atoms are attached to an aromatic ring. In some embodiments, the deuterium atoms are attached to a saturated carbon atom, such as an alkyl or cycloalkyl carbon atom. In some other embodiments, the deuterium atoms are attached to a heteroatom, such as Si, or Ge atom.

[0437] It is understood that the various embodiments described herein are by way of example only and are not intended to limit the scope of the invention. For example, many of the materials and structures described herein may be substituted with other materials and structures without deviating from the spirit of the invention. The present invention as claimed may therefore include variations from the particular examples and preferred embodiments described herein, as will be apparent to one of skill in the art. It is understood that various theories as to why the invention works are not intended to be limiting.

E. Experimental Data

##STR01009##

1-(3-Methoxynaphthalen-1-yl)-2,2-dimethylpropan-1-one (1)

[0438] In a flow reactor at -25°C . were mixed solutions of 1-bromo-3-methoxynaphthalene (1.00 g, 1 Eq, 4.218 mmol), delivered at 8.5 mL/min and hexyllithium (2.64 mL, 1.760 molar, 1.1 Eq, 4.639 mmol) at -25°C . delivered at 1.75 mL/min, giving an in-reactor residence time of 1 s. The outgoing stream was quenched with pivaloyl chloride (763 mg, 0.8 mL, 1.5 Eq, 6.326 mmol) in a separate reactor loop also at -25°C . going to room temperature, giving a residence time of 2 s. The material was collected into sat. NH_4Cl (30 mL). The mixture was diluted with ethyl acetate (50 mL) and water (20 mL) and the two phases were separated. The aqueous layer was extracted with ethyl acetate (2 $\times$ 40 mL). The combined organic extracts were washed with brine (50 mL), dried over MgSO_4 , filtered and concentrated in vacuo to get the crude. The crude product was purified by chromatography on silica gel (12 g cartridge, 0-70% dichloromethane/iso-hexane, dry load) to afford 1-(3-methoxynaphthalen-1-yl)-2,2-dimethylpropan-1-one (900 mg, 62%) as a colorless solid.

##STR01010##

2-(3-Methoxynaphthalen-1-yl)-3,3-dimethylbutan-2-ol (2)

[0439] To a stirring solution of 1-(3-methoxynaphthalen-1-yl)-2,2-dimethylpropan-1-one (8.500 g, 1 Eq, 35.08 mmol) in tetrahydrofuran (100 mL) at 0°C . was added methyllithium (23.0 mL, 3.100 molar, 2.033 Eq). The resulting reaction was stirred 30 min at 0°C . and then 1 h at rt. After this time the reaction was quenched with ammonium chloride (100 mL, sat sol). and extracted with ethyl acetate (3 $\times$ 50 mL). the combined organic layers were dried with magnesium sulfate, filtered and concentrated under reduced pressure. The crude product was purified by chromatography on silica gel (330 g cartridge, 0-60% dichloromethane/iso-hexane) to afford 2-(3-methoxynaphthalen-1-yl)-3,3-dimethylbutan-2-ol (6.8 g, 26 mmol, 75%) as a colorless solid.

##STR01011##

4-Methoxy-1,1,2,2-tetramethyl-1,2-dihydroacenaphthylene (3)

[0440] To a stirring solution of 2-(3-methoxynaphthalen-1-yl)-3,3-dimethylbutan-2-ol (7.00 g, 1 Eq, 27.09 mmol) in Eaton's reagent (7.7% wt P.sub.2O.sub.5 in MsOH) (210 mL, 1 Eq, 27.09 mmol) was stirred at 40° C. for 2 h. The reaction mixture was added slowly to an ice cold stirring mixture of TBME (300 mL) and 2 M NaOH(aq) (300 mL) such that the temperature was maintained <30° C. The layers were separated, and the organic layer was washed with brine (300 mL), dried over MgSO.sub.4, filtered and concentrated. The crude product was purified by chromatography on silica gel (330 g cartridge, 0-40% dichloromethane/iso-hexane) to afford 4-methoxy-1,1,2,2-tetramethyl-1,2-dihydroacenaphthylene (4.8 g, 20 mmol, 75%).

##STR01012##

1,1,2,2-Tetramethyl-1,2-dihydroacenaphthylene-4-ol (4)

[0441] BBr.sub.3 (43.7 mL, 1.00 molar, 1.5 Eq, 43.69 mmol) was added to a solution of 4-methoxy-1,1,2,2-tetramethyl-1,2-dihydroacenaphthylene (7.00 g, 1 Eq, 29.12 mmol) in dichloromethane (110 mL) at 0° C. The mixture was stirred at RT for 20 hours. The reaction mixture was poured slowly into saturated aqueous NaHCO.sub.3 (300 mL, chilled in an ice-water bath) and extracted with dichloromethane (2×300 mL). The combined organic extracts were dried over magnesium sulfate, filtered and concentrated to leave a brown solid. The crude product was purified by chromatography on silica gel (24 g cartridge, 0-40% ethyl acetate/iso-hexane) to afford 1,1,2,2-tetramethyl-1,2-dihydroacenaphthylene-4-ol (6.500 g, 26 mmol, 89%).

##STR01013##

1,1,2,2-Tetramethyl-1,2-dihydroacenaphthylene-4-yl trifluoromethanesulfonate (5)

[0442] To a solution of 1,1,2,2-tetramethyl-1,2-dihydroacenaphthylene-4-ol (6.50 g, 1 Eq, 28.72 mmol) and DIPEA (5.568 g, 7.50 mL, 1.5 Eq, 43.08 mmol) in dichloromethane (100 mL) was added triflic anhydride (9.72 g, 5.82 mL, 1.2 Eq, 34.46 mmol) at 0° C. The resulting solution was left to warm to room temperature overnight. The reaction was quenched with ammonium chloride (100 mL, sat sol) and the two phases were separated. The aqueous solution was extracted with dichloromethane (2×30 mL). The combined organic solutions were dried with magnesium sulfate, filtered and concentrated under reduced pressure. The crude product was purified by chromatography on silica gel (0-20% ethyl acetate/iso-hexane) to afford 1,1,2,2-tetramethyl-1,2-dihydroacenaphthylene-4-yl trifluoromethanesulfonate (11.00 g, 29 mmol, 98%) as colourless oil.

##STR01014##

4,4,5,5-Tetramethyl-2-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylene-4-yl)-1,3,2-dioxaborolane (6)

[0443] 1,1,2,2-Tetramethyl-1,2-dihydroacenaphthylene-4-yl trifluoromethanesulfonate (4.20 g, 1 Eq, 11.72 mmol) and bis(pinacolato)diboron (3.57 g, 1.2 Eq, 14.06 mmol) were suspended in 2-methyltetrahydrofuran (40 mL), potassium 2-ethylhexanoate (0.5 M in THF) (1.406 mL, 0.50 molar, 0.06 Eq, 703.2 µmol) and potassium carbonate (3.56 g, 2.2 Eq, 25.78 mmol) were added. The reaction mixture was degassed with nitrogen for 15 min. XPhosPd(crotyl)Cl (395 mg, 0.05 Eq, 586.0 µmol) and XPhos (279.3 mg, 0.05 Eq, 586.0 µmol) were added and the reaction mixtures heated to 60° C. (baseplate) overnight. The reaction mixture was quenched with saturated aqueous ammonium chloride (50 mL) and extracted with ethyl acetate (5×50 mL). The combined organic layers were dried with magnesium sulfate, filtered and concentrated under reduced pressure. The crude product was purified by chromatography on silica gel (0-20% ethyl acetate/iso-hexane) to afford 4,4,5,5-tetramethyl-2-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylene-4-yl)-1,3,2-dioxaborolane (2.90 g, 8.4 mmol, 71%).

##STR01015##

7-chloro-8-methyl-1-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylene-4-yl)benzo[4,5]thieno[2,3-c]pyridine (7)

[0444] A 250 mL round bottom flask with a stir bar was charged with 4,7-dichloro-6-methylbenzo[4,5]thieno[3,2-d]pyrimidine (2.527 g, 1 Eq, 9.391 mmol) in 1,4-Dioxane (30 mL) and

water (10 mL). The solution was degassed with nitrogen for 10 minutes followed by the addition of 4,4,5,5-tetramethyl-2-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylen-4-yl)-1,3,2-dioxaborolane (3.0 g, 0.95 Eq, 8.921 mmol), potassium carbonate (2.6 g, 2 Eq, 18.78 mmol), and Bis(triphenylphosphine)palladium(II) chloride (329.6 mg, 0.05 Eq, 469.5 μ mol). The mixture was then heated to 90° C. and stirred at that temperature for 18 hours. The reaction was cooled to room temperature, diluted with dichloromethane (50 mL), the layers separated, and the organic layer was washed with saturated aqueous brine. The aqueous layer was extracted with dichloromethane (3 $\times$ 50 mL). All the organic portions were combined, dried over sodium sulfate, filtered, and the filtrate was concentrated in vacuo. The crude product was purified on a chromatography system, eluting with 0-70% ethyl acetate in hexanes to provide 7-chloro-6-methyl-4-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylen-4-yl)benzo[4,5]thieno[3,2-d]pyrimidine (3.910 g, 93.99% yield) as a white crystallin solid.

8-methyl-7-neopentyl-1-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylen-4-yl)benzo[4,5]thieno[2,3-c]pyridine (8)

[0445] To a 250 mL round bottom flask was added 7-chloro-6-methyl-4-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylen-4-yl)benzo[4,5]thieno[3,2-d]pyrimidine (3.91 g, 1 Eq, 8.826 mmol) in THF (50 mL). The solvent was purged with nitrogen for 10 minutes and then palladium acetate (99.08 mg, 0.05 Eq, 441.3 μ mol) and dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphane (362.3 mg, 0.1 Eq, 882.6 μ mol) were added. The reaction was heated to 35° C. and neopentylzinc(II) bromide (2.865 g, 26.48 mL, 0.500 molar, 1.5 Eq, 13.24 mmol) was added dropwise. Upon complete addition of the zincate, the reaction temperature was increased to 50° C. The reaction was cooled to 0° C. in an ice bath and water (50 mL) was added slowly to quench any excess zincate. The layers were separated, and the aqueous layer extracted with ethyl acetate (3 $\times$ 30 mL). The combined organic layers were combined and dried over sodium sulfate, filtered, and the solvent removed in vacuo to provide 6-methyl-7-neopentyl-4-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylen-4-yl)benzo[4,5]thieno[3,2-d]pyrimidine (4.120 g, 97.51% yield) as an off-white solid.

##STR01016##

Inventive Example 1-Dimer

[0446] To a 250 mL round-bottom flask equipped with a stir bar was added 6-methyl-7-neopentyl-4-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylen-4-yl)benzo[4,5]thieno[3,2-d]pyrimidine (2.600 g, 2 Eq, 5.431 mmol), 2-ethoxy-ethanol (25.000 mL), and DI water (6.2500 mL). The solvent was sparged with nitrogen for 10 minutes before 2,6-dimethylpyridine (1.164 g, 1.26 mL, 4 Eq, 10.86 mmol) and iridium(III) chloride hydrate (810.8 mg, 1 Eq, 2.716 mmol) were added. The vessel was equipped with a reflux condenser and heated at 98° C. for 18 hours. The reaction mixture was cooled to room temperature and methanol (50 mL) was added. The resulting solid was filtered under vacuum to provide di- μ -chloro-tetrakis[8-methyl-7-neopentyl-1-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylen-4-yl)-1'-yl]benzo[4,5]thieno[2,3-c]pyridine-2-yl]diiridium(III) (4.850 g, 2.050 mmol, 75.48%) as a red solid. This material was carried forward to the final step without purification.

Inventive Example 1

[0447] To a 250 mL round bottom flask equipped with a stir bar was added di- μ -chloro-tetrakis[8-methyl-7-neopentyl-1-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylen-4-yl)-1'-yl]benzo[4,5]thieno[2,3-c]pyridine-2-yl]diiridium(III) (4.850 g, 1 Eq, 2.050 mmol) in dichloromethane (10.000 mL) and methanol (10.000 mL). 3,7-diethylnonane-4,6-dione (1.741 g, 2.0 mL, 4 Eq, 8.199 mmol) and powdered potassium carbonate (1.133 g, 4 Eq, 8.199 mmol) were added and the vessel was sealed. The material was heated to 45° C. for 18 hours. Methanol (25 mL) was added to the reaction mixture and the suspension was filtered under reduced pressure. The red solid was collected, and the solid material was purified on a Biotage automated chromatography system, eluting with 0-40% methylene chloride in hexanes to give bis[8-methyl-7-neopentyl-1-

(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylen-4-yl-1'-yl)benzo[4,5]thieno[2,3-c]pyridine-2-yl]-[3,7-diethyl-nonane-4,6-dionato-k2O,O']iridium(III) (1.805 g, 67% yield) as a red solid.

##STR01017##

Inventive Example 2

[0448] In a 40 mL amber vial equipped with a stir bar, a mixture of Inventive compound 2-Cl (700.0 mg, 1 Eq, 728.0 μ mol), potassium phosphate hydrate (502.9 mg, 3.0 Eq, 2.184 mmol), and 4,4,5,5-tetramethyl-2-(1,1,2,2-tetramethyl-1,2-dihydroacenaphthylen-4-yl)-1,3,2-dioxaborolane (734.4 mg, 3.0 Eq, 2.184 mmol) in a 10 to 1 mixture of 1,4-Dioxane (6.618 mL) and Water (661.8 μ L) was sparged with nitrogen for 3 minutes. SPhos Pd Gen2 (62.95 mg, 0.12 Eq, 87.35 μ mol) was then added and solution was sparged with nitrogen for an additional minute. Reaction was heated at 100° C. overnight. The product was purified by column chromatography on silica to afford Inventive example 2 (797 mg, 87% yield).

##STR01018##

9-methoxy-4,4,5,5-tetramethyl-5,7-dihydro-4H-indeno[1,7-bc]carbazole (10)

[0449] Compound 10 could be synthesized starting from 5 via Pd-mediated C—N coupling following by ring closure using the procedures reported in *J. Org. Chem.* 2009, 74, 4490-4498.

##STR01019##

Inventive Example 3—Ligand

[0450] Inventive example 3-ligand could be synthesized using our previously reported method, KR2022128311 A, from 13, via copper-mediated C—O coupling. 13 could be obtained from 11 with Pd-mediated C—N coupling followed by demethylation (US20200140471 A1).

##STR01020##

Inventive Example 3

[0451] Inventive example 3 could be synthesized via the common metalation condition described in KR2024022989 A.

##STR01021##

Inventive Example 5

[0452] Inventive example 5 could be synthesized by using the boronic ester, 6, for Suzuki coupling with Platinum, [2-[3-[3-(2,6-dibromophenyl)-2,3-dihydro-1H-benzimidazol-1-yl- κ C.sup.2]phenoxy- κ C]-9-[4-(1,1-dimethylethyl)-2-pyridinyl- κ N]-9H-carbazolato(4-)- κ C.sup.1]—, (SP-4-4)-(ACI) (Intermediary-Br, CAS: 2448398-24-9) as reported in US20210284672 A1, which is incorporated herein by reference.

##STR01022##

N6-(5-(tert-butyl)-[1,1'-biphenyl]-2-yl)-1,1,3,3-tetramethyl-2,3-dihydro-1H-phenalene-5,6-diamine (11)

[0453] Compound 16 could be synthesized starting from Compound 5 to convert the triflate into amine using the literature procedure, *Org. Lett.* 2005, 7, 1169-1172, followed by bromination (Tetrahedron Lett. 2006, 47, 8693-8697), then Pd-mediated C—N coupling using reported procedures in CN113493448 A and KR2021097470 A.

##STR01023##

Inventive Example 7—Ligand

[0454] Inventive example 7—ligand could be synthesized using our previously reported method, KR2022128311 A, from 18, via copper-mediated C—O coupling, wherein 18 could be obtained by condensation between 17 and 3-Bromo-5-(1,1-dimethylethyl)benzaldehyde.

##STR01024##

Inventive Example 7

[0455] Inventive example 7 could be synthesized via the common metalation condition described in KR2024022989 A.

##STR01025##

Intermediary-Cl for Inventive Example 10

[0456] Intermediary-Cl could be synthesized using our previously reported method, KR2022128311 A, from 19, via copper-mediated C—O coupling followed by platination, wherein compound 19 can be obtained by the method described in KR2024022989 A.

##STR01026##

Inventive Example 10

[0457] Inventive example 10 could be synthesized by using the boronic ester, 6, for Suzuki coupling with Intermediary-Cl using reported condition in US20220396596 A1. All the above referenced publications are incorporated herein by reference.

##STR01027##

TABLE-US-00009 TABLE 1 PL properties of the Inventive Example 1 and comparative example 1. λ_{max} (nm) PLQY Inventive example 1 639 1.03 Comparative example 1 615 1.00

[0458] The photoluminescence (PL) properties of the inventive and comparative examples were measured in 2-MeTHF solutions, and the results are shown in Table 1. The inventive example showed a λ_{max} at 639 nm, which is a 24 nm redshift compared to the comparative example. PLQY of the inventive example was 3% higher than the comparative example. The results suggested that the inventive structure provides a significant redshift while maintaining high PLQY, making these compounds suitable display and lighting applications.

TABLE-US-00010 TABLE 2 PL properties of the Inventive Example 2 and comparative example 2. λ_{max} (nm) PLQY τ Inventive example 2 507 1.04 0.94 Comparative example 2 505 1.00 1.00

[0459] The photoluminescence (PL) properties of the Inventive Example 2 and comparative example 2 were measured PMMA, the results are shown in Table 2. The inventive example showed a slightly redshift by 2 nm compared to the comparative example. PLQY and excited lifetime of the inventive example 2 were 4% higher and 6% shorter than those of the comparative example 2, respectively. The results suggested that the inventive structure 2 provides preferred PL properties in terms of a higher PLQY and a shorter excited lifetime, making these compounds suitable for display and lighting applications. All the improvements shown Table 1 and Table 2 above are beyond any value that could be attributed to experimental error and the observed improvements are significant and unexpected.

[0460] Emission spectra were collected on a Horiba Fluorolog-3 spectrofluorometer equipped with a Synapse Plus CCD detector. All samples were excited at 340 nm. Transient data was measured by time correlated single photon counting (TCSPC) in the Fluorolog-3 using a 335 nm NanoLED pulsed excitation source. PLQY values were measured using a Hamamatsu Quantaaurus-QY Plus UV-NIR absolute PL quantum yield spectrometer with an excitation wavelength of 340 nm. Solutions of 1% emitter with PMMA in toluene were prepared, filtered, and dropcast onto Quartz substrates.

TABLE-US-00011 TABLE 3 DFT calculations T1 HOMO LUMO Structure (nm) (eV) (eV)

Inventive example 3 [01028]  558 -5.01 -1.73 Inventive example 4 [01029]  553 -5.05 -1.72 Inventive example 5 [01030]  533 -5.26 -1.69 Inventive example 6 [01031]  511 -5.13 -1.62 Inventive example 7 [01032]  506 -5.13 -1.61 Inventive example 8 [01033]  528 -5.17 -1.71 Inventive example 9 [01034]  514 -5.16 -1.70 Inventive example 10 [01035]  549 -5.16 -1.73 Inventive example 11 [01036]  535 -5.16 -1.73

[0461] Table 3 summarizes the DFT calculations of T1, HOMO, and LUMO for Inventive examples 3 to 11. The calculated values of T1 show these compounds have the emission in the saturated green to yellowish green region (506 to 558 nm). The HOMO and LUMO levels can be turned within a range of ca. 0.2 eV (-5.01 to -5.26 eV for HOMO and -1.61 to -1.73 eV for LUMO). These properties demonstrate the tunability of Pt tetradentate compounds incorporating the inventive piece to fit the commercial need for OLED applications.

[0462] Calculations were performed using the B3LYP functional with a CEP-31G basis set.

Geometry optimizations were performed in vacuum. Excitation energies were obtained at these optimized geometries using time-dependent density functional theory (TDDFT). A continuum solvent model was applied in the TDDFT calculation to simulate tetrahydrofuran solvent. All calculations were carried out using the program Gaussian.

[0463] The calculations obtained with the above-identified DFT functional set and basis set are theoretical. Computational composite protocols, such as Gaussian with the CEP-31G basis set used herein, rely on the assumption that electronic effects are additive and, therefore, larger basis sets can be used to extrapolate to the complete basis set (CBS) limit. However, when the goal of a study is to understand variations in HOMO, LUMO, S.sub.1, T.sub.1, bond dissociation energies, etc. over a series of structurally-related compounds, the additive effects are expected to be similar. Accordingly, while absolute errors from using the B3LYP may be significant compared to other computational methods, the relative differences between the HOMO, LUMO, S.sub.1, T.sub.1, and bond dissociation energy values calculated with B3LYP protocol are expected to reproduce experiment quite well. See, e.g., Hong et al., *Chem. Mater.* 2016, 28, 5791-98, 5792-93 and Supplemental Information (discussing the reliability of DFT calculations in the context of OLED materials). Moreover, with respect to iridium or platinum complexes that are useful in the OLED art, the data obtained from DFT calculations correlates very well to actual experimental data. See Tavasli et al., *J. Mater Chem.* 2012, 22, 6419-29, 6422 (Table 3) (showing DFT calculations closely correlating with actual data for a variety of emissive complexes); Morello, G. R., *J. Mol. Model.* 2017, 23:174 (studying of a variety of DFT functional sets and basis sets and concluding the combination of B3LYP and CEP-31G is particularly accurate for emissive complexes). [0464] 11. The compound of claim 1, wherein the ligand L.sub.A is selected from the group consisting of L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) and L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein ia is an integer from 1 to 178 and 263 to 290, ib is an integer from 179 to 262 and 291 to 296, R.sup.J is selected from R469 to R774 for L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) and R.sup.J is selected from R1 to R774 for L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) and each of R.sup.K, R.sup.L, and R.sup.M, is independently selected from R1 to R774, wherein each of L.sub.A1(R469)(R1)(R1)(R1) to L.sub.A296(R774)(R774)(R774)(R744) is defined as follows:

TABLE-US-00012 L.sub.A Structure of L.sub.A L.sub.A1(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A1(R469)(R1)(R1)(R1) to L.sub.A1(R774)(R774)(R774)(R774) have the structure [01037]  embedded image L.sub.A2(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A2(R469)(R1)(R1)(R1) to L.sub.A2(R774)(R774)(R774)(R774) have the structure [01038]  embedded image L.sub.A3(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A3(R469)(R1)(R1)(R1) to L.sub.A3(R774)(R774)(R774)(R774) have the structure [01039]  embedded image L.sub.A4(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A4(R469)(R1)(R1)(R1) to L.sub.A4(R774)(R774)(R774)(R774) have the structure [01040]  embedded image L.sub.A5(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A5(R469)(R1)(R1)(R1) to L.sub.A5(R774)(R774)(R774)(R774) have the structure [01041]  embedded image L.sub.A6(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A6(R469)(R1)(R1)(R1) to L.sub.A6(R774)(R774)(R774)(R774) have the structure [01042]  embedded image L.sub.A7(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A7(R469)(R1)(R1)(R1) to L.sub.A7(R774)(R774)(R774)(R774) have the structure [01043]  embedded image L.sub.A8(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A8(R469)(R1)(R1)(R1) to L.sub.A8(R774)(R774)(R774)(R774) have the structure [01044]  embedded image L.sub.A9(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A9(R469)(R1)(R1)(R1) to L.sub.A9(R774)(R774)(R774)(R774) have the structure [01045]  embedded image L.sub.A10(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A10(R469)(R1)(R1)(R1) to L.sub.A10(R774)(R774)(R774)(R774) have the structure [01046]  embedded image L.sub.A11(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A11(R469)(R1)(R1)(R1) to

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

L.sub.A286(R774)(R774)(R774) have the structure [01322]  embedded image
L.sub.A287(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A287(R469)(R1)(R1)(R1) to
L.sub.A287(R774)(R774)(R774)(R774) have the structure [01323]  embedded image
L.sub.A288(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A288(R469)(R1)(R1)(R1) to
L.sub.A288(R774)(R774)(R774)(R774) have the structure [01324]  embedded image
L.sub.A289(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A289(R469)(R1)(R1)(R1) to
L.sub.A289(R774)(R774)(R774)(R774) have the structure [01325]  embedded image
L.sub.A290(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A290(R469)(R1)(R1)(R1) to
L.sub.A290(R774)(R774)(R774)(R774) have the structure [01326]  embedded image
L.sub.A291(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A291(R1)(R1)(R1)(R1) to
L.sub.A291(R774)(R774)(R774)(R774) have the structure [01327]  embedded image
L.sub.A292(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A292(R1)(R1)(R1)(R1) to
L.sub.A292(R774)(R774)(R774)(R774) have the structure [01328]  embedded image
L.sub.A293(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A293(R1)(R1)(R1)(R1) to
L.sub.A293(R774)(R774)(R774)(R774) have the structure [01329]  embedded image
L.sub.A294(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A294(R1)(R1)(R1)(R1) to
L.sub.A294(R774)(R774)(R774)(R774) have the structure [01330]  embedded image
L.sub.A295(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A295(R1)(R1)(R1)(R1) to
L.sub.A295(R774)(R774)(R774)(R774) have the structure [01331]  embedded image
L.sub.A296(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), wherein L.sub.A296(R1)(R1)(R1)(R1) to
L.sub.A296(R774)(R774)(R774)(R774) have the structure [01332]  embedded image [0465]
wherein R1 to R468 have the structures defined in the following LIST 8:

```
##STR01333## ##STR01334## ##STR01335## ##STR01336## ##STR01337## ##STR01338##  
##STR01339## ##STR01340## ##STR01341## ##STR01342## ##STR01343## ##STR01344##  
##STR01345## ##STR01346## ##STR01347## ##STR01348## ##STR01349## ##STR01350##  
##STR01351##  
##STR01352## ##STR01353## ##STR01354## ##STR01355## ##STR01356## ##STR01357##  
##STR01358## ##STR01359## ##STR01360## ##STR01361## ##STR01362## ##STR01363##  
##STR01364## ##STR01365## ##STR01366## ##STR01367## ##STR01368## ##STR01369##  
##STR01370## ##STR01371##  
##STR01372## ##STR01373## ##STR01374## ##STR01375## ##STR01376## ##STR01377##  
##STR01378## ##STR01379## ##STR01380## ##STR01381## ##STR01382## ##STR01383##  
##STR01384## [0466] wherein R469 to R774 have the structures defined in the following LIST 9:  
##STR01385## ##STR01386## ##STR01387## ##STR01388## ##STR01389## ##STR01390##  
##STR01391## ##STR01392## ##STR01393## ##STR01394## ##STR01395## ##STR01396##  
##STR01397## ##STR01398## ##STR01399## ##STR01400##  
##STR01401## ##STR01402## ##STR01403## ##STR01404## ##STR01405## ##STR01406##  
##STR01407## ##STR01408## ##STR01409## ##STR01410## ##STR01411## ##STR01412##  
##STR01413## ##STR01414## ##STR01415## ##STR01416## ##STR01417##  
##STR01418## ##STR01419## ##STR01420## ##STR01421## ##STR01422## ##STR01423##  
##STR01424## ##STR01425## ##STR01426## ##STR01427## ##STR01428## ##STR01429##  
##STR01430## ##STR01431## ##STR01432## ##STR01433## ##STR01434##  
##STR01435## ##STR01436## ##STR01437## ##STR01438## ##STR01439## ##STR01440##  
##STR01441## ##STR01442## ##STR01443## ##STR01444## ##STR01445## ##STR01446##  
##STR01447##
```

[0467] In some embodiments, the ligand L.sub.A is selected from the group consisting of
L.sub.A'ib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) and L.sub.A'ib(R.sup.J)(R.sup.K)(R.sup.L)
(R.sup.M) wherein ia is an integer selected from 1 to 178 and 263 to 290, ib is an integer selected
from 179 to 262 and 291 to 296, RK is selected from R469 to R774 for L.sub.A'291(R.sup.J)
(R.sup.K)(R.sup.L)(R.sup.M), and R.sup.K is selected from R1 to R774 for L.sub.A'ib(R.sup.J)

(R.sup.K)(R.sup.L)(R.sup.M) and each of R.sup.J, R.sup.L, and R.sup.M, is independently selected from R1 to R774, wherein each structure of L.sub.A'ia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) to L.sub.A'179(R774)(R774)(R774)(R774) and L.sub.A'263(R1)(R469)(R1)(R1) to L.sub.A'290(R774)(R774)(R774)(R774) for L.sub.A'ia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), each structure of L.sub.A'179(R1)(R1)(R1)(R1) to L.sub.A'262(R774)(R774)(R774)(R774) and L.sub.A'291(R1)(R1)(R1)(R1) to L.sub.A'296(R774)(R774)(R774)(R774) for L.sub.A'ib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) as defined in LIST 7.

[0468] In some embodiments, the ligand L.sub.A is selected from the group consisting of L.sub.A'ia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) and L.sub.A'ib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) wherein ia is an integer selected from 1 to 178 and 263 to 290, ib is an integer selected from 179 to 262 and 291 to 296, R.sup.L is selected from R469 to R774 for L.sub.A'ia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), and R.sup.L is selected from R1 to R774 for L.sub.A'ib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) and each of R.sup.J, R.sup.K, and R.sup.M, is independently selected from R1 to R774, wherein each structure of L.sub.A''1(R1)(R1)(R469)(R1) to L.sub.A''178(R774)(R774)(R774)(R774) and L.sub.A''263(R1)(R1)(R469)(R1) to L.sub.A''290(R774)(R774)(R774)(R774) for L.sub.A''ia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), each structure of L.sub.A''179(R1)(R1)(R1)(R1) to L.sub.A''262(R774)(R774)(R774)(R774) and L.sub.A''290(R1)(R1)(R1)(R1) to L.sub.A''296(R774)(R774)(R774)(R774) for L.sub.A''ib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) as defined in LIST 7.

[0469] In some embodiments, the ligand L.sub.A is selected from the group consisting of L.sub.A'''ia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) and L.sub.A'''ib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) wherein ia is an integer selected from 1 to 178 and 263 to 290, ib is an integer selected from 179 to 262 and 291 to 296, R.sup.M is selected from R469 to R774 for L.sub.A'''ia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) and R.sup.L is selected from R1 to R774 for L.sub.A'''ib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), and each of R.sup.J, R.sup.K, and R.sup.L, is independently selected from R1 to R774, wherein each structure of L.sub.A'''1(R1)(R1)(R1)(R469) to L.sub.A'''178(R774)(R774)(R774)(R774) and L.sub.A'''263(R1)(R1)(R1)(R469) to L.sub.A'''290(R774)(R774)(R774)(R774) for L.sub.A'''ia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), each structure of L.sub.A'''179(R1)(R1)(R1)(R1) to L.sub.A'''262(R774)(R774)(R774)(R774) and L.sub.A'''291(R1)(R1)(R1)(R1) to L.sub.A'''296(R774)(R774)(R774)(R774) for L.sub.A'''ib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) as defined in LIST 7.

[0470] In some embodiments, wherein the compound has a formula of M(L.sub.A).sub.p(L.sub.B).sub.q(L.sub.C).sub.r wherein L.sub.B and L.sub.C are each a bidentate ligand; and wherein p is 1, 2, or 3; q is 0, 1, or 2; r is 0, 1, or 2; and p+q+r is the oxidation state of the metal M.

[0471] In some embodiments, the compound has a formula selected from the group consisting of Ir(L.sub.A).sub.3, Ir(L.sub.A)(L.sub.B).sub.2, Ir(L.sub.A).sub.2(L.sub.B), Ir(L.sub.A).sub.2(L.sub.C), and Ir(L.sub.A)(L.sub.B)(L.sub.C); and wherein L.sub.A, L.sub.B, and L.sub.C are different from each other.

[0472] In some embodiments, L.sub.B is a substituted or unsubstituted phenylpyridine, and L.sub.C is a substituted or unsubstituted acetylacetonate.

[0473] In some embodiments, the compound has a formula of Pt(L.sub.A)(L.sub.B); and wherein L.sub.A and L.sub.B can be same or different. In some embodiments, L.sub.A and L.sub.B are connected to form a tetradentate ligand.

[0474] In some embodiments, L.sub.B and L.sub.C are each independently selected from the group consisting of the structures of the following LIST 10:

##STR01448## ##STR01449## ##STR01450## ##STR01451##

[0475] wherein: [0476] T is selected from the group consisting of B, Al, Ga, and In; [0477] K.sup.1' is selected from the group consisting of a single bond, 0, 5, NR.sub.e, PR.sub.e, BR.sub.e, CR.sub.eR.sub.f, and SiR.sub.eR.sub.f; [0478] each of Y.sup.1 to Y.sup.13 is independently

selected from the group consisting of C and N; [0479] Y^{sup.1} is selected from the group consisting of BR.sub.e, BR.sub.eR.sub.f, NR.sub.e, PR.sub.e, P(O)R.sub.e, O, S, Se, C=O, C=S, C=Se, C=NR.sub.e, C=CR.sub.eR.sub.f, S=O, SO.sub.2, CR.sub.eR.sub.f, SiR.sub.eR.sub.f, and GeR.sub.eR.sub.f; [0480] R.sub.e and R.sub.f can be fused or joined to form a ring; [0481] each R.sub.a, R.sub.b, R.sub.c, and R.sub.d independently represents from mono to the maximum allowed number of substitutions, or no substitution; [0482] each of R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1, R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sub.e, and R.sub.f is independently a hydrogen or a substituent selected from the group consisting of the General Substituents defined herein; and [0483] any two substituents of R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1, R.sub.a, R.sub.b, R.sub.c, and R.sub.d can be fused or joined to form a ring or form a multidentate ligand.

[0484] In some embodiments when a structure is selected from LIST 10, at least one of R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1, R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sub.e, or R.sub.f is or comprises Formula I, or is or comprises a structure selected from a structure form LIST 1 or LIST 2.

[0485] In some embodiments, L.sub.B and L.sub.C are each independently selected from the group consisting of the structures of the following List 11:

##STR01452## ##STR01453## ##STR01454## ##STR01455## ##STR01456## ##STR01457##
##STR01458## ##STR01459## ##STR01460## ##STR01461## ##STR01462## ##STR01463##
##STR01464## ##STR01465## ##STR01466## ##STR01467##

[0486] wherein: [0487] R.sub.a', R.sub.b', R.sub.c', R.sub.d', and R.sub.e' each independently represents zero, mono, or up to a maximum allowed number of substitution to its associated ring;

[0488] R.sub.a', R.sub.b', R.sub.c', R.sub.d', and R.sub.e' each independently hydrogen or a substituent selected from the group consisting of the General Substituents defined herein; and

[0489] two substituents of R.sub.a', R.sub.b', R.sub.c', R.sub.d', and R.sub.e' can be fused or joined to form a ring or form a multidentate ligand.

[0490] In some embodiments when a structure is selected from LIST 11, at least one of R.sub.a', R.sub.b', R.sub.c', R.sub.d', or R.sub.e' is or comprises Formula I, or is or comprises a structure selected from a structure form LIST 1 or LIST 2.

[0491] In some embodiments, L.sub.B comprises a structure of
##STR01468##

wherein the variables are the same as previously defined. In some embodiments, each of Y^{sup.1} to Y^{sup.4} is independently carbon. In some embodiments, at least one of Y^{sup.1} to Y^{sup.4} is N. In some embodiments, exactly one of Y^{sup.1} to Y^{sup.4} is N. In some embodiments, Y^{sup.1} is N. In some embodiments, Y^{sup.2} is N. In some embodiments, Y^{sup.3} is N. In some embodiments, Y^{sup.4} is N. In some embodiments, at least one of R.sub.a is a tertiary alkyl, silyl or germyl. In some embodiments, at least one of R.sub.a is a tertiary alkyl. In some embodiments, Y^{sup.3} is C and the R.sub.a attached thereto is a tertiary alkyl, silyl or germyl. In some embodiments, Y^{sup.1} to Y^{sup.3} is C, Y^{sup.4} is N, and the R.sub.a attached to Y^{sup.3} is a tertiary alkyl, silyl or germyl. In some embodiments, Y^{sup.1} to Y^{sup.3} is C, Y^{sup.4} is N, and the R.sub.a attached to Y^{sup.2} is a tertiary alkyl, silyl or germyl. In some embodiments, at least one of R.sub.b is a tertiary alkyl, silyl, or germyl. In some embodiments, the tertiary alkyl is tert-butyl. In some embodiments, at least one pair of R.sub.a and R.sub.b are joined or fused to form a ring.

[0492] In some embodiments, the compound has formula Ir(L.sub.A).sub.3, formula Ir(L.sub.A)(L.sub.Bk).sub.2, formula Ir(L.sub.A).sub.2(L.sub.Bk), formula Ir(L.sub.A).sub.2(L.sub.Cj-I), or Ir(L.sub.A).sub.2(L.sub.Cj-II), [0493] wherein L.sub.A is according to any one of claims 1-95, including L.sub.Aia(R^{sup.J})(R^{sup.K})(R^{sup.L})(R^{sup.M}), L.sub.Aib(R^{sup.J})(R^{sup.K})(R^{sup.L})(R^{sup.M}), L.sub.A'ia(R^{sup.J})(R^{sup.K})(R^{sup.L})(R^{sup.M}), L.sub.A'ib(R^{sup.J})(R^{sup.K})(R^{sup.L})(R^{sup.M}), L.sub.A'''ia(R^{sup.J})(R^{sup.K})(R^{sup.L})(R^{sup.M}), L.sub.A''ib(R^{sup.J})(R^{sup.K})(R^{sup.L})(R^{sup.M}), L.sub.A'''ia(R^{sup.J})(R^{sup.K})(R^{sup.L})(R^{sup.M}), and L.sub.A''ib(R^{sup.J})(R^{sup.K})(R^{sup.L})(R^{sup.M}); [0494] wherein k is an integer from 1 to 541, and each

L.sub.Bk has the structure defined in the following LIST 12:

##STR01469## ##STR01470## ##STR01471## ##STR01472## ##STR01473## ##STR01474##
##STR01475## ##STR01476## ##STR01477## ##STR01478## ##STR01479## ##STR01480##
##STR01481## ##STR01482## ##STR01483## ##STR01484## ##STR01485## ##STR01486##
##STR01487## ##STR01488## ##STR01489## ##STR01490## ##STR01491## ##STR01492##
##STR01493## ##STR01494## ##STR01495## ##STR01496## ##STR01497## ##STR01498##
##STR01499## ##STR01500## ##STR01501## ##STR01502## ##STR01503## ##STR01504##
##STR01505## ##STR01506## ##STR01507## ##STR01508## ##STR01509## ##STR01510##
##STR01511## ##STR01512## ##STR01513## ##STR01514## ##STR01515## ##STR01516##
##STR01517## ##STR01518## ##STR01519## ##STR01520## ##STR01521## ##STR01522##
##STR01523## ##STR01524## ##STR01525## ##STR01526## ##STR01527## ##STR01528##
##STR01529## ##STR01530## ##STR01531## ##STR01532## ##STR01533## ##STR01534##
##STR01535## ##STR01536## ##STR01537## ##STR01538## ##STR01539## ##STR01540##
##STR01541## ##STR01542## ##STR01543## ##STR01544## ##STR01545## ##STR01546##
##STR01547## ##STR01548## ##STR01549## ##STR01550## ##STR01551## ##STR01552##
##STR01553## ##STR01554## ##STR01555## ##STR01556## ##STR01557## ##STR01558##
##STR01559## ##STR01560## ##STR01561## ##STR01562## ##STR01563## ##STR01564##
##STR01565## ##STR01566## ##STR01567## ##STR01568## ##STR01569## ##STR01570##
##STR01571## ##STR01572## ##STR01573## ##STR01574## ##STR01575## ##STR01576##
##STR01577## ##STR01578## ##STR01579##

##STR01580## ##STR01581## ##STR01582## ##STR01583## ##STR01584## ##STR01585##

wherein each L.sub.Cj-I has a structure based on formula

##STR01586##

and

each L.sub.Cj-II has a structure based on formula

##STR01587##

wherein for each L.sub.Cj in L.sub.Cj-I and L.sub.Cj-II, R.sup.201 and R.sup.202 are defined in the following LIST 13:

TABLE-US-00013 L.sub.Cj R.sup.201 R.sup.202 L.sub.C1 R.sup.D1 R.sup.D1 L.sub.C2 R.sup.D2
R.sup.D2 L.sub.C3 R.sup.D3 R.sup.D3 L.sub.C4 R.sup.D4 R.sup.D4 L.sub.C5 R.sup.D5 R.sup.D5
L.sub.C6 R.sup.D6 R.sup.D6 L.sub.C7 R.sup.D7 R.sup.D7 L.sub.C8 R.sup.D8 R.sup.D8 L.sub.C9
R.sup.D9 R.sup.D9 L.sub.C10 R.sup.D10 R.sup.D10 L.sub.C11 R.sup.D11 R.sup.D11 L.sub.C12
R.sup.D12 R.sup.D12 L.sub.C13 R.sup.D13 R.sup.D13 L.sub.C14 R.sup.D14 R.sup.D14
L.sub.C15 R.sup.D15 R.sup.D15 L.sub.C16 R.sup.D16 R.sup.D16 L.sub.C17 R.sup.D17
R.sup.D17 L.sub.C18 R.sup.D18 R.sup.D18 L.sub.C19 R.sup.D19 R.sup.D19 L.sub.C20
R.sup.D20 R.sup.D20 L.sub.C21 R.sup.D21 R.sup.D21 L.sub.C22 R.sup.D22 R.sup.D22
L.sub.C23 R.sup.D23 R.sup.D23 L.sub.C24 R.sup.D24 R.sup.D24 L.sub.C25 R.sup.D25
R.sup.D25 L.sub.C26 R.sup.D26 R.sup.D26 L.sub.C27 R.sup.D27 R.sup.D27 L.sub.C28
R.sup.D28 R.sup.D28 L.sub.C29 R.sup.D29 R.sup.D29 L.sub.C30 R.sup.D30 R.sup.D30
L.sub.C31 R.sup.D31 R.sup.D31 L.sub.C32 R.sup.D32 R.sup.D32 L.sub.C33 R.sup.D33
R.sup.D33 L.sub.C34 R.sup.D34 R.sup.D34 L.sub.C35 R.sup.D35 R.sup.D35 L.sub.C36
R.sup.D36 R.sup.D36 L.sub.C37 R.sup.D37 R.sup.D37 L.sub.C38 R.sup.D38 R.sup.D38
L.sub.C39 R.sup.D39 R.sup.D39 L.sub.C40 R.sup.D40 R.sup.D40 L.sub.C41 R.sup.D41
R.sup.D41 L.sub.C42 R.sup.D42 R.sup.D42 L.sub.C43 R.sup.D43 R.sup.D43 L.sub.C44
R.sup.D44 R.sup.D44 L.sub.C45 R.sup.D45 R.sup.D45 L.sub.C46 R.sup.D46 R.sup.D46
L.sub.C47 R.sup.D47 R.sup.D47 L.sub.C48 R.sup.D48 R.sup.D48 L.sub.C49 R.sup.D49
R.sup.D49 L.sub.C50 R.sup.D50 R.sup.D50 L.sub.C51 R.sup.D51 R.sup.D51 L.sub.C52
R.sup.D52 R.sup.D52 L.sub.C53 R.sup.D53 R.sup.D53 L.sub.C54 R.sup.D54 R.sup.D54
L.sub.C55 R.sup.D55 R.sup.D55 L.sub.C56 R.sup.D56 R.sup.D56 L.sub.C57 R.sup.D57
R.sup.D57 L.sub.C58 R.sup.D58 R.sup.D58 L.sub.C59 R.sup.D59 R.sup.D59 L.sub.C60

[illegible]

R.sup.D3 L.sub.C194 R.sup.D1 R.sup.D4 L.sub.C195 R.sup.D1 R.sup.D5 L.sub.C196 R.sup.D1
R.sup.D9 L.sub.C197 R.sup.D1 R.sup.D10 L.sub.C198 R.sup.D1 R.sup.D17 L.sub.C199 R.sup.D1
R.sup.D18 L.sub.C200 R.sup.D1 R.sup.D20 L.sub.C201 R.sup.D1 R.sup.D22 L.sub.C202
R.sup.D1 R.sup.D37 L.sub.C203 R.sup.D1 R.sup.D40 L.sub.C204 R.sup.D1 R.sup.D41
L.sub.C205 R.sup.D1 R.sup.D42 L.sub.C206 R.sup.D1 R.sup.D43 L.sub.C207 R.sup.D1
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L.sub.C213 R.sup.D1 R.sup.D59 L.sub.C214 R.sup.D1 R.sup.D78 L.sub.C215 R.sup.D1
R.sup.D79 L.sub.C216 R.sup.D1 R.sup.D81 L.sub.C217 R.sup.D1 R.sup.D87 L.sub.C218
R.sup.D1 R.sup.D88 L.sub.C219 R.sup.D1 R.sup.D89 L.sub.C220 R.sup.D1 R.sup.D93
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L.sub.C248 R.sup.D4 R.sup.D22 L.sub.C249 R.sup.D4 R.sup.D37 L.sub.C250 R.sup.D4
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L.sub.C264 R.sup.D4 R.sup.D87 L.sub.C265 R.sup.D4 R.sup.D88 L.sub.C266 R.sup.D4
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L.sub.C291 R.sup.D9 R.sup.D17 L.sub.C292 R.sup.D9 R.sup.D18 L.sub.C293 R.sup.D9
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R.sup.D9 R.sup.D149 L.sub.C329 R.sup.D9 R.sup.D151 L.sub.C330 R.sup.D9 R.sup.D154
L.sub.C331 R.sup.D9 R.sup.D155 L.sub.C332 R.sup.D9 R.sup.D161 L.sub.C333 R.sup.D9
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L.sub.C379 R.sup.D17 R.sup.D3 L.sub.C380 R.sup.D17 R.sup.D5 L.sub.C381 R.sup.D17
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R.sup.D17 R.sup.D37 L.sub.C385 R.sup.D17 R.sup.D40 L.sub.C386 R.sup.D17 R.sup.D41
L.sub.C387 R.sup.D17 R.sup.D42 L.sub.C388 R.sup.D17 R.sup.D43 L.sub.C389 R.sup.D17
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R.sup.D17 R.sup.D54 L.sub.C393 R.sup.D17 R.sup.D55 L.sub.C394 R.sup.D17 R.sup.D58
L.sub.C395 R.sup.D17 R.sup.D59 L.sub.C396 R.sup.D17 R.sup.D78 L.sub.C397 R.sup.D17
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R.sup.D118 L.sub.C406 R.sup.D17 R.sup.D119 L.sub.C407 R.sup.D17 R.sup.D120 L.sub.C408
R.sup.D17 R.sup.D133 L.sub.C409 R.sup.D17 R.sup.D134 L.sub.C410 R.sup.D17 R.sup.D135
L.sub.C411 R.sup.D17 R.sup.D136 L.sub.C412 R.sup.D17 R.sup.D143 L.sub.C413 R.sup.D17
R.sup.D144 L.sub.C414 R.sup.D17 R.sup.D145 L.sub.C415 R.sup.D17 R.sup.D146 L.sub.C416
R.sup.D17 R.sup.D147 L.sub.C417 R.sup.D17 R.sup.D149 L.sub.C418 R.sup.D17 R.sup.D151
L.sub.C419 R.sup.D17 R.sup.D154 L.sub.C420 R.sup.D17 R.sup.D155 L.sub.C421 R.sup.D17
R.sup.D161 L.sub.C422 R.sup.D17 R.sup.D175 L.sub.C423 R.sup.D50 R.sup.D3 L.sub.C424
R.sup.D50 R.sup.D5 L.sub.C425 R.sup.D50 R.sup.D18 L.sub.C426 R.sup.D50 R.sup.D20
L.sub.C427 R.sup.D50 R.sup.D22 L.sub.C428 R.sup.D50 R.sup.D37 L.sub.C429 R.sup.D50
R.sup.D40 L.sub.C430 R.sup.D50 R.sup.D41 L.sub.C431 R.sup.D50 R.sup.D42 L.sub.C432
R.sup.D50 R.sup.D43 L.sub.C433 R.sup.D50 R.sup.D48 L.sub.C434 R.sup.D50 R.sup.D49
L.sub.C435 R.sup.D50 R.sup.D54 L.sub.C436 R.sup.D50 R.sup.D55 L.sub.C437 R.sup.D50
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L.sub.C507 R.sup.D55 R.sup.D175 L.sub.C508 R.sup.D116 R.sup.D3 L.sub.C509 R.sup.D116
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R.sup.D116 R.sup.D20 L.sub.C513 R.sup.D116 R.sup.D22 L.sub.C514 R.sup.D116 R.sup.D37
L.sub.C515 R.sup.D116 R.sup.D40 L.sub.C516 R.sup.D116 R.sup.D41 L.sub.C517 R.sup.D116
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L.sub.C563 R.sup.D143 R.sup.D54 L.sub.C564 R.sup.D143 R.sup.D58 L.sub.C565 R.sup.D143
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L.sub.C571 R.sup.D143 R.sup.D89 L.sub.C572 R.sup.D143 R.sup.D93 L.sub.C573 R.sup.D143
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L.sub.C619 R.sup.D144 R.sup.D120 L.sub.C620 R.sup.D144 R.sup.D133 L.sub.C621 R.sup.D144
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R.sup.D220 L.sub.C1121 R.sup.D9 R.sup.D221 L.sub.C1122 R.sup.D9 R.sup.D222 L.sub.C1123
R.sup.D9 R.sup.D223 L.sub.C1124 R.sup.D9 R.sup.D224 L.sub.C1125 R.sup.D9 R.sup.D225
L.sub.C1126 R.sup.D9 R.sup.D226 L.sub.C1127 R.sup.D9 R.sup.D227 L.sub.C1128 R.sup.D9
R.sup.D228 L.sub.C112 R.sup.D9 R.sup.D229 L.sub.C1130 R.sup.D9 R.sup.D230 L.sub.C1131
R.sup.D9 R.sup.D231 L.sub.C1132 R.sup.D9 R.sup.D232 L.sub.C1133 R.sup.D9 R.sup.D233
L.sub.C1134 R.sup.D9 R.sup.D234 L.sub.C1135 R.sup.D9 R.sup.D235 L.sub.C1136 R.sup.D9
R.sup.D236 L.sub.C1137 R.sup.D9 R.sup.D237 L.sub.C1138 R.sup.D9 R.sup.D238 L.sub.C1139
R.sup.D9 R.sup.D239 L.sub.C1140 R.sup.D9 R.sup.D240 L.sub.C1141 R.sup.D9 R.sup.D241
L.sub.C1142 R.sup.D9 R.sup.D242 L.sub.C1143 R.sup.D9 R.sup.D243 L.sub.C1144 R.sup.D9
R.sup.D244 L.sub.C1145 R.sup.D9 R.sup.D245 L.sub.C1146 R.sup.D9 R.sup.D246 L.sub.C1147
R.sup.D168 R.sup.D193 L.sub.C1148 R.sup.D168 R.sup.D194 L.sub.C1149 R.sup.D168
R.sup.D195 L.sub.C1150 R.sup.D168 R.sup.D196 L.sub.C1151 R.sup.D168 R.sup.D197
L.sub.C1152 R.sup.D168 R.sup.D198 L.sub.C1153 R.sup.D168 R.sup.D199 L.sub.C1154
R.sup.D168 R.sup.D200 L.sub.C1155 R.sup.D168 R.sup.D201 L.sub.C1156 R.sup.D168
R.sup.D202 L.sub.C1157 R.sup.D168 R.sup.D203 L.sub.C1158 R.sup.D168 R.sup.D204
L.sub.C1159 R.sup.D168 R.sup.D205 L.sub.C1160 R.sup.D168 R.sup.D206 L.sub.C1161
R.sup.D168 R.sup.D207 L.sub.C1162 R.sup.D168 R.sup.D208 L.sub.C1163 R.sup.D168
R.sup.D209 L.sub.C1164 R.sup.D168 R.sup.D210 L.sub.C1165 R.sup.D168 R.sup.D211
L.sub.C1166 R.sup.D168 R.sup.D212 L.sub.C1167 R.sup.D168 R.sup.D213 L.sub.C1168
R.sup.D168 R.sup.D214 L.sub.C1169 R.sup.D168 R.sup.D215 L.sub.C1170 R.sup.D168
R.sup.D216 L.sub.C1171 R.sup.D168 R.sup.D217 L.sub.C1172 R.sup.D168 R.sup.D218
L.sub.C1173 R.sup.D168 R.sup.D219 L.sub.C1174 R.sup.D168 R.sup.D220 L.sub.C1175
R.sup.D168 R.sup.D221 L.sub.C1176 R.sup.D168 R.sup.D222 L.sub.C1177 R.sup.D168
R.sup.D223 L.sub.C1178 R.sup.D168 R.sup.D224 L.sub.C1179 R.sup.D168 R.sup.D225
L.sub.C1180 R.sup.D168 R.sup.D226 L.sub.C1181 R.sup.D168 R.sup.D227 L.sub.C1182
R.sup.D168 R.sup.D228 L.sub.C1183 R.sup.D168 R.sup.D229 L.sub.C1184 R.sup.D168
R.sup.D230 L.sub.C1185 R.sup.D168 R.sup.D231 L.sub.C1186 R.sup.D168 R.sup.D232
L.sub.C1187 R.sup.D168 R.sup.D233 L.sub.C1188 R.sup.D168 R.sup.D234 L.sub.C1189
R.sup.D168 R.sup.D235 L.sub.C1190 R.sup.D168 R.sup.D236 L.sub.C1191 R.sup.D168
R.sup.D237 L.sub.C1192 R.sup.D168 R.sup.D238 L.sub.C1193 R.sup.D168 R.sup.D239
L.sub.C1194 R.sup.D168 R.sup.D240 L.sub.C1195 R.sup.D168 R.sup.D241 L.sub.C1196
R.sup.D168 R.sup.D242 L.sub.C1197 R.sup.D168 R.sup.D243 L.sub.C1198 R.sup.D168
R.sup.D244 L.sub.C1199 R.sup.D168 R.sup.D245 L.sub.C1200 R.sup.D168 R.sup.D246
L.sub.C1201 R.sup.D10 R.sup.D193 L.sub.C1202 R.sup.D10 R.sup.D194 L.sub.C1203 R.sup.D10
R.sup.D195 L.sub.C1204 R.sup.D10 R.sup.D196 L.sub.C1205 R.sup.D10 R.sup.D197
L.sub.C1206 R.sup.D10 R.sup.D198 L.sub.C1207 R.sup.D10 R.sup.D199 L.sub.C1208 R.sup.D10
R.sup.D200 L.sub.C1209 R.sup.D10 R.sup.D201 L.sub.C1210 R.sup.D10 R.sup.D202
L.sub.C1211 R.sup.D10 R.sup.D203 L.sub.C1212 R.sup.D10 R.sup.D204 L.sub.C1213 R.sup.D10
R.sup.D205 L.sub.C1214 R.sup.D10 R.sup.D206 L.sub.C1215 R.sup.D10 R.sup.D207
L.sub.C1216 R.sup.D10 R.sup.D208 L.sub.C1217 R.sup.D10 R.sup.D209 L.sub.C1218 R.sup.D10
R.sup.D210 L.sub.C1219 R.sup.D10 R.sup.D211 L.sub.C1220 R.sup.D10 R.sup.D212
L.sub.C1221 R.sup.D10 R.sup.D213 L.sub.C1222 R.sup.D10 R.sup.D214 L.sub.C1223 R.sup.D10
R.sup.D215 L.sub.C1224 R.sup.D10 R.sup.D216 L.sub.C1225 R.sup.D10 R.sup.D217
L.sub.C1226 R.sup.D10 R.sup.D218 L.sub.C1227 R.sup.D10 R.sup.D219 L.sub.C1228 R.sup.D10
R.sup.D220 L.sub.C1229 R.sup.D10 R.sup.D221 L.sub.C1230 R.sup.D10 R.sup.D222
L.sub.C1231 R.sup.D10 R.sup.D223 L.sub.C1232 R.sup.D10 R.sup.D224 L.sub.C1233 R.sup.D10
R.sup.D225 L.sub.C1234 R.sup.D10 R.sup.D226 L.sub.C1235 R.sup.D10 R.sup.D227
L.sub.C1236 R.sup.D10 R.sup.D228 L.sub.C1237 R.sup.D10 R.sup.D229 L.sub.C1238 R.sup.D10
R.sup.D230 L.sub.C1239 R.sup.D10 R.sup.D231 L.sub.C1240 R.sup.D10 R.sup.D232
L.sub.C1241 R.sup.D10 R.sup.D233 L.sub.C1242 R.sup.D10 R.sup.D234 L.sub.C1243 R.sup.D10

[illegible]

R.sup.D198 L.sub.C1369 R.sup.D143 R.sup.D199 L.sub.C1370 R.sup.D143 R.sup.D200 L.sub.C1371 R.sup.D143 R.sup.D201 L.sub.C1372 R.sup.D143 R.sup.D202 L.sub.C1373 R.sup.D143 R.sup.D203 L.sub.C1374 R.sup.D143 R.sup.D204 L.sub.C1375 R.sup.D143 R.sup.D205 L.sub.C1376 R.sup.D143 R.sup.D206 L.sub.C1377 R.sup.D143 R.sup.D207 L.sub.C1378 R.sup.D143 R.sup.D208 L.sub.C1379 R.sup.D143 R.sup.D209 L.sub.C1380 R.sup.D143 R.sup.D210 L.sub.C1381 R.sup.D143 R.sup.D211 L.sub.C1382 R.sup.D143 R.sup.D212 L.sub.C1383 R.sup.D143 R.sup.D213 L.sub.C1384 R.sup.D143 R.sup.D214 L.sub.C1385 R.sup.D143 R.sup.D215 L.sub.C1386 R.sup.D143 R.sup.D216 L.sub.C1387 R.sup.D143 R.sup.D217 L.sub.C1388 R.sup.D143 R.sup.D218 L.sub.C1389 R.sup.D143 R.sup.D219 L.sub.C1390 R.sup.D143 R.sup.D220 L.sub.C1391 R.sup.D143 R.sup.D221 L.sub.C1392 R.sup.D143 R.sup.D222 L.sub.C1393 R.sup.D143 R.sup.D223 L.sub.C1394 R.sup.D143 R.sup.D224 L.sub.C1395 R.sup.D143 R.sup.D225 L.sub.C1396 R.sup.D143 R.sup.D226 L.sub.C1397 R.sup.D143 R.sup.D227 L.sub.C1398 R.sup.D143 R.sup.D228 L.sub.C1399 R.sup.D143 R.sup.D229 L.sub.C1400 R.sup.D143 R.sup.D230 L.sub.C1401 R.sup.D143 R.sup.D231 L.sub.C1402 R.sup.D143 R.sup.D232 L.sub.C1403 R.sup.D143 R.sup.D233 L.sub.C1404 R.sup.D143 R.sup.D234 L.sub.C1405 R.sup.D143 R.sup.D235 L.sub.C1406 R.sup.D143 R.sup.D236 L.sub.C1407 R.sup.D143 R.sup.D237 L.sub.C1408 R.sup.D143 R.sup.D238 L.sub.C1409 R.sup.D143 R.sup.D239 L.sub.C1410 R.sup.D143 R.sup.D240 L.sub.C1411 R.sup.D143 R.sup.D241 L.sub.C1412 R.sup.D143 R.sup.D242 L.sub.C1413 R.sup.D143 R.sup.D243 L.sub.C1414 R.sup.D143 R.sup.D244 L.sub.C1415 R.sup.D143 R.sup.D245 L.sub.C1416 R.sup.D143 R.sup.D246 [0495] wherein R.sup.D1 to R.sup.D216 have the structures defined in the following LIST 14:

##STR01588## ##STR01589## ##STR01590## ##STR01591## ##STR01592## ##STR01593##
##STR01594## ##STR01595## ##STR01596## ##STR01597## ##STR01598## ##STR01599##
##STR01600## ##STR01601## ##STR01602## ##STR01603## ##STR01604## ##STR01605##
##STR01606## ##STR01607## ##STR01608## ##STR01609## ##STR01610## ##STR01611##
##STR01612##

[0496] In some embodiments, the compound is selected from the group consisting of only those compounds whose L.sub.Bk corresponds to one of the following: L.sub.B1, L.sub.B30, L.sub.B31, L.sub.B109, L.sub.B110, L.sub.B112, L.sub.B113, L.sub.B114, L.sub.B125, L.sub.B127, L.sub.B138, L.sub.B140, L.sub.B149, L.sub.B150, L.sub.B170, L.sub.B171, L.sub.B172, L.sub.B174, L.sub.B208, L.sub.B241, L.sub.B312, L.sub.B315, L.sub.B356, L.sub.B36, L.sub.B371, L.sub.B382, L.sub.B439, L.sub.B440, L.sub.B455, L.sub.B456, L.sub.B457, L.sub.B458, L.sub.B461, L.sub.B462, L.sub.B463, L.sub.B469, and L.sub.B476.

[0497] In some embodiments, compound is selected from the group consisting of only those compounds whose L.sub.Bk corresponds to one of the following: L.sub.B1, L.sub.B30, L.sub.B31, L.sub.B125, L.sub.B138, L.sub.B171, L.sub.B172, L.sub.B356, L.sub.B357, L.sub.B367, L.sub.B371, L.sub.B382, L.sub.B455, and L.sub.B456.

[0498] In some embodiments, the compound is selected from the group consisting of only those compounds having L.sub.Cj-I or L.sub.Cj-II ligand whose corresponding R.sup.201 and R.sup.202 are defined to be one of the following structures: R.sup.D1, R.sup.D3, R.sup.D4, R.sup.D5, R.sup.D9, R.sup.D10, R.sup.D17, R.sup.D18, R.sup.D20, R.sup.D22, R.sup.D37, R.sup.D40, R.sup.D41, R.sup.D42, R.sup.D43, R.sup.D48, R.sup.D49, R.sup.D50, R.sup.D54, R.sup.D55, R.sup.D58, R.sup.D59, R.sup.D78, R.sup.D79, R.sup.D81, R.sup.D87, R.sup.D88, R.sup.D89, R.sup.D93, R.sup.D116, R.sup.D117, R.sup.D118, R.sup.D119, R.sup.D120, R.sup.D133, R.sup.D134, R.sup.D135, R.sup.D136, R.sup.D143, R.sup.D144, R.sup.D145, R.sup.D146, R.sup.D147, R.sup.D149, R.sup.D151, R.sup.D154, R.sup.D155, R.sup.D161, R.sup.D175, R.sup.D190, R.sup.D193, R.sup.D200, R.sup.D201, R.sup.D206, R.sup.D210, R.sup.D214, R.sup.D215, R.sup.D216, R.sup.D218, R.sup.D219, R.sup.D220, R.sup.D227, R.sup.D237, R.sup.D241, R.sup.D242, R.sup.D245, and R.sup.D246.

[0499] In some embodiments, the compound is selected from the group consisting of only those compounds having L.sub.Cj-I or L.sub.Cj-II ligand whose corresponding R.sup.201 and R.sup.202 are defined to be one of selected from the following structures: R.sup.D1, R.sup.D3, R.sup.D4, R.sup.D5, R.sup.D9, R.sup.D10, R.sup.D17, R.sup.D22, R.sup.D43, R.sup.D50, R.sup.D78, R.sup.D116, R.sup.D118, R.sup.D133, R.sup.D134, R.sup.D135, R.sup.D136, R.sup.D143, R.sup.D144, R.sup.D145, R.sup.D146, R.sup.D149, R.sup.D151, R.sup.D154, R.sup.D155, R.sup.D190, R.sup.D193, R.sup.D200, R.sup.D201, R.sup.D206, R.sup.D210, R.sup.D214, R.sup.D215, R.sup.D216, R.sup.D218, R.sup.D219, R.sup.D220, R.sup.D227, R.sup.D237, R.sup.D241, R.sup.D242, R.sup.D245, and R.sup.D246.

[0500] In some embodiments, the compound is selected from the group consisting of only those compounds having one of the structures of the following LIST 15 for the L.sub.Cj-I ligand:

##STR01613## ##STR01614## ##STR01615## ##STR01616## ##STR01617## ##STR01618##

[0501] In some embodiments, L.sub.A is selected from the group consisting of the structures of LIST 2, LIST 5, and LIST 7. In some embodiments, L.sub.B is selected from the group consisting of the structures of LIST 10, LIST 11, and LIST 11. In some embodiments, L.sub.A is selected from the group consisting of the structures of LIST 7 of L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) consisting of L.sub.A1(R469)(R1)(R1)(R1) to L.sub.A178(R774)(R774)(R774)(R774) and L.sub.A263(R469)(R1)(R1)(R1) to L.sub.A290(R774)(R774)(R774)(R774) and L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) consisting of L.sub.A179(R1)(R1)(R1)(R1) to L.sub.A262(R774)(R774)(R774)(R774) and L.sub.A291(R1)(R1)(R1)(R1) to L.sub.A296(R774)(R774)(R774)(R774) defined herein and L.sub.B is selected from the group consisting of LIST 12 of L.sub.Bk consisting of L.sub.B1 to L.sub.B541 as defined herein.

[0502] In some embodiments, the compound can be Ir(L.sub.A).sub.3, Ir(L.sub.A).sub.2(L.sub.B), Ir(L.sub.A)(L.sub.B).sub.2, Ir(L.sub.A).sub.2(L.sub.C), Ir(L.sub.A)(L.sub.C).sub.2, or Ir(L.sub.A)(L.sub.B)(L.sub.C).

[0503] In some of these embodiments, the compound can be Ir(L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M)).sub.3 consisting of the compounds of Ir(L.sub.A1(R469)(R1)(R1)(R1)).sub.3 to Ir(L.sub.A178(R774)(R774)(R774)(R774)).sub.3 and Ir(L.sub.A263(R469)(R1)(R1)(R1)).sub.3 to Ir(L.sub.A290(R774)(R774)(R774)(R774)).sub.3, Ir(L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M)).sub.3 consisting of the compounds of Ir(L.sub.A179(R1)(R1)(R1)(R1)).sub.3 to Ir(L.sub.A262(R774)(R774)(R774)(R774)).sub.3 and Ir(L.sub.A291(R1)(R1)(R1)(R1)).sub.3 to Ir(L.sub.A296(R774)(R774)(R774)(R774)).sub.3, Ir(L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M)).sub.2(L.sub.B), Ir(L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M)).sub.2(L.sub.B), Ir(L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M))(L.sub.B).sub.2, Ir(L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M))(L.sub.B).sub.2, Ir(L.sub.A).sub.2(L.sub.Bk), Ir(L.sub.A)(L.sub.Bk).sub.2, Ir(L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M)).sub.2(L.sub.Bk) consisting of the compounds from Ir(L.sub.A1(R469)(R1)(R1)(R1)).sub.2(L.sub.B1) to Ir(L.sub.A178(R774)(R774)(R774)(R774)).sub.2(L.sub.B541) and Ir(L.sub.A263(R469)(R1)(R1)(R1)).sub.2(L.sub.B1) to Ir(L.sub.A290(R774)(R774)(R774)(R774)).sub.2(L.sub.B541), Ir(L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M)).sub.2(L.sub.Bk) consisting of the compounds from Ir(L.sub.A179(R1)(R1)(R1)(R1)).sub.2(L.sub.B1) to Ir(L.sub.A262(R774)(R774)(R774)(R774)).sub.2(L.sub.B541) and Ir(L.sub.A291(R1)(R1)(R1)(R1)).sub.2(L.sub.B1) to Ir(L.sub.A296(R774)(R774)(R774)(R774)).sub.2(L.sub.B541), Ir(L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M))(L.sub.Bk).sub.2 consisting of the compounds from Ir(L.sub.A1(R469)(R1)(R1)(R1))(L.sub.B1).sub.2 to Ir(L.sub.A178(R774)(R774)(R774)(R774))(L.sub.B541).sub.2 and Ir(L.sub.A263(R469)(R1)(R1)(R1))(L.sub.B1).sub.2 to Ir(L.sub.A290(R774)(R774)(R774)(R774))(L.sub.B541).sub.2, Ir(L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M))(L.sub.Bk).sub.2 consisting of the compounds from Ir(L.sub.A179(R1)(R1)(R1)(R1))(L.sub.B1).sub.2 to Ir(L.sub.A262(R774)(R774)(R774)(R774))(L.sub.B541).sub.2 and Ir(L.sub.A291(R1)(R1)(R1)(R1))(L.sub.B1).sub.2 to Ir(L.sub.A296(R774)(R774)(R774)(R774))(L.sub.B541).sub.2, Ir(L.sub.Aia(R.sup.J)(R.sup.K)

(R.sup.L)(R.sup.M)).sub.2(L.sub.Cj-I), consisting of the compounds of Ir(L.sub.A1(R469)(R1)(R1)(R1)).sub.2(L.sub.C1-I) to Ir(L.sub.A178(R774)(R774)(R774)(R774)).sub.2(L.sub.C1416-I) and Ir(L.sub.A263(R469)(R1)(R1)(R1)).sub.2(L.sub.C1-I) to Ir(L.sub.A290(R774)(R774)(R774)(R774)).sub.2(L.sub.C1416-I), Ir(L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M)).sub.2(L.sub.Cj-I) consisting of the compounds of Ir(L.sub.A179(R1)(R1)(R1)(R1)).sub.2(L.sub.C1-II) to Ir(L.sub.A262(R774)(R774)(R774)(R774)).sub.2(L.sub.C1416-I) and Ir(L.sub.A291(R1)(R1)(R1)(R1)).sub.2(L.sub.C1-II) to Ir(L.sub.A296(R774)(R774)(R774)(R774)).sub.2(L.sub.C1416-II), Ir(L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M)).sub.2(L.sub.Cj-II) consisting of the compounds of Ir(L.sub.A1(R469)(R1)(R1)(R1)).sub.2(L.sub.C1-II) to Ir(L.sub.A178(R774)(R774)(R774)(R774)).sub.2(L.sub.C1416-II) and Ir(L.sub.A263(R469)(R1)(R1)(R1)).sub.2(L.sub.C1-II) to Ir(L.sub.A290(R774)(R774)(R774)(R774)).sub.2(L.sub.C1416-II), Ir(L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M)).sub.2(L.sub.Cj-II) consisting of the compounds of Ir(L.sub.A179(R1)(R1)(R1)(R1)).sub.2(L.sub.C1-II) to Ir(L.sub.A262(R774)(R774)(R774)(R774)).sub.2(L.sub.C1416-II) and Ir(L.sub.A291(R1)(R1)(R1)(R1)).sub.2(L.sub.C1-II) to Ir(L.sub.A296(R774)(R774)(R774)(R774)).sub.2(L.sub.C1416-II), Ir(L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M))(L.sub.Bk)(L.sub.C-II) consisting of the compounds of Ir(L.sub.A1(R469)(R1)(R1)(R1))(L.sub.B1)(L.sub.C1-I) to Ir(L.sub.A178(R774)(R774)(R774)(R774))(L.sub.B541)(L.sub.C1416-I) and Ir(L.sub.A263(R469)(R1)(R1)(R1))(L.sub.B1)(L.sub.C1-I) to Ir(L.sub.A290(R774)(R774)(R774)(R774))(L.sub.B541)(L.sub.C1416-I), Ir(L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M))(L.sub.Bk)(L.sub.Cj-II) consisting of the compounds of Ir(L.sub.A179(R1)(R1)(R1)(R1))(L.sub.B1)(L.sub.C1-I) to Ir(L.sub.A262(R774)(R774)(R774)(R774))(L.sub.B541)(L.sub.C1416-I) and Ir(L.sub.A291(R1)(R1)(R1)(R1))(L.sub.B1)(L.sub.C1-I) to Ir(L.sub.A296(R774)(R774)(R774)(R774))(L.sub.B541)(L.sub.C1416-I), Ir(L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M))(L.sub.Bk)(L.sub.C1416-II) consisting of the compounds of Ir(L.sub.A1(R469)(R1)(R1)(R1))(L.sub.B1)(L.sub.C1-II) to Ir(L.sub.A178(R774)(R774)(R774)(R774))(L.sub.B541)(L.sub.C1416-II) and Ir(L.sub.A263(R469)(R1)(R1)(R1))(L.sub.B1)(L.sub.C1-II) to Ir(L.sub.A290(R774)(R774)(R774)(R774))(L.sub.B541)(L.sub.C1416-II), or Ir(L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M))(L.sub.Bk)(L.sub.Cj-II) consisting of the compounds of Ir(L.sub.A179(R1)(R1)(R1)(R1))(L.sub.B1)(L.sub.C1-II) to Ir(L.sub.A262(R774)(R774)(R774)(R774))(L.sub.B541)(L.sub.C1416-II) and Ir(L.sub.A291(R1)(R1)(R1)(R1))(L.sub.B1)(L.sub.C1-II) to Ir(L.sub.A296(R774)(R774)(R774)(R774))(L.sub.B541)(L.sub.C1416-II), wherein L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), L.sub.Bk, L.sub.Cj-II and L.sub.Cj-II are all defined herein.

[0504] It should be understood that all the above formulas related to L.sub.Aia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) and L.sub.Aib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) can be equally applied to the equivalent ones of L.sub.A'ia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), L.sub.A'ib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), L.sub.A''ia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M), L.sub.A''ib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) L.sub.A'''ia(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) (R.sup.M), and L.sub.A'''ib(R.sup.J)(R.sup.K)(R.sup.L)(R.sup.M) respectively.

[0505] In some embodiments, the compound is selected from the group consisting of the structures of the following LIST 16:

##STR01619## ##STR01620## ##STR01621## ##STR01622## ##STR01623## ##STR01624##
##STR01625## ##STR01626## ##STR01627## ##STR01628##

[0506] In some embodiments, the compound comprises a structure of Formula III:

##STR01629##

wherein: [0507] M.sup.1 is Pd or Pt; [0508] moieties E and F are each independently monocyclic or polycyclic ring structure, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring; [0509] Z.sup.3 and Z.sup.4 are each independently C or N; [0510] K.sup.1, K.sup.2, K.sup.3, and K.sup.4 are each independently selected from the group consisting of a direct bond, O, and S, wherein at

least two of them are direct bonds; [0511] L^{sup.1}, L^{sup.2}, and L^{sup.3} are each independently absent or selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO₂, CR, CRR', SiRR', GeRR', alkylene, cycloalkyl, aryl, cycloalkylene, arylene, heteroarylene, and combinations thereof, wherein at least one of L' and L^{sup.2} is present; [0512] R^{sup.E} and R^{sup.F} each independently represent zero, mono, or up to a maximum allowed number of substitutions; [0513] each of R, R', R^{sup.E}, and R^{sup.F} is independently a hydrogen or a substituent selected from the group consisting of the General Substituents defined herein; and [0514] two adjacent R^{sup.A}, R^{sup.B}, R^{sup.C}, R^{sup.E}, and R^{sup.F} can be joined or fused together to form a ring.

[0515] In some embodiments of Formula III, each of R^{sup.1}, R^{sup.2}, R^{sup.3}, R^{sup.4}, R, R', R^{sup.A}, R^{sup.B}, R^{sup.C}, R^{sup.D}, R^{sup.E}, and R^{sup.F} is independently a hydrogen or a substituent selected from the group consisting of the Preferred General Substituents.

[0516] In some embodiments, moieties E and F are each independently monocyclic or polycyclic ring structures, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered or 6-membered carbocyclic or heterocyclic rings.]

[0517] In some embodiments of Formula III, at least one R^{sup.A}, R^{sup.B}, R^{sup.C}, R^{sup.D}, R^{sup.E}, or R^{sup.F} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.A} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.B} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.C} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.D} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.E} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.F} is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^{sup.A}, R^{sup.B}, R^{sup.C}, R^{sup.D}, R^{sup.E}, or R^{sup.F} is selected from the group consisting of the Preferred General Substituents defined herein.

[0518] In some embodiments of Formula III, at least one of R, R', R^{sup.A}, R^{sup.B}, R^{sup.C}, R^{sup.D}, R^{sup.E}, or R^{sup.F} is partially or fully deuterated. In some embodiments, at least one R^{sup.A} is partially or fully deuterated. In some embodiments, at least one R^{sup.B} is partially or fully deuterated. In some embodiments, at least one R^{sup.C} is partially or fully deuterated. In some embodiments, at least one R^{sup.D} is partially or fully deuterated. In some embodiments, at least one R^{sup.E} is partially or fully deuterated. In some embodiments, at least one R^{sup.F} is partially or fully deuterated. In some embodiments, at least one of R or R' is partially or fully deuterated.

[0519] In some embodiments of Formula III, at least one R^{sup.A} is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R^{sup.A} is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R^{sup.A} is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R^{sup.A} is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R^{sup.A} is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0520] In some embodiments of Formula III, at least one R^{sup.B} is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R^{sup.B} is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R^{sup.B} is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R^{sup.B} is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R^{sup.B} is or comprises an electron-withdrawing group from the Pi-EWG LIST as

defined herein.

[0521] In some embodiments of Formula III, at least one R^{sup.C} is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R^{sup.C} is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R^{sup.C} is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R^{sup.C} is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R^{sup.C} is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0522] In some embodiments of Formula III, at least one R^{sup.D} is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R^{sup.D} is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R^{sup.D} is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R^{sup.D} is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R^{sup.D} is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0523] In some embodiments of Formula III, at least one R^{sup.E} is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R^{sup.E} is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R^{sup.E} is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R^{sup.E} is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R^{sup.E} is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0524] In some embodiments of Formula III, at least one R^{sup.F} is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R^{sup.F} is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R^{sup.F} is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R^{sup.F} is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R^{sup.F} is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0525] In some embodiments, Formula III comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, Formula III comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, Formula III comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, Formula III comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, Formula III comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0526] In some embodiments of Formula III, moiety E and moiety F are both 6-membered aromatic rings.

[0527] In some embodiments of Formula III, moiety F is a 5-membered or 6-membered heteroaromatic ring.

[0528] In some embodiments of Formula III, L^{sup.1} is O or CRR'.

[0529] In some embodiments of Formula III, Z^{sup.4} is N and Z^{sup.3} is C. In some embodiments of Formula III, Z^{sup.4} is C and Z^{sup.3} is N.

[0530] In some embodiments of Formula III, L^{sup.2} is a direct bond. In some embodiments of Formula III, L^{sup.2} is NR.

[0531] In some embodiments of Formula III, K^{sup.1}, K^{sup.2}, K^{sup.3}, and K^{sup.4} are all direct

bonds. In some embodiments of Formula III, one of K^{sup.1}, K^{sup.2}, K^{sup.3}, and K^{sup.4} is O. [0532] In some embodiments of Formula III, the compound is selected from the group consisting of compounds having the formula of Pt(L_{sub}.A')(Ly):

##STR01630## [0533] wherein L_{sub}.A' is selected from the group consisting of the structures shown in the following LIST 17:

##STR01631## ##STR01632## ##STR01633## [0534] wherein L_{sub}.y is selected from the group consisting of the structures shown in the following LIST 18:

##STR01634## ##STR01635## ##STR01636## ##STR01637## ##STR01638## ##STR01639## ##STR01640## [0535] wherein at least one of R^{sup.A}, R^{sup.B}, R^{sup.C}, R^{sup.E}, R^{sup.F}, R^{sup.G}, R, R', R'', R''', R^{sup.X}, R' is selected from the structures of LIST 1, wherein each R^{sup.A}, R^{sup.B}, R^{sup.C}, R^{sup.E}, R^{sup.F}, R^{sup.G}, R, R', R'', R''', R^{sup.X}, R' is independently hydrogen or a substituents selected from the group consisting of the General Substituents defined herein.

[0536] In some embodiments of Formula III, each R^{sup.A}, R^{sup.B}, R^{sup.C}, R^{sup.E}, R^{sup.F}, R^{sup.G}, R, R', R'', R''', R^{sup.X}, and R^{sup.Y} is independently selected from the list consisting of structures of Formula I defined herein and the following LIST 19:

##STR01641## ##STR01642## ##STR01643## ##STR01644## ##STR01645## ##STR01646## ##STR01647## ##STR01648## ##STR01649## ##STR01650## ##STR01651## ##STR01652## ##STR01653## ##STR01654## ##STR01655## ##STR01656## ##STR01657## ##STR01658## ##STR01659## ##STR01660## ##STR01661## ##STR01662## ##STR01663## ##STR01664## ##STR01665## ##STR01666## ##STR01667## ##STR01668## ##STR01669## ##STR01670## ##STR01671## ##STR01672## ##STR01673##

##STR01674## ##STR01675## ##STR01676## ##STR01677## ##STR01678## ##STR01679## ##STR01680## ##STR01681## ##STR01682## ##STR01683## ##STR01684## ##STR01685## ##STR01686## ##STR01687## ##STR01688## ##STR01689## ##STR01690## ##STR01691##

In some embodiments, at least one of R^{sup.A}, R^{sup.B}, R^{sup.C}, R^{sup.E}, R^{sup.F}, R^{sup.G}, R, R', R'', R''', R^{sup.X}, and R^{sup.Y} is selected from the group consisting of the structures of the following LIST 20:

##STR01692## ##STR01693## ##STR01694## [0537] wherein each of Q^{sup.A}, Q^{sup.B}, Q^{sup.C}, Q^{sup.D} and Q^{sup.E} independently represents mono to the maximum allowable substitution, or no substitution; [0538] wherein each Q^{sup.A}, Q^{sup.B}, Q^{sup.C}, Q^{sup.D}, Q^{sup.E}, Q^{sup.A1}, Q^{sup.B1}, Q^{sup.C1}, Q^{sup.D1} and Q^{sup.E1} is independently a hydrogen, or a substituent selected from the group consisting of the General Substituents defined herein; [0539] each Y^{sup.aa} and Y^{sup.bb} is independently selected from the group consisting of a direct bond, BR, BRR', NR, PR, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO_{sub}.2, CR, CRR', SiRR', GeRR', alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof; and [0540] any two substituents can be joined or fused to form a ring.

Claims

1. A compound having a first ligand L_{sub}.A comprising a structure of Formula I: ##STR01695## wherein: each of ring A and ring B is independently a 5-membered or 6-membered carbocyclic or heterocyclic ring; each of X^{sup.1} to X^{sup.3} is independently C or N; each of Y^{sup.1} and Y^{sup.2} is independently C, Si, or Ge; each  independently represents a single bond or a double bond; L is selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO_{sub}.2, CRR', SiRR', and GeRR'; each of R^{sup.A} and R^{sup.B} independently represent mono to the maximum allowable substitutions, or no substitutions; each R^{sup.1}, R^{sup.2}, R^{sup.3}, R^{sup.4}, R, R', R^{sup.A}, and R^{sup.B} is independently hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl,

boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and combinations thereof; L.sub.A is coordinated to a metal M selected from the group consisting of Ir, Rh, R.sub.e, Ru, Os, Pt, Pd, Ag, Au, and Cu; M can be coordinated to other ligands; L.sub.A can be a monodentate, bidentate, tridentate, tetradentate, pentadentate, or hexadentate ligand; any two substituents can be joined or fused to form a ring; when M is Ir, at least one of the following statements is true: (1) Ir is not directly coordinated to ring A or ring B; or (2) at least one of R.sup.1 to R.sup.4 is not H; or (3) at least one of Y.sup.1 or Y.sup.2 is Si or Ge; or (4) L is not CRR'; with the proviso that when M is Ir, L.sub.A does not comprise ##STR01696## and with the proviso that the compound is not ##STR01697##

2. The compound of claim 1, wherein each R.sup.1, R.sup.2, R.sup.3, R.sup.4, R, R', R.sub.A, and R.sub.B is independently hydrogen, or a substituent selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, heteroalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, aryl, heteroaryl, nitrile, isonitrile, sulfanyl, and combinations thereof.

3. The compound of claim 1, wherein each of ring A and ring B is independently selected from the group consisting of benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, and triazole; and/or wherein each of X.sup.1, X.sup.2, and X.sup.3 is C or wherein at least one of X.sup.1, X.sup.2, or X.sup.3 is N; and/or wherein L is a direct bond, O, S, Se, BRR', CRR', SiRR', or GeRR'; and/or wherein the metal is Ir, Pt or Pd.

4. The compound of claim 1, wherein the first ligand L.sub.A comprises a structure of Formula II: ##STR01698## wherein: each of moiety C and moiety D is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring; each of X.sup.4 to X.sup.7, Z.sup.1, and Z.sup.2 is independently C or N; L.sub.X is selected from the group consisting of a direct bond BR, BRR', NR, PR, P(O)R, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO.sub.2, CR, CRR', SiRR', and GeRR'; K.sup.1 and K.sup.2 are each independently selected from the group consisting of a direct bond, O, S, N(R.sub.α), P(R.sub.α), B(R.sub.α), C(R.sub.α)(R.sub.β), and Si(R.sub.α)(R.sub.β); each of R.sub.C and R.sub.D independently represents mono to the maximum allowable substitutions, or no substitutions; each R, R', R.sub.α, R.sub.β, R.sub.C, and R.sub.D is independently hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and combinations thereof; any two substituents can be joined or fused to form a ring; L.sub.A is coordinated to the metal M by the dashed lines; M can be coordinated to other ligands; L.sub.A may be joined with other ligands to comprise a tridentate, tetradentate, pentadentate, or hexadentate ligand; and at least one of the following conditions is true: (1) at least one R.sub.C or R.sub.D comprises a structure of Formula I; (2) moiety C comprises a structure of Formula I; (3) moiety D comprises a structure of Formula I; and (4) at least one R.sub.C and at least one R.sub.D are joined to form a structure comprising Formula I.

5. The compound of claim 4, wherein each of moiety C and moiety D is independently selected from the group consisting of the following Cyclic Moiety List: benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, imidazole derived carbene, Pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, triazole, naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, phenanthro[3,2-b]benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, benzimidazole derived carbene, aza-benzimidazole derived carbene, benzobenzimidazole, aza-benzobenzimidazole,

carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene; and/or wherein L.sup.X is a direct bond, S, O, or Se; and/or wherein Z.sup.1 is N or carbene C and Z.sup.2 is C; and/or wherein K.sup.1 is a direct bond, S, or O; and/or wherein K.sup.2 is a direct bond, S, or O; and/or wherein the structure of Formula I is not fused to the structure of Formula II; and/or wherein one of R.sup.A or R.sup.B is bonded to moiety C or moiety D.

6. The compound of claim 4, wherein at least one of ring A or ring B is part of moiety C or moiety D, but neither ring A nor ring B is the ring containing Z.sup.1 or the ring containing Z.sup.2.

7. The compound of claim 1, wherein Formula I has a structure selected from the group consisting of: ##STR01699## wherein: each of X.sup.10 to X.sup.15 is independently C or N; and any two substituents can be joined or fused to form a ring.

8. The compound of claim 1, wherein Formula I has a structure selected from the group consisting of: ##STR01700## ##STR01701## ##STR01702## ##STR01703## ##STR01704## ##STR01705## ##STR01706## ##STR01707## ##STR01708## ##STR01709## ##STR01710## ##STR01711## ##STR01712## ##STR01713## ##STR01714##

9. The compound of claim 1, wherein the ligand L.sub.A is selected from the group consisting of: ##STR01715## ##STR01716## ##STR01717## ##STR01718## ##STR01719## ##STR01720## ##STR01721## ##STR01722## ##STR01723## ##STR01724## ##STR01725## ##STR01726## ##STR01727## ##STR01728## ##STR01729## ##STR01730## ##STR01731## ##STR01732## ##STR01733## ##STR01734## ##STR01735## ##STR01736## ##STR01737## ##STR01738## ##STR01739## ##STR01740## ##STR01741## ##STR01742## ##STR01743## ##STR01744## ##STR01745## ##STR01746## ##STR01747## ##STR01748## wherein: T is selected from the group consisting of B, Al, Ga, and In; X.sub.1 to X.sub.58 are each independently C or N; K.sup.1' is selected from the group consisting of a direct bond, BR.sub.e, NR.sub.e, PR.sub.e, O, S, Se, C=O, S=O, SO.sub.2, CR.sub.eR.sub.f, SiR.sub.eR.sub.f, and GeR.sub.eR.sub.f; each occurrence of Y', Y'', Y.sub.1, Y.sup.A, Y.sup.B, and Y.sup.C is independently selected from the group consisting of BR.sub.e, NR.sub.e, PR.sub.e, O, S, Se, C=O, S=O, SO.sub.2, CR.sub.eR.sub.f, SiR.sub.eR.sub.f, and GeR.sub.eR.sub.f; each of Y.sup.1 to Y.sup.23 is independently selected from the group consisting of carbon and nitrogen; W.sup.1 is selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C—O, C=S, C=Se, C=NR, C=CRR', S=O, SO.sub.2, CRR', P(O)R, SiRR', and GeRR'; W.sup.2 is B, N, or P; each R.sup.A1, R.sup.B1, R.sup.B2, R.sup.B3, R.sub.a, R.sub.b, R.sub.c, and R.sub.d independently represent from mono to the maximum possible number of substitutions, or no substitution; each R, R', R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1, R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sub.e, R.sub.f, R.sup.A1, R.sup.B1, R.sup.B2, and R.sup.B3 is independently a hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and combinations thereof; and any two substituents can be joined or fused to form a ring; and one of the following conditions (1) to (3) is true: (1) at least one of R, R', R.sub.a1, R.sub.b1, R.sub.c1, R.sub.d1, R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sup.A1, R.sup.B1, R.sup.B2, R.sup.B3, R.sub.e, and R.sub.f comprises a structure of Formula I or is selected from the structures of LIST 1 or LIST 2; (2) at least one pair selected from (X.sub.50, X.sub.51), (X.sub.52, X.sub.53), (X.sub.51, X.sub.54), (X.sub.55, X.sub.56), or (X.sub.57, X.sub.58) is joined by a moiety selected from the group consisting of the structures of the following LIST 4: ##STR01749## (3) two R.sup.A1, R.sup.B1, R.sup.B2, R.sup.B3, R.sub.a, R.sub.b, R.sub.c, or R.sub.d are fused or fused to form a structure of Formula I, or a structure of LIST 1 or LIST 2; wherein LIST 1 consists of ##STR01750## wherein each of X.sup.10 to X.sup.15 is independently C or N; and any two substituents can be joined or fused to form a ring; and LIST 2 consists of ##STR01751##

##STR01752## ##STR01753## ##STR01754## ##STR01755## ##STR01756## ##STR01757##
##STR01758## ##STR01759## ##STR01760## ##STR01761## ##STR01762## ##STR01763##
##STR01764## ##STR01765## ##STR01766##

10. The compound of claim 1, wherein the ligand L.sub.A is selected from the group consisting of:
##STR01767## ##STR01768## ##STR01769## ##STR01770## ##STR01771## ##STR01772##
##STR01773## ##STR01774## ##STR01775## ##STR01776## ##STR01777## ##STR01778##
##STR01779## ##STR01780## ##STR01781## ##STR01782## ##STR01783## ##STR01784##
##STR01785## ##STR01786## ##STR01787## ##STR01788## ##STR01789## ##STR01790##
##STR01791## ##STR01792## ##STR01793## ##STR01794## ##STR01795## ##STR01796##
##STR01797## ##STR01798## ##STR01799## ##STR01800## ##STR01801## ##STR01802##
##STR01803## ##STR01804## ##STR01805## ##STR01806## ##STR01807## ##STR01808##
##STR01809## ##STR01810## ##STR01811## ##STR01812## ##STR01813## ##STR01814##
##STR01815## ##STR01816## ##STR01817## ##STR01818## ##STR01819## ##STR01820##
##STR01821## ##STR01822## ##STR01823## ##STR01824## ##STR01825## ##STR01826##
##STR01827## ##STR01828## ##STR01829## ##STR01830## ##STR01831## ##STR01832##
##STR01833## ##STR01834## ##STR01835## ##STR01836## ##STR01837## ##STR01838##
##STR01839## ##STR01840## ##STR01841## wherein: K.sup.1' is selected from the group
consisting of a direct bond, NR.sub.e, PR.sub.e, O, S, and Se; each of X.sub.50 to X.sub.59, and
Y.sub.2 to Y.sub.7 is independently selected from the group consisting of carbon and nitrogen; each
occurrence of X, Y, Y', Y'', and W.sup.1 is independently selected from the group consisting of a
direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O,
SO.sub.2, CRR', SiRR', and GeRR'; each of R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sup.AA,
R.sup.BB, and R.sup.C independently represents mono to the maximum allowable substitutions, or
no substitutions; each R, R', R.sub.a, R.sub.b, R.sub.c, R.sub.d, R.sup.AA, R.sup.BB, R.sup.C, and
R.sup.N is independently hydrogen or a substituent selected from the group consisting of
deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy,
amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl,
carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and
combinations thereof; the remaining variables are the same as previously defined; any two
substituents can be fused or joined to form a ring; any two substituents can be joined or fused to
form a ring; and one of the following conditions (1) to (3) is true: (1) at least one of R, R', R.sub.a,
R.sub.b, R.sub.c, R.sub.d, R.sup.AA, R.sup.BB, R.sup.C, and R.sup.N comprises a structure of
Formula I or is selected from a structure of LIST 1 or LIST 2 defined herein; or (2) at least one pair
of (X.sub.50, X.sub.51), (X.sub.52, X.sub.53), (X.sub.51, X.sub.54), (X.sub.55, X.sub.56),
(X.sub.57, X.sub.58), or (X.sub.53, X.sub.59) is joined to a moiety selected from the group
consisting of ##STR01842## or (3) two R.sup.AA, R.sup.BB, R.sup.C, R.sub.a, R.sub.b, R.sub.c,
or R.sub.d are joined or fused to form a structure of Formula I, or to a structure of LIST 1 or LIST
2; wherein LIST 1 consists of ##STR01843## wherein each of X.sub.10 to X.sub.15 is
independently C or N; and any two substituents can be joined or fused to form a ring; and LIST 2
consists of ##STR01844## ##STR01845## ##STR01846## ##STR01847## ##STR01848##
##STR01849## ##STR01850## ##STR01851## ##STR01852## ##STR01853## ##STR01854##
##STR01855## ##STR01856## ##STR01857##
